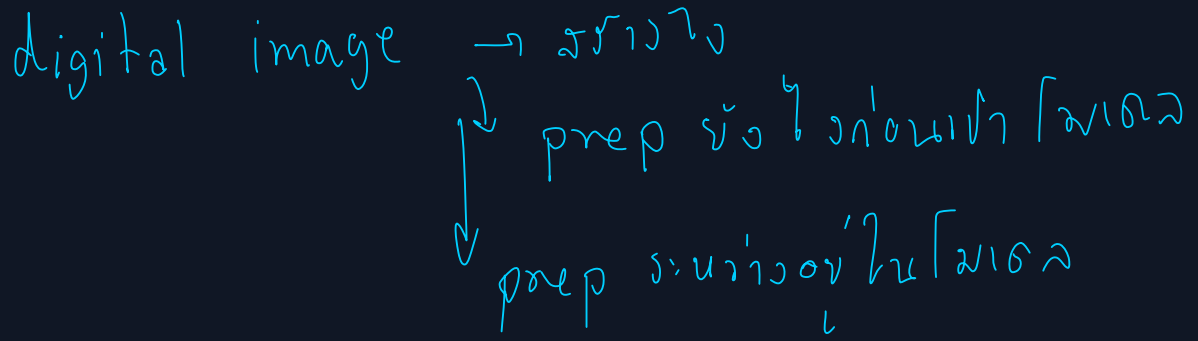
How a digital image is formed

Image data, class 1 of 2. From photons to pixels: cameras, sensors, medical scanners, sampling, quantization, and the first preprocessing decisions.



Asst. Prof. Thanapong Intharah, PhD

Visual Intelligence Laboratory · Khon Kaen University

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Today, in three hours

0:00–0:20	**A1 · Pinhole and lens cameras.** Why a bare sensor sees nothing, and what a camera really does.	CMU 16-385, Lecture 9
0:20–0:35	**A2 · The imaging sensor.** Photons to electrons to numbers: photodiodes, CCD vs CMOS.	CMU 16-385, Lectures 24–25
0:35–0:55	**A3 · Medical image formation.** X-ray, ultrasound, CT, MRI: very different physics, same arrays.	Bagci Lab BME 495, Lectures 1–2
0:55–1:10	**A4 · Sampling and quantization.** Bit depth, and what 8 bits throw away.	MIT OCW 6.003, Lecture 22
1:10–1:30	**A5 · Aliasing and downsampling.** Moire, Nyquist, and why naive resizing lies.	CMU 16-385, Lecture 3
1:30–1:40	*Break*	
1:40–2:30	**B · Representation and color.** Images as arrays, DICOM, color sensing, and the in-camera pipeline your JPEGs already went through.	CMU L2, Bagci, CMU L24–25
2:30–3:00	**C · First preprocessing.** Resizing and normalization: the two operations every pipeline starts with.	authored for this course

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PART A1 · 0:00–0:20

A sensor alone sees nothing: pinhole and lens cameras

Every dataset image began as light through an aperture. Focal length, aperture size, and lenses decide what the array will even contain.

SOURCE DECK

Geometric camera models, 16-385 Computer Vision, Fall 2023, Lecture 9 · Ioannis Gkioulekas, Carnegie Mellon University.
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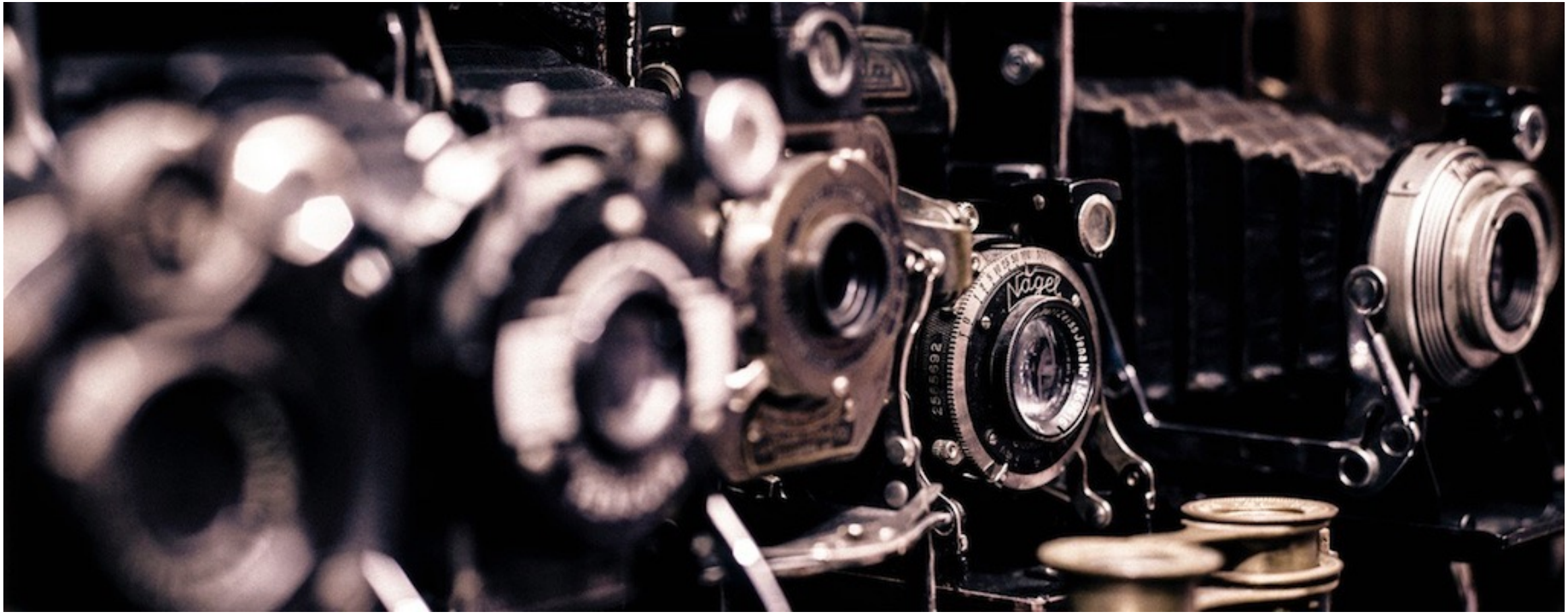
Geometric camera models



16-385 Computer Vision
Fall 2023, Lecture 9

http://16385.courses.cs.cmu.edu/

Some motivational imaging experiments

Let's say we have a sensor…

รูปภาพมาไง ?

digital sensor
(CCD or CMOS)

… and an object we like to photograph

real-world object

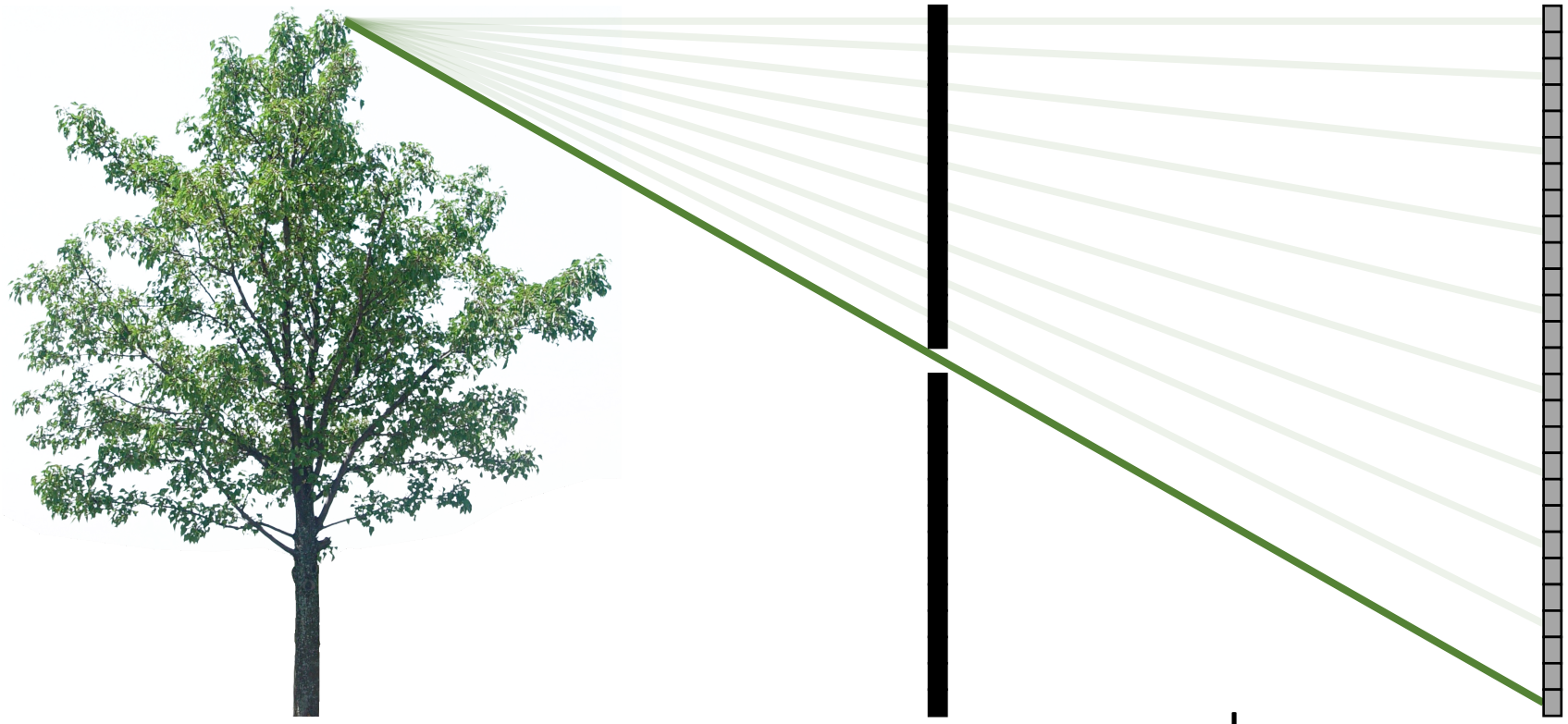
digital sensor (CCD or CMOS)

What would an image taken like this look like?

Bare-sensor imaging

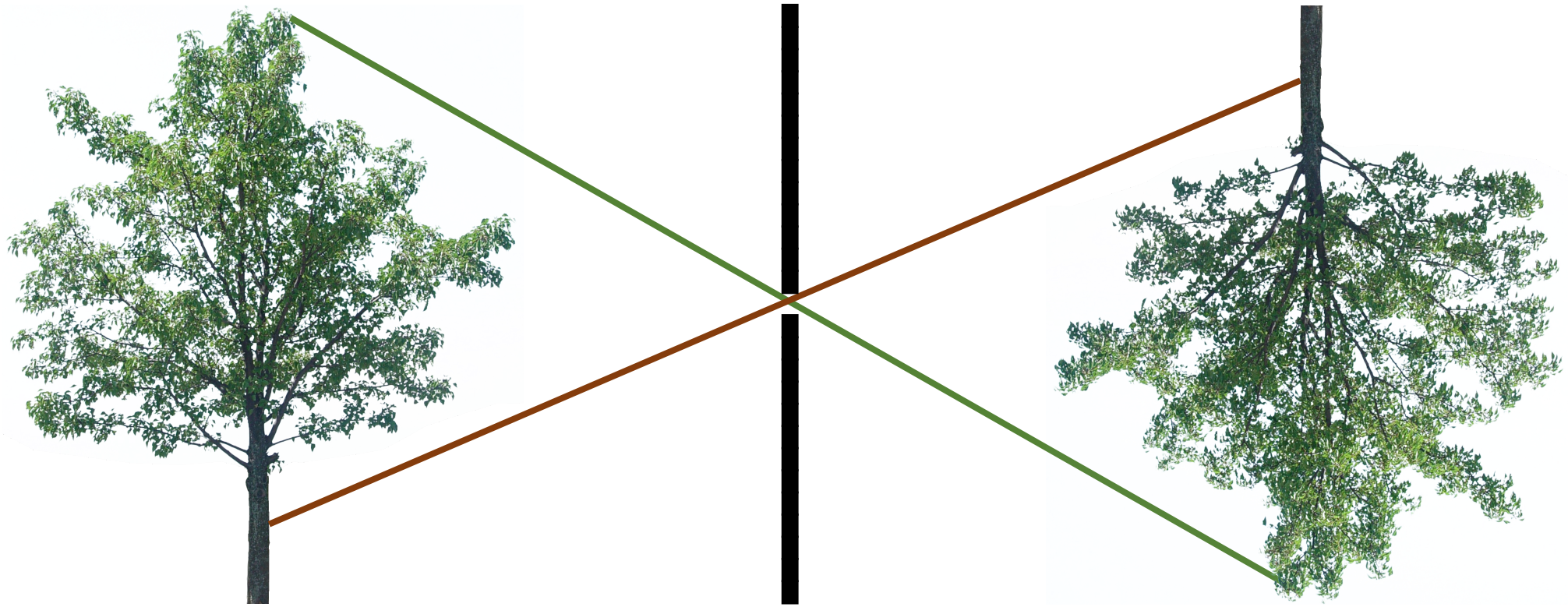
All scene points contribute to all sensor pixels

Pinhole imaging



Pinhole imaging

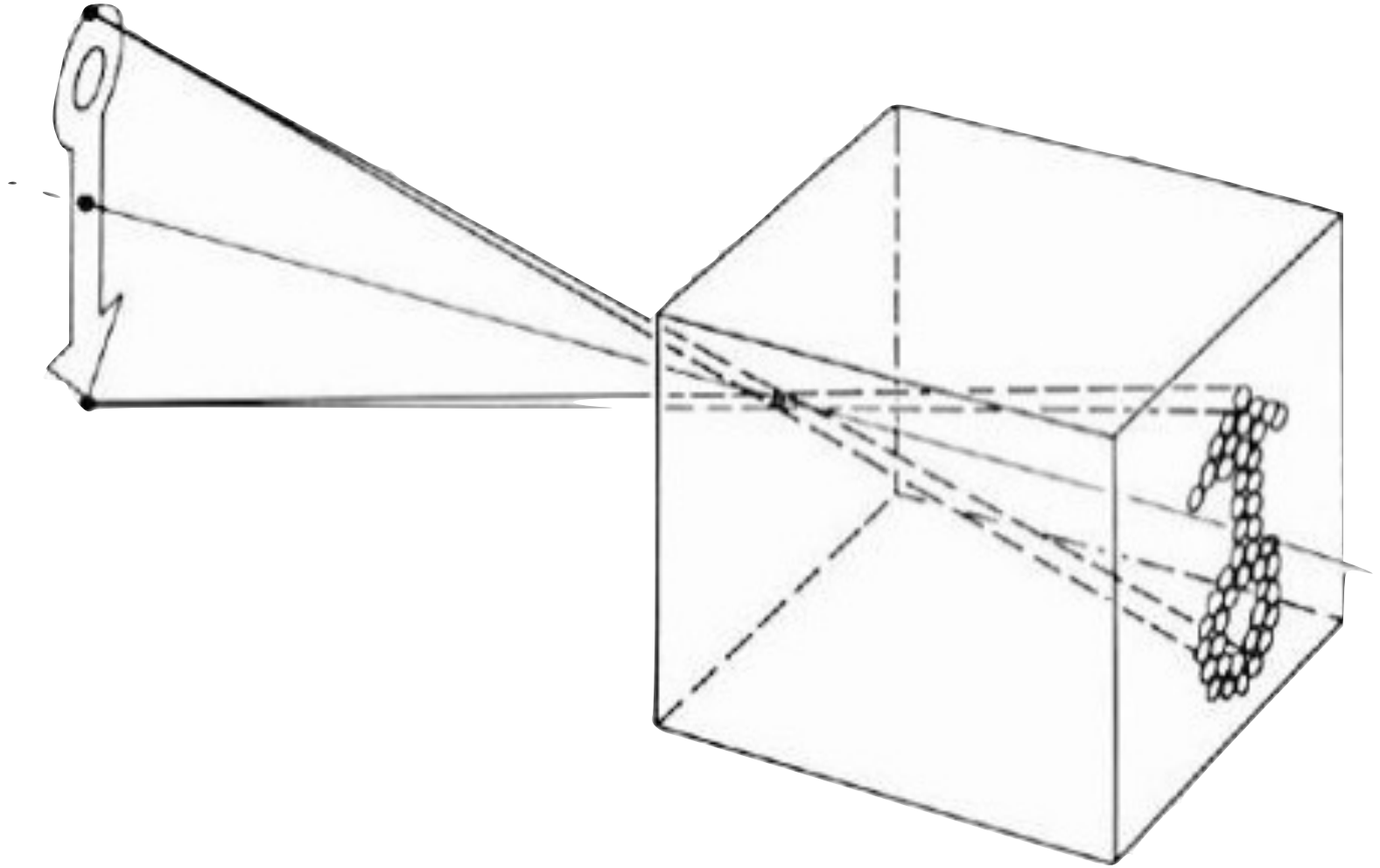
real-world object



copy of real-world object (inverted and scaled)

Pinhole camera

Pinhole camera a.k.a. camera obscura



Pinhole camera terms

barrier (diaphragm)

real-world object

pinhole (aperture)

digital sensor (CCD or CMOS)

Pinhole camera terms

real-world object

barrier (diaphragm)

pinhole (aperture)

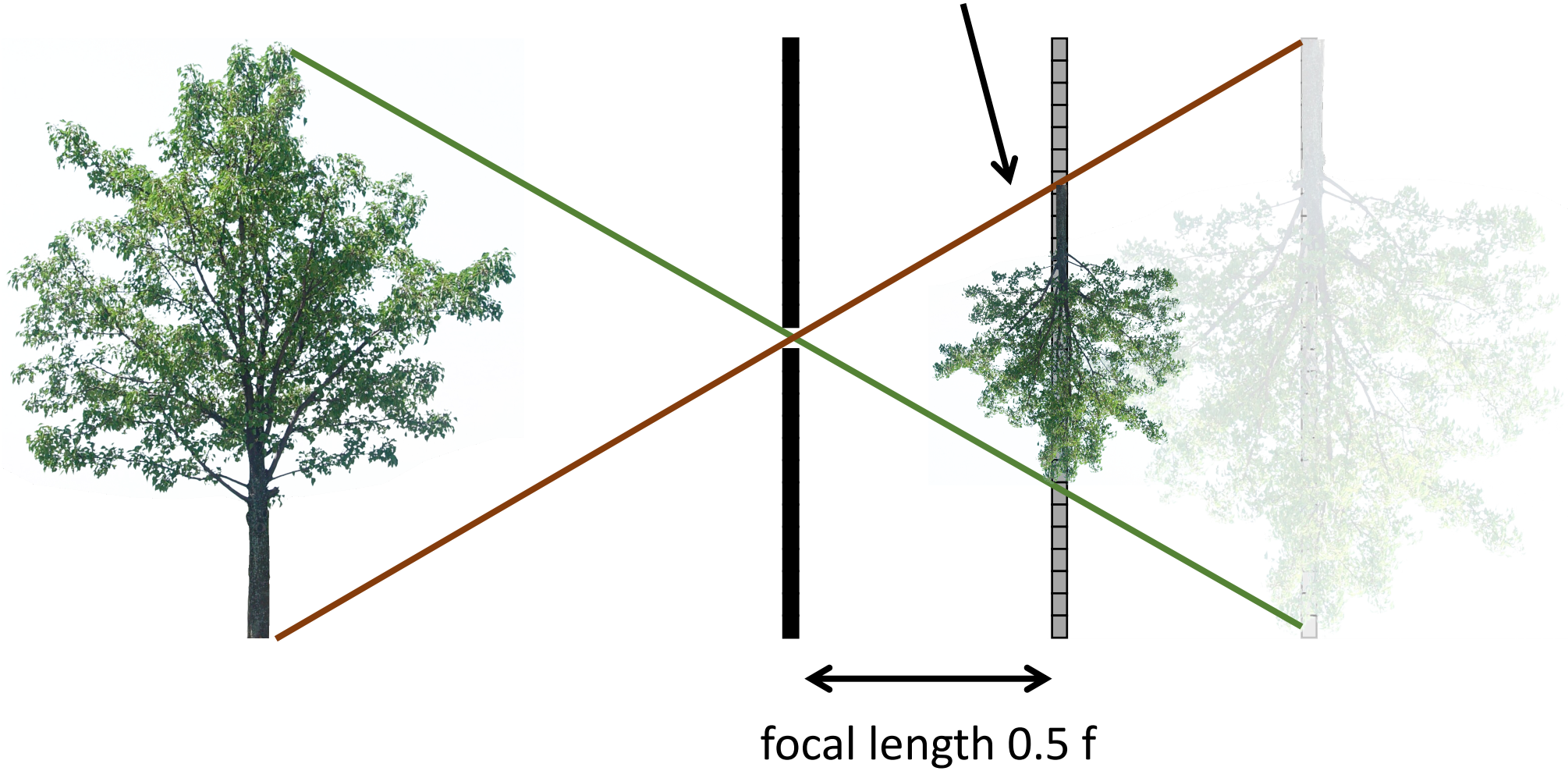
camera center (center of projection)

image plane

digital sensor (CCD or CMOS)

Focal length

What happens as we change the focal length?

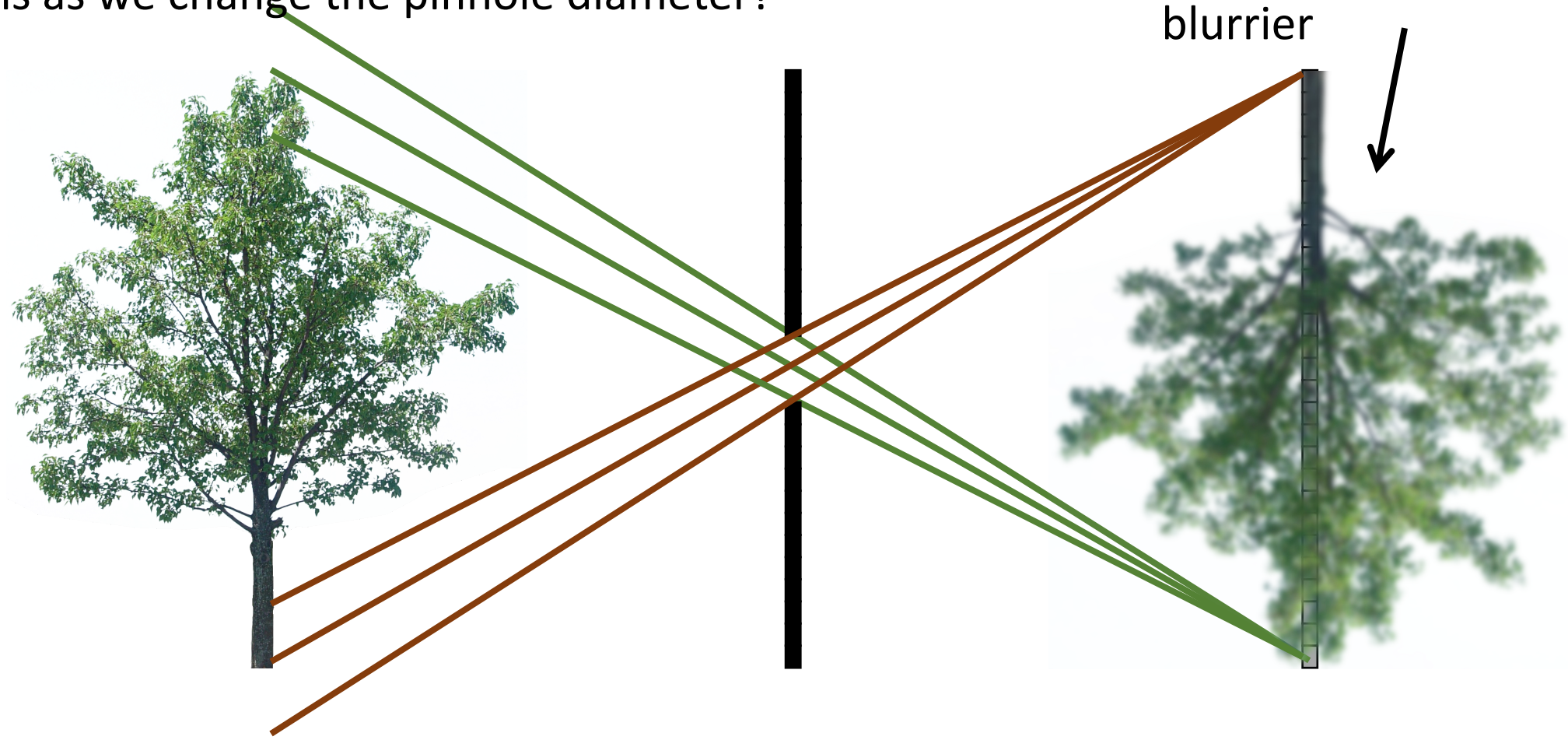


Pinhole size

What happens as we change the pinhole diameter?

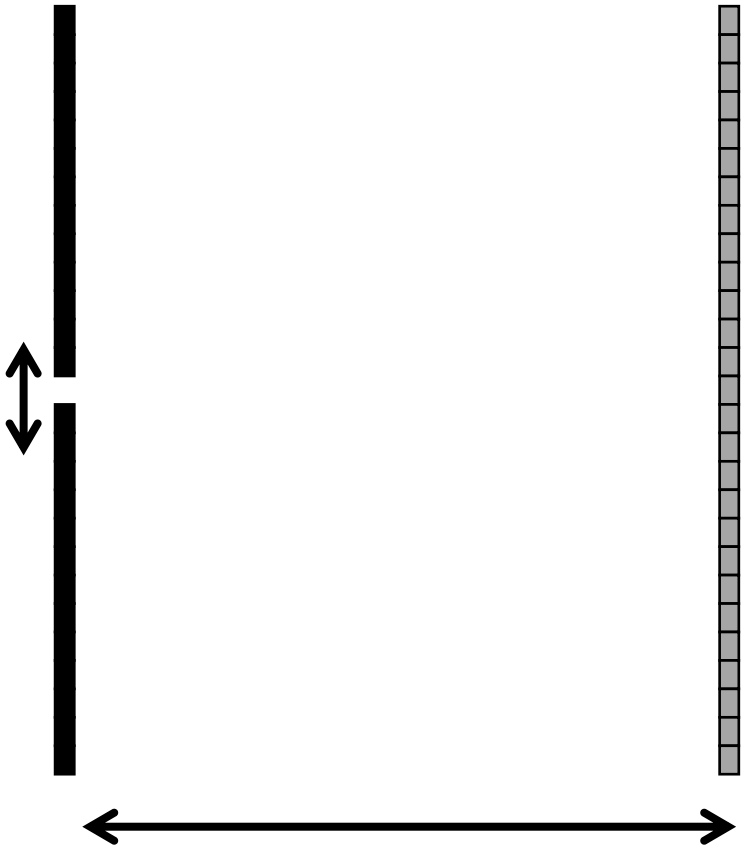
object projection becomes blurrier

real-world object



What about light efficiency?

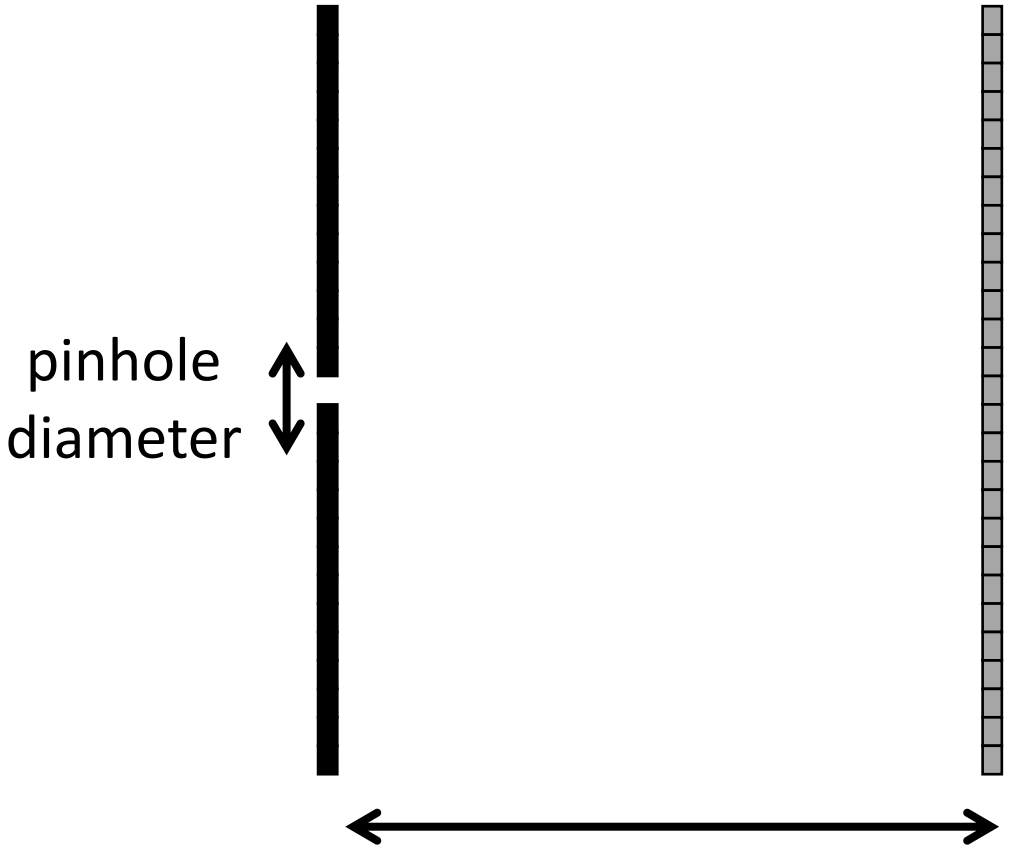
real-world object



- What is the effect of doubling the pinhole diameter?
- What is the effect of doubling the focal length?

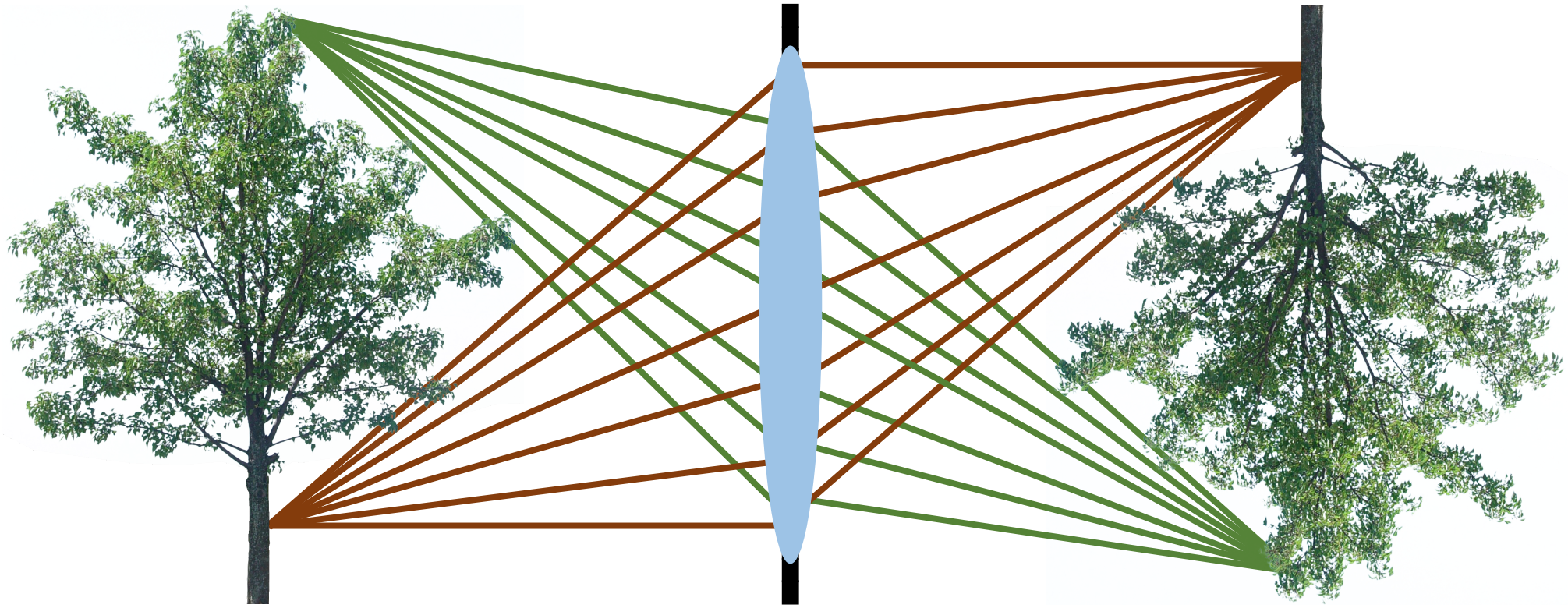
What about light efficiency?

real-world object



- 2x pinhole diameter → 4x light
- 2x focal length → ¼x light

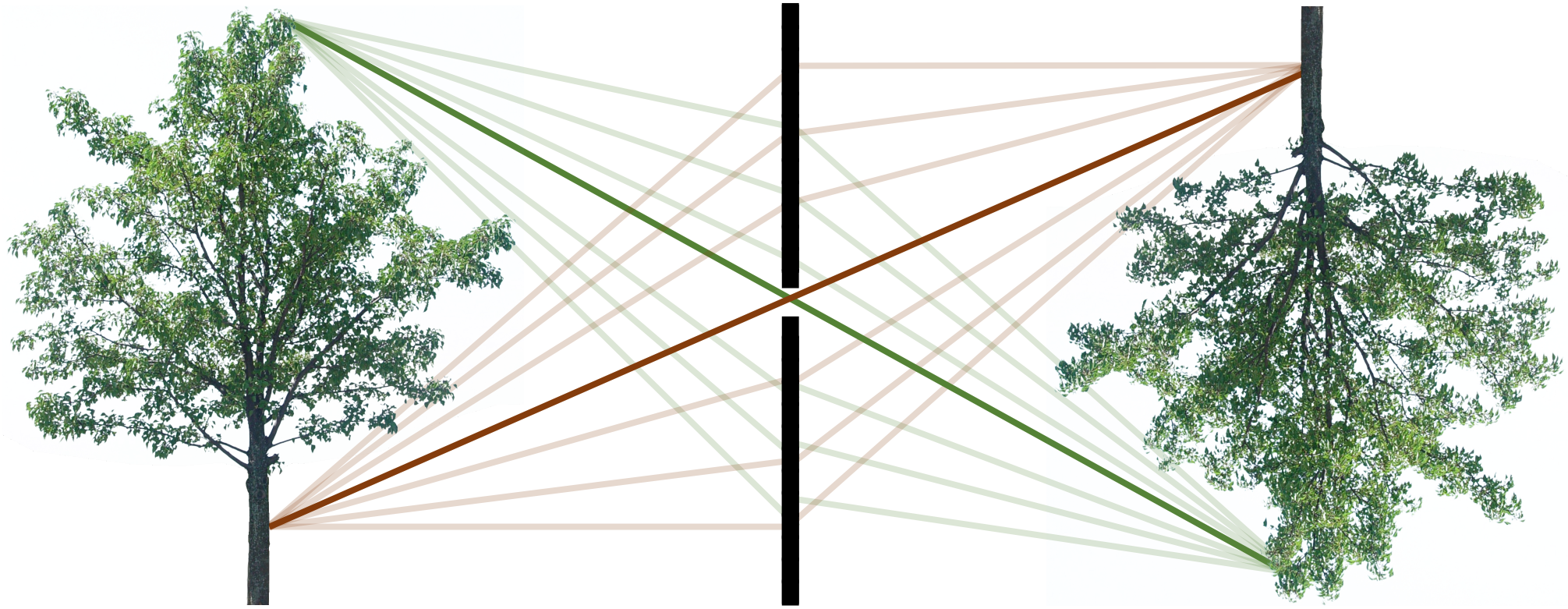
The lens camera



Lenses map “bundles” of rays from points on the scene to the sensor.

How does this mapping work exactly?

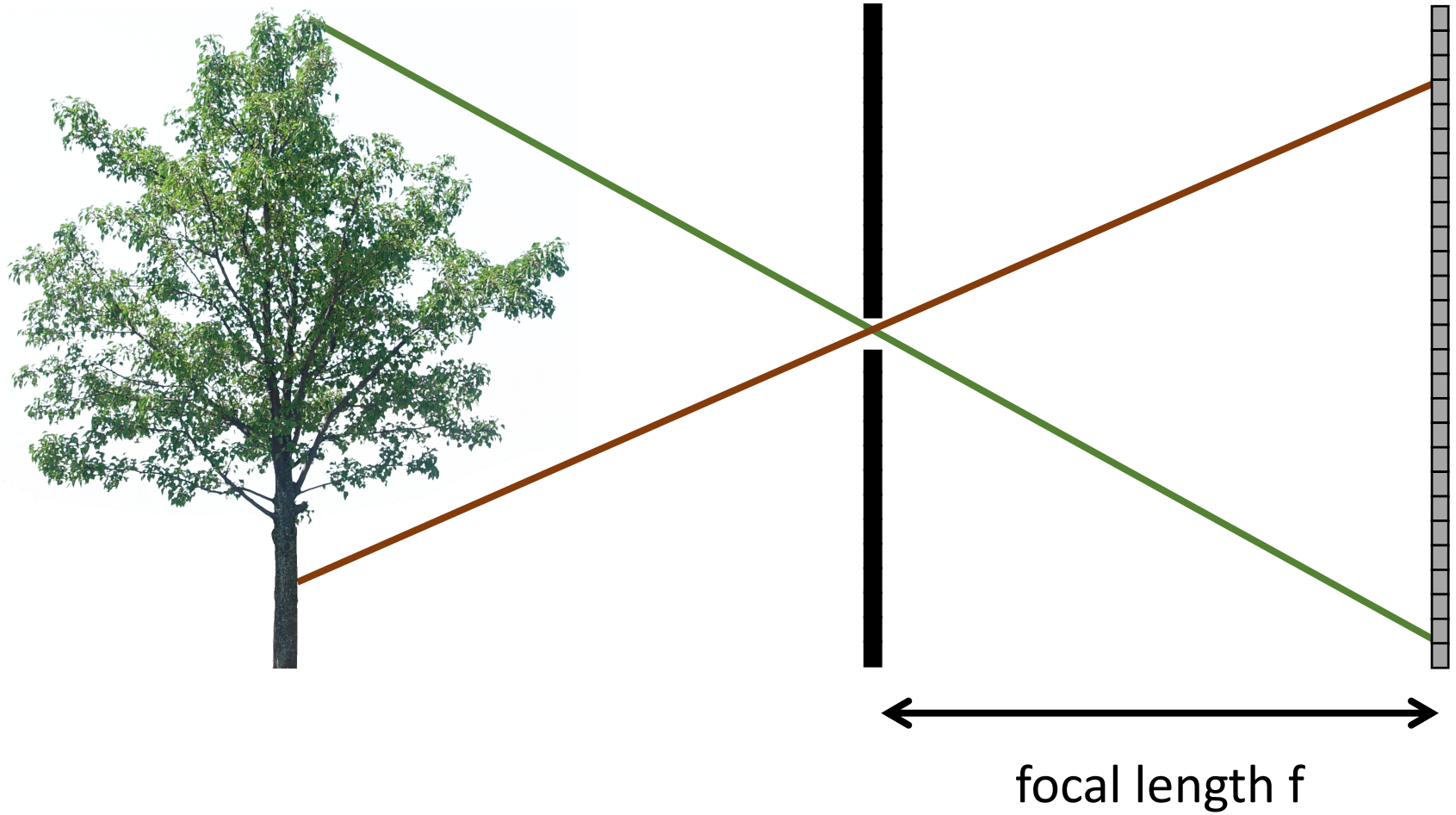
The pinhole camera



Central rays propagate in the same way for both models!

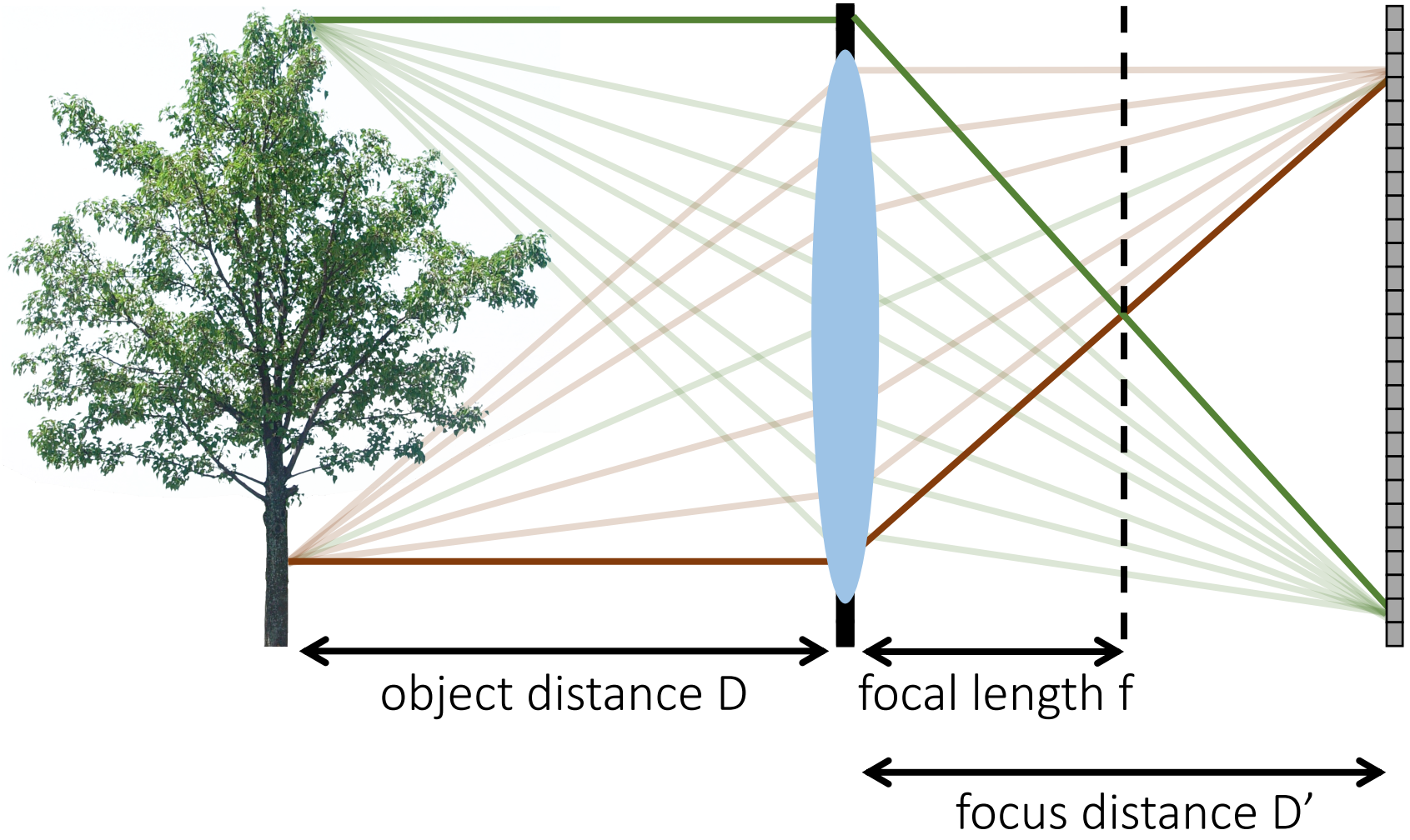
Important difference: focal length

In a pinhole camera, focal length is distance between aperture and sensor



Important difference: focal length

In a lens camera, focal length is distance where parallel rays intersect



PART A2 · 0:20–0:35

The imaging sensor: photons in, numbers out

Quantum efficiency, full-well capacity, and readout noise are the physical reasons your pixels are noisy, clipped, and not quite linear.

SOURCE DECK

Digital photography, 16-385 Computer Vision, Fall 2023, Lectures 24–25 · Ioannis Gkioulekas, Carnegie Mellon University.
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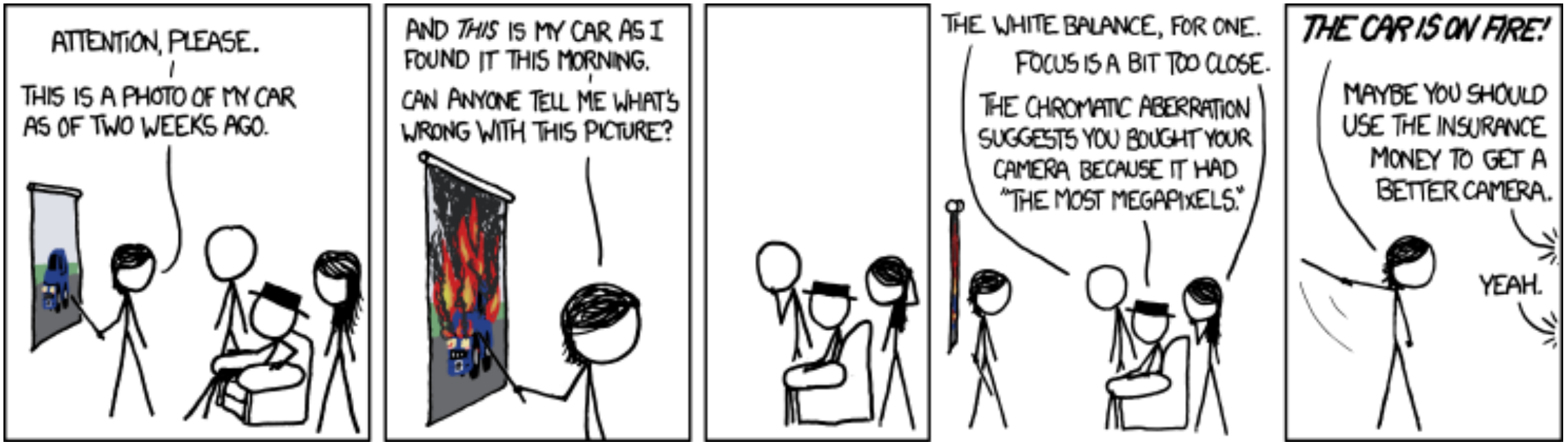
Digital photography



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16-385 Computer Vision
Fall 2023, Lecture 24 & 25

The modern photography pipeline

The modern photography pipeline

post-capture processing
(16-385, 15-463)

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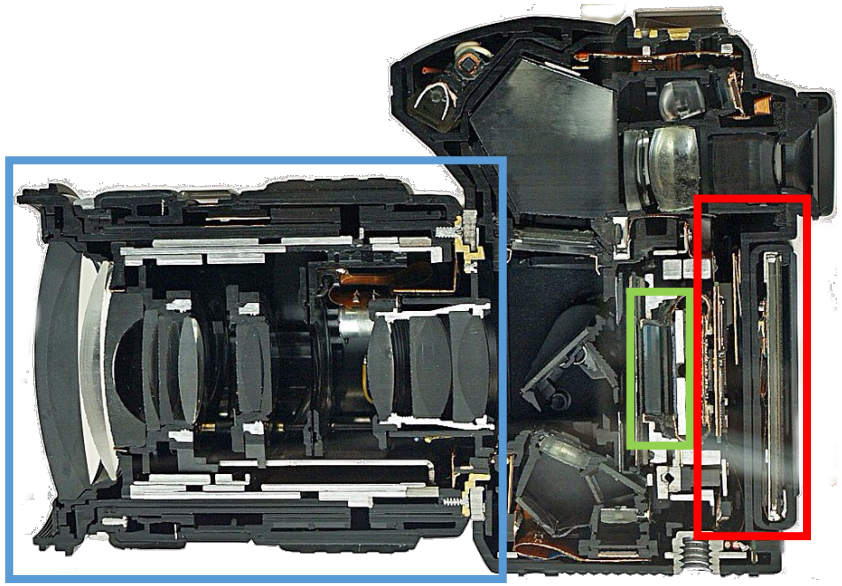
0 - 255



optics and optical controls

(15-463)

sensor, analog front-end, and color filter array

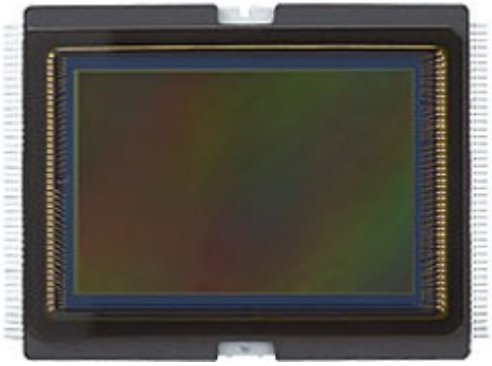
(this lecture)

in-camera image processing pipeline

(this lecture)

Imaging sensors

- Very high-level overview of digital imaging sensors.
- We could spend an entire course covering imaging sensors.

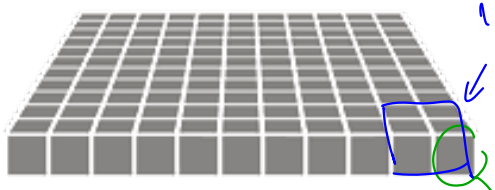


Canon 6D sensor
(20.20 MP, full-frame)

What does an imaging sensor do?

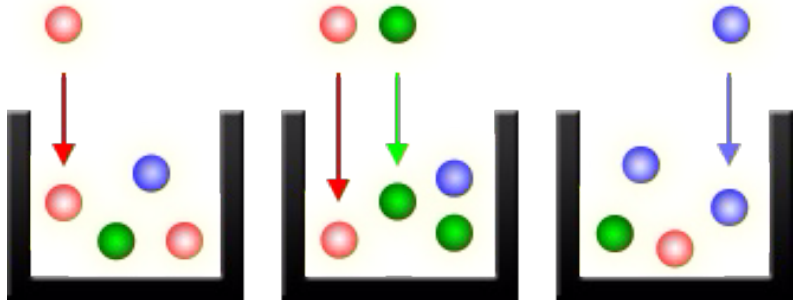
When the camera shutter opens…

… exposure begins…



array of photon buckets

photons



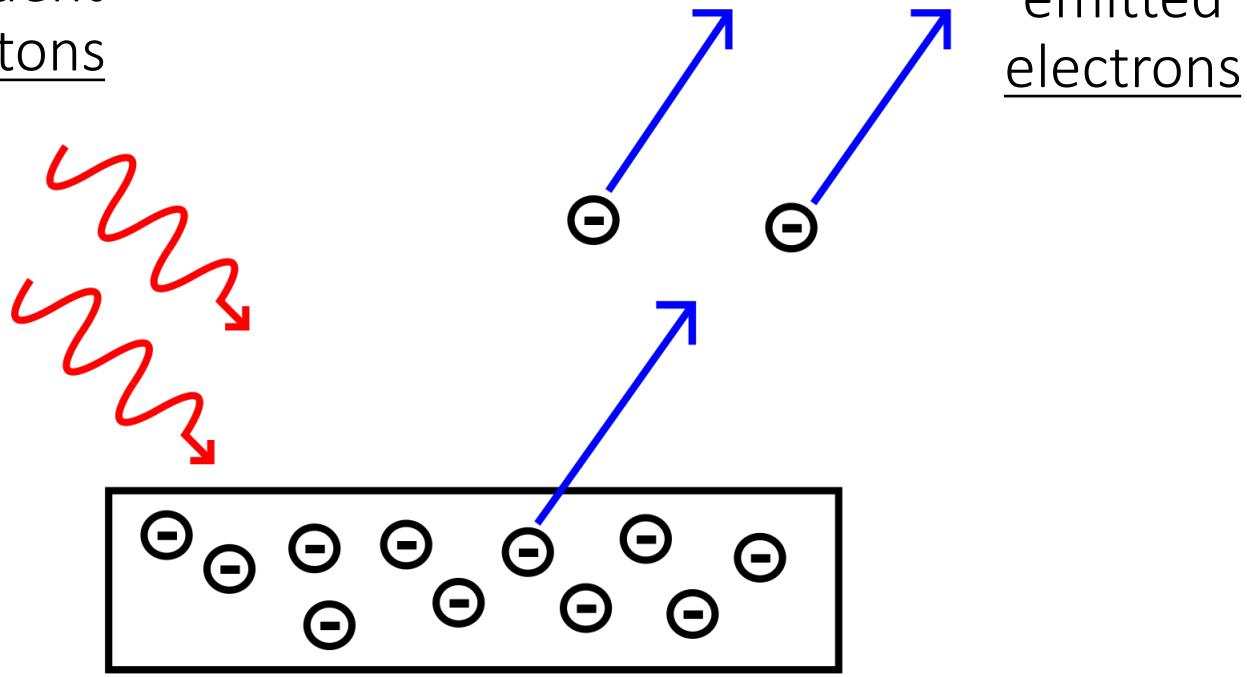
close-up view of photon buckets

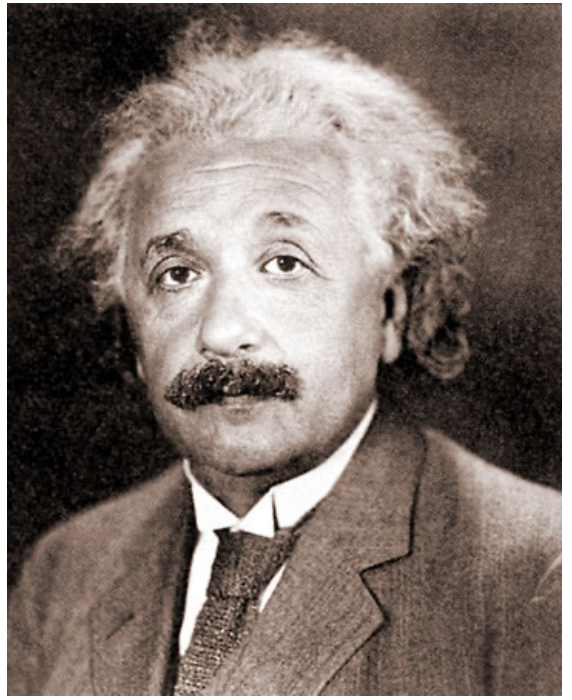
… photon buckets begin to store photons…

… until the camera shutter closes. Then, they convert stored photons to intensity values.

DILR
Full frame

Photoelectric effect

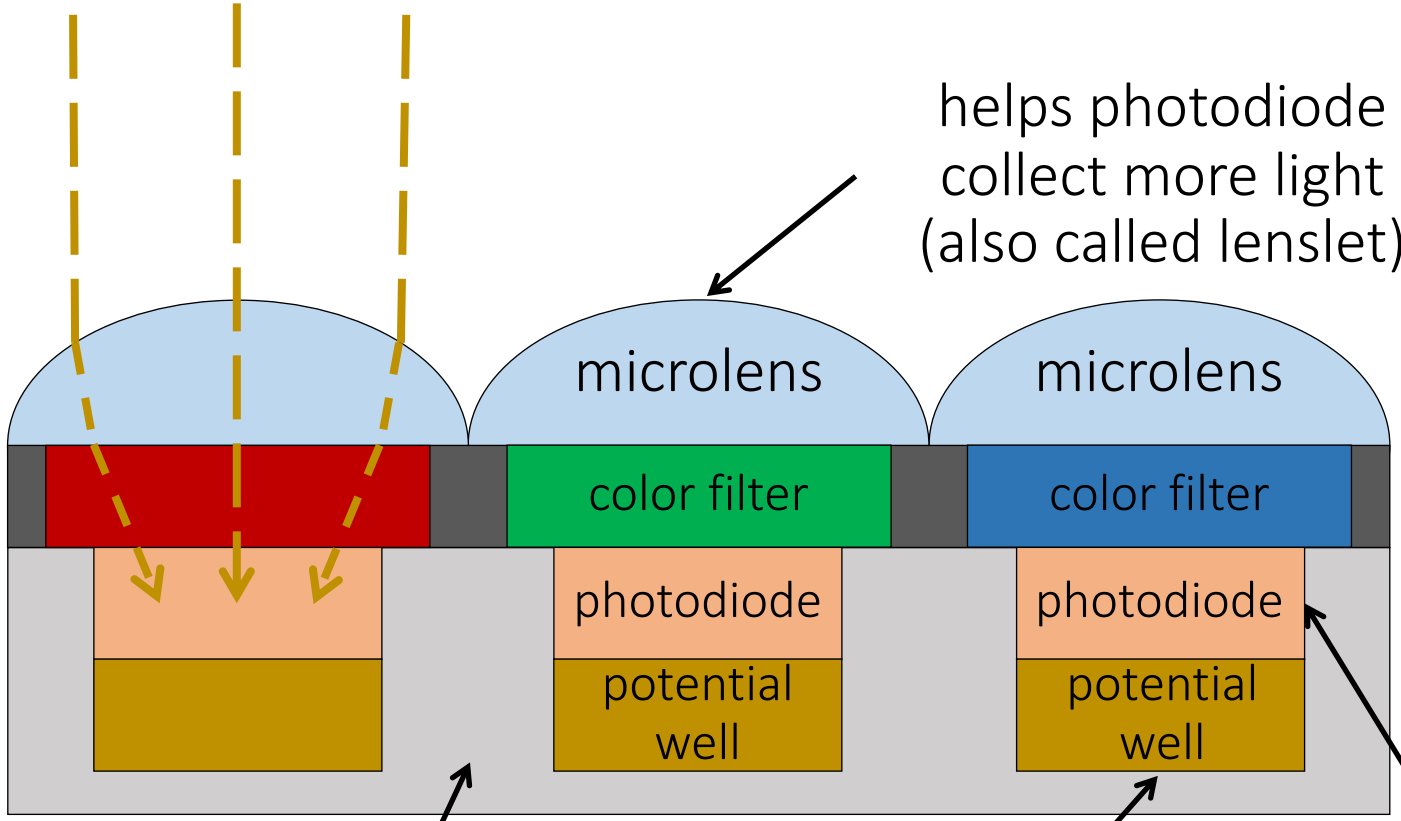




Albert Einstein

Einstein's Nobel Prize in 1921 "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"

Basic imaging sensor design



- Lenslets also filter the image to avoid resolution artifacts.
- Lenslets are problematic when working with coherent light.
- Many modern cameras do not have lenslet arrays.

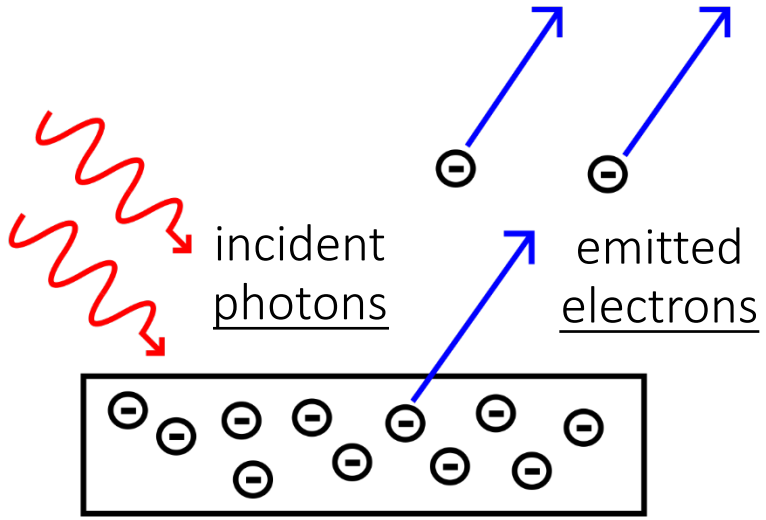
We will discuss these issues in more detail at a later lecture.

We will see what the color filters are for later in this lecture.

Photodiode quantum efficiency (QE)

How many of the incident photons will the photodiode convert into electrons?

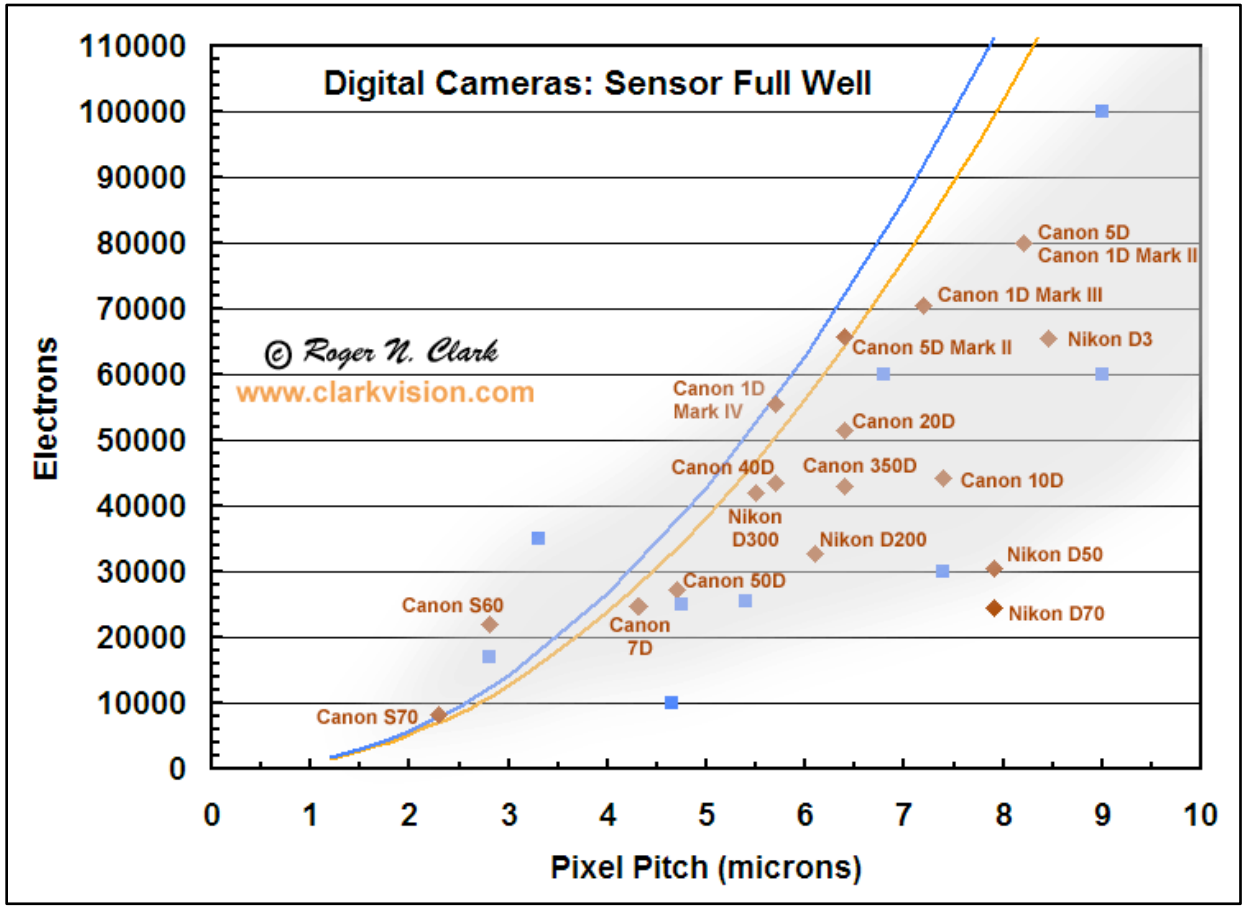
$$QE = \frac{\#\ \text{electrons}}{\#\ \text{photons}}$$



- Fundamental optical performance metric of imaging sensors.
- Not the only important optical performance metric!

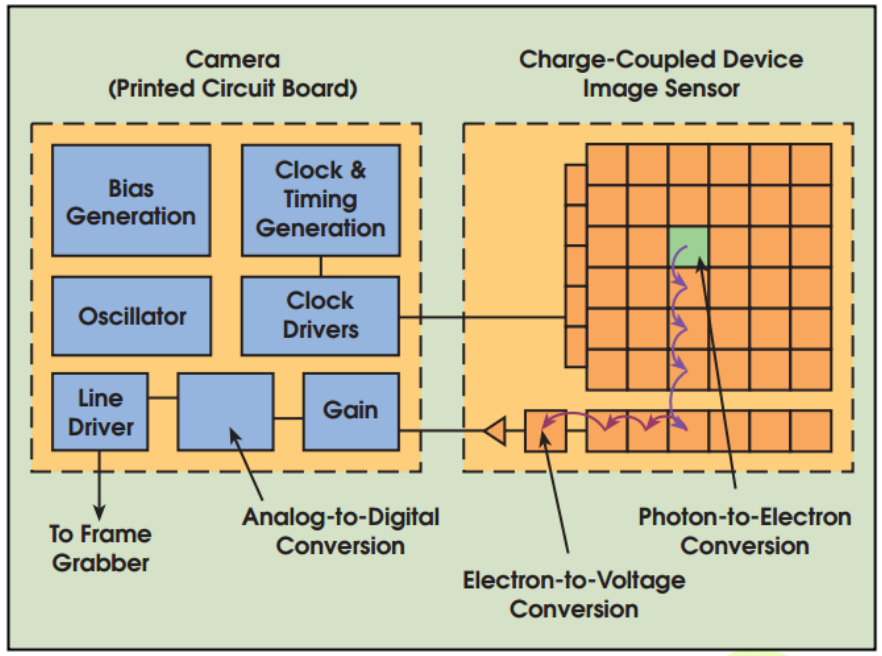
Photodiode full well capacity

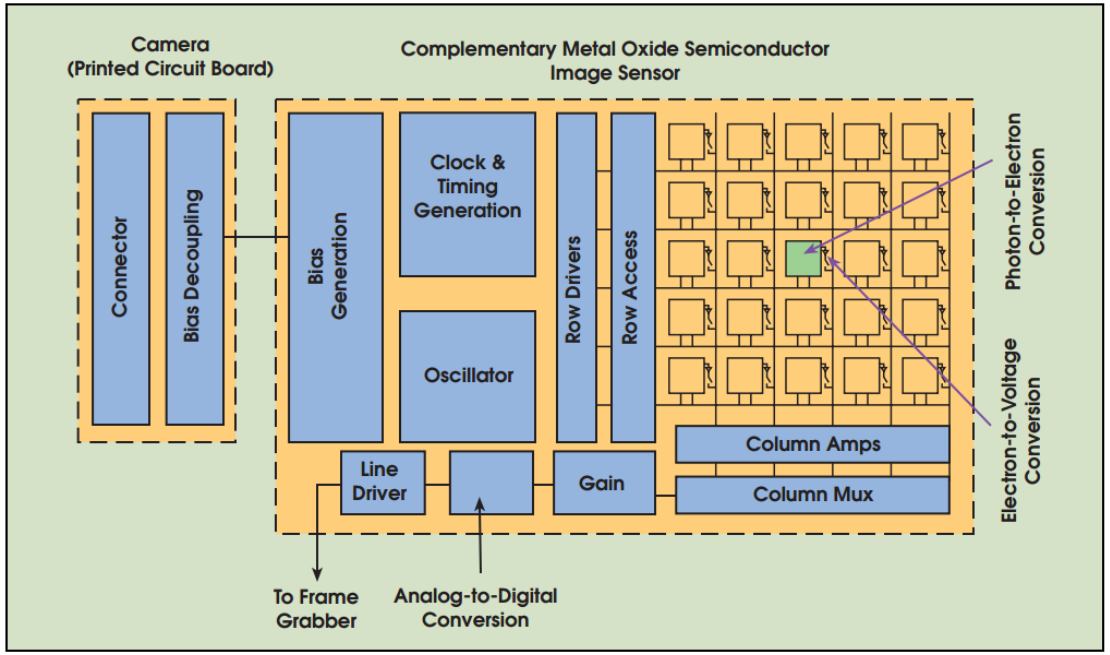
How many electrons can photodiode store before saturation?



- Another important optical performance metric of imaging sensors.

Two main types of imaging sensors





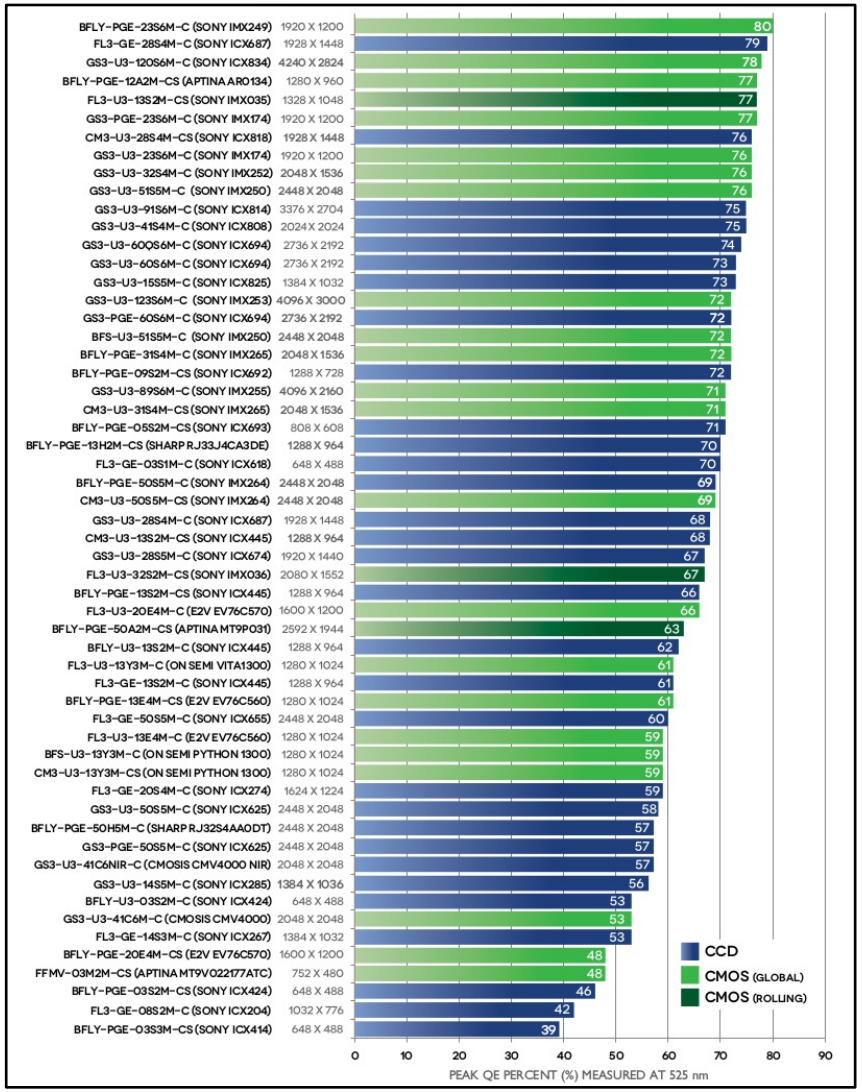
Charged Coupled Device (CCD): converts electrons to voltage using readout circuitry separate from pixel

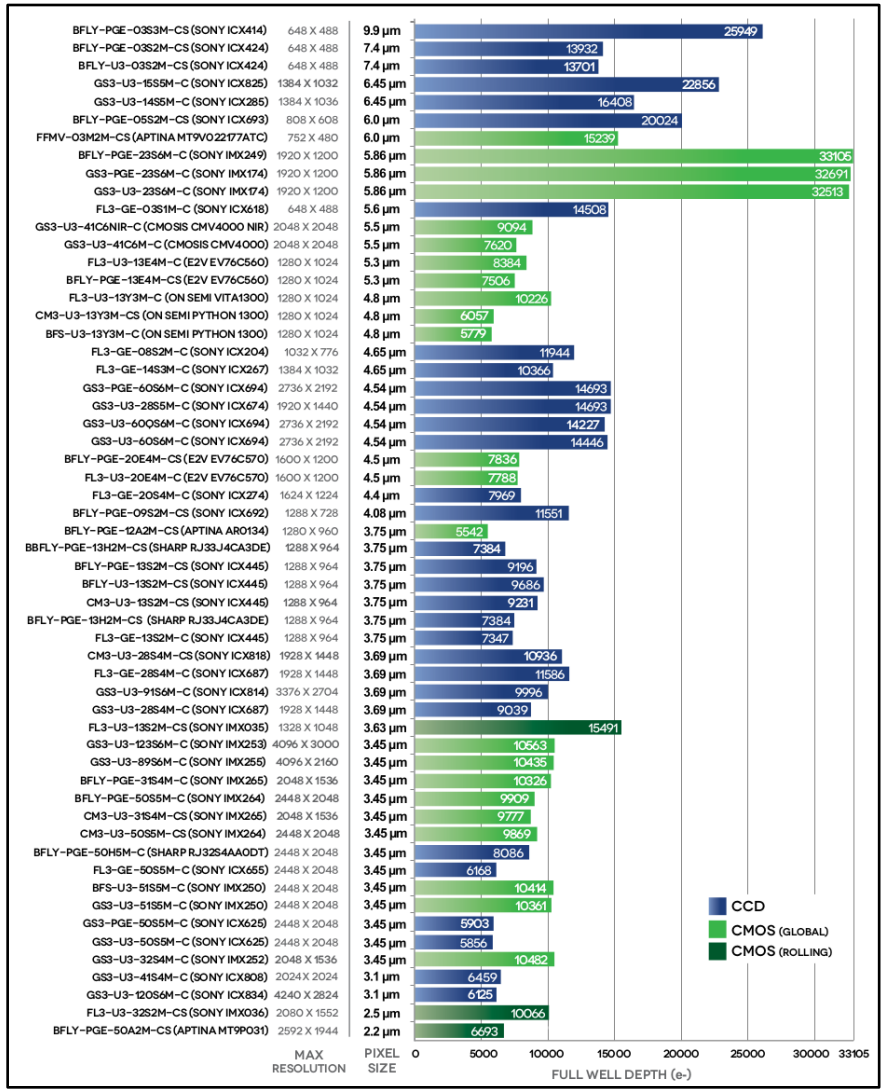
Complementary Metal Oxide Semiconductor (CMOS): converts electrons to voltage using per-pixel readout circuitry

Can you think of advantages and disadvantages of each type?

CCD vs CMOS

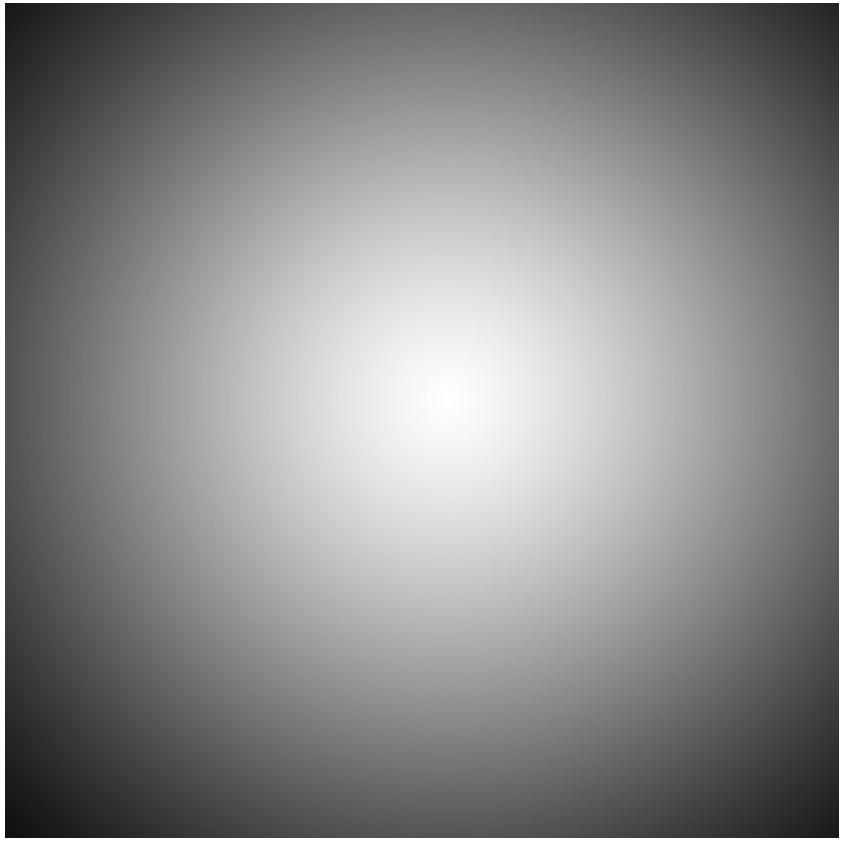
- Modern CMOS sensors have optical performance comparable to CCD sensors.
- Most modern commercial and industrial cameras use CMOS sensors.





Vignetting

Fancy word for: pixels far off the center receive less light



white wall under uniform light

more interesting example of vignetting

What does an imaging sensor do?

When the camera shutter opens, the sensor:

- at every photodiode, converts incident photons into electrons
- stores electrons into the photodiode's potential well until it is full

… until camera shutter closes. Then, the analog front-end:

- reads out photodiodes' wells, row-by-row, and converts them to analog signals
- applies a (possibly non-uniform) gain to these analog signals
- converts them to digital signals
- corrects non-linearities

… and finally returns an image.

PART A3 · 0:35–0:55

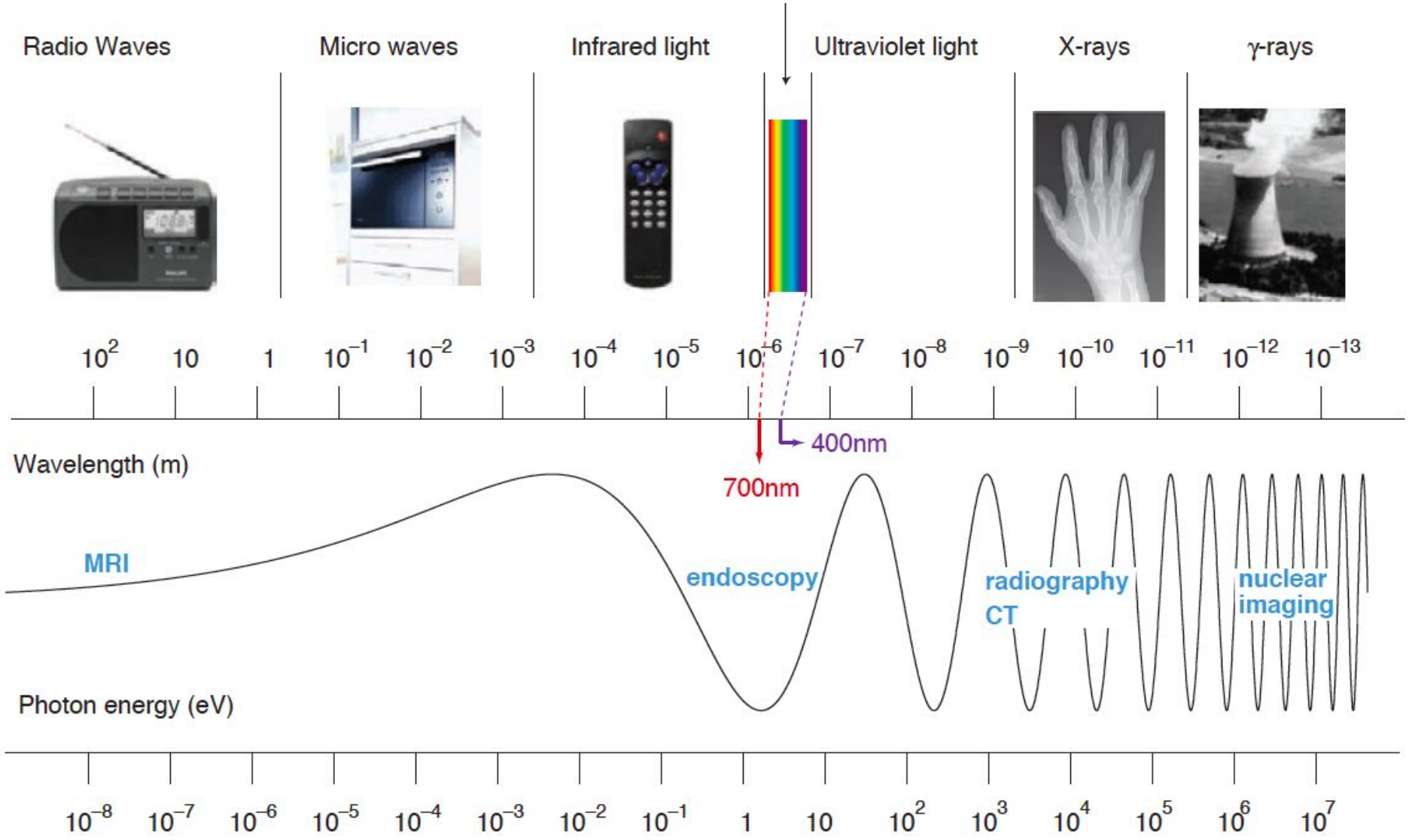
Medical image formation: four physics, one array

X-ray attenuation, sound echoes, reconstructed slices, magnetic resonance. The physics decides intensity meaning, noise, and artifacts, which is exactly what preprocessing must respect.

SOURCE DECK

Introduction to medical imaging and modalities, BME 495 Deep Learning for Biomedical Imaging, Winter 2025, Lectures 1–2 · Ulas Bagci, Northwestern University.

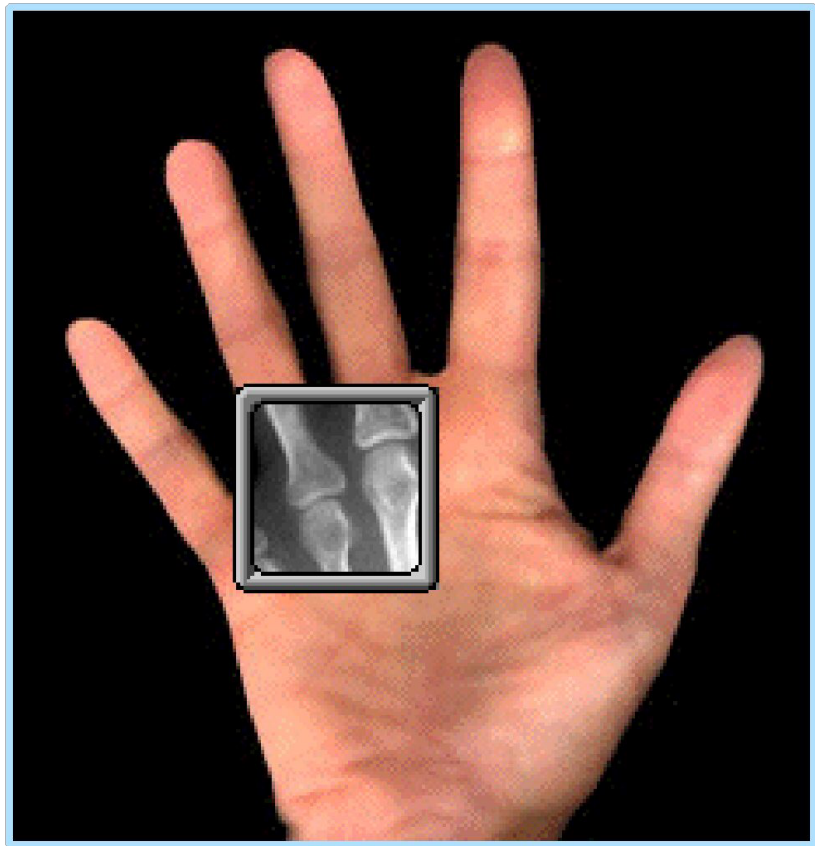
https://bagcilab.com/courses/ publicly shared by the author · shown unmodified · selected pages



ELECTROMAGNETIC SPECTRUM (P. Suetens)

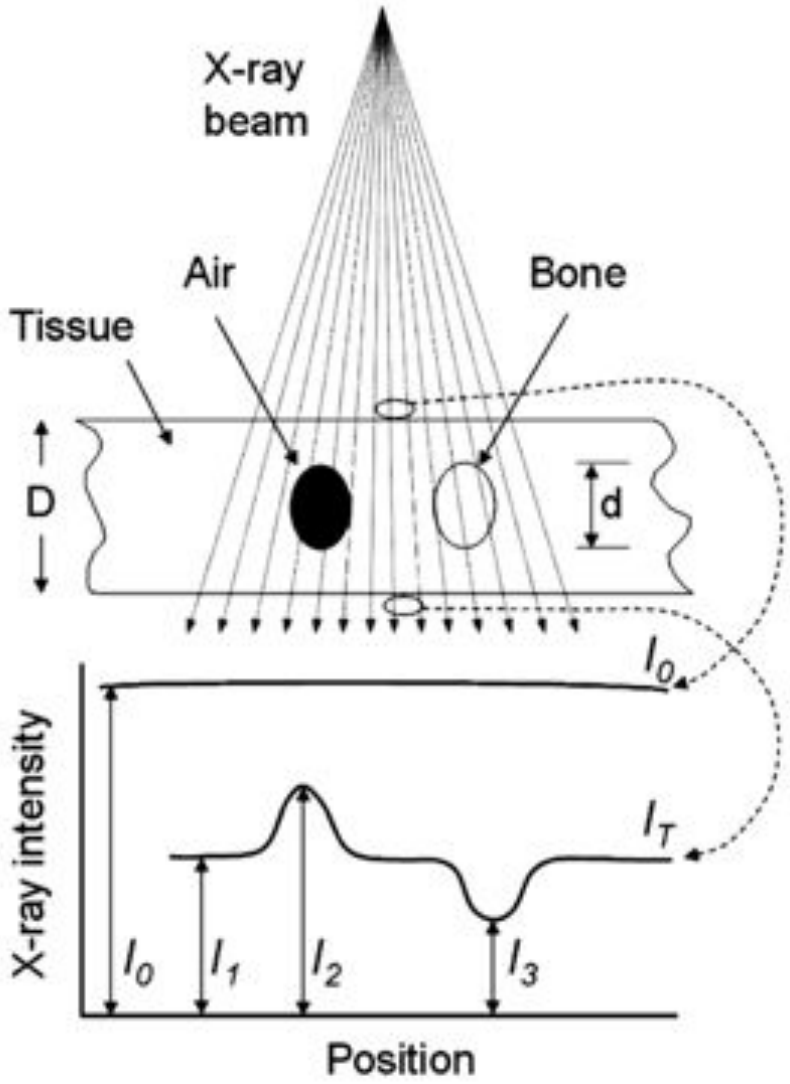
X-Ray Imaging / Radiography

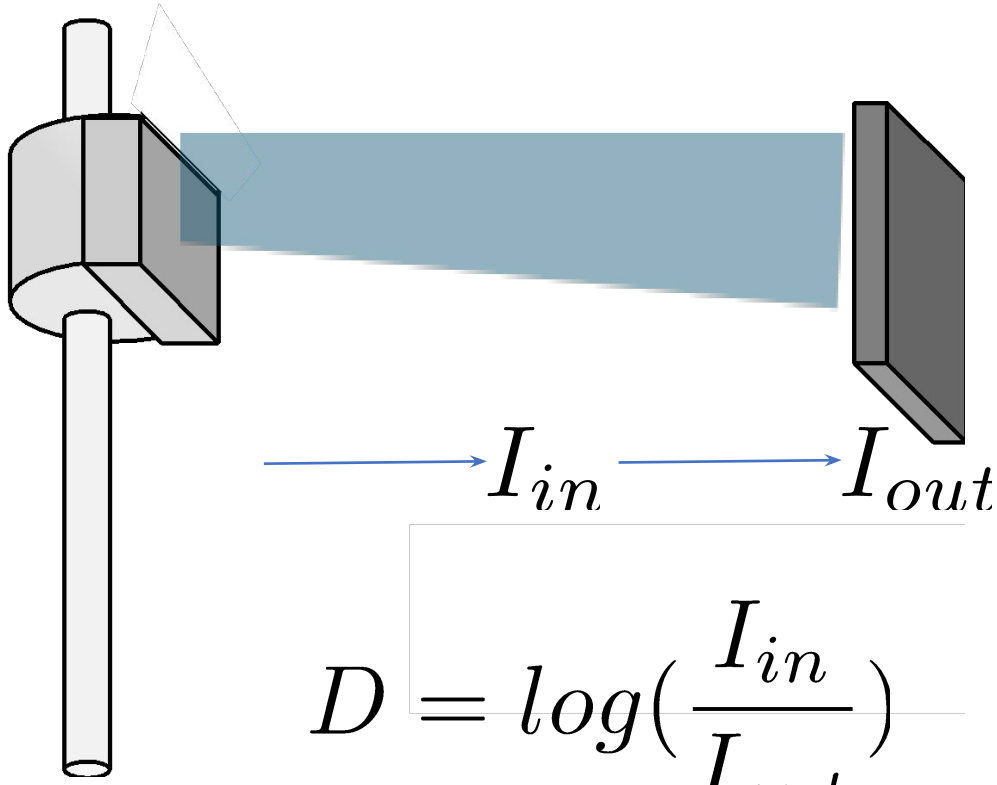
- The first published medical image was a radiograph of the hand of Wilhelm Conrad Roentgen's wife in 1895. *Nobel Prize in Physics 1901.*



routine diagnostic radiography (2D images):
chest x-rays, fluoroscopy, mammography, motion tomography, angiography, …

X-Ray Imaging / Radiography





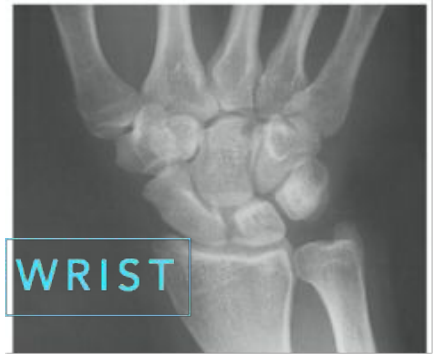
$$D = log(\frac{I_{in}}{I_{out}})$$

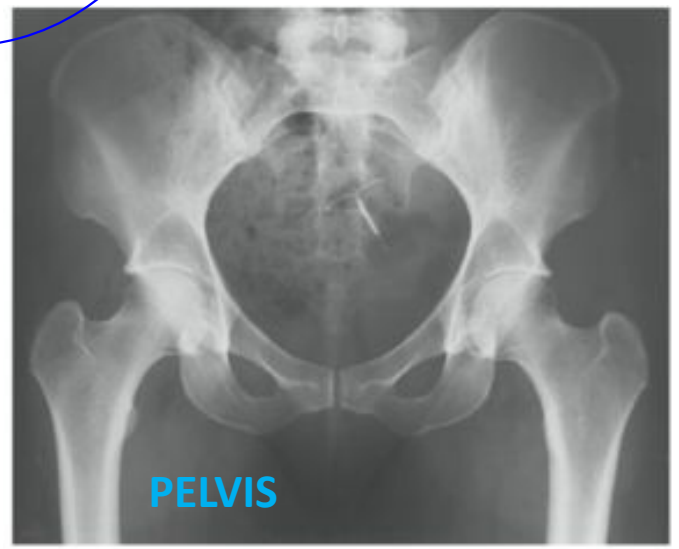
D=Optical density
E=exposure (Iin/Iout)
Iin=incoming light intensity
Iout=outgoing light intensity

ทุกอย่างขัดกันหมด 2 ข้อ

Clinical Examples – X-Rays

→ ใช่ยาก → ทำยังไง เป็นมะเร็ง





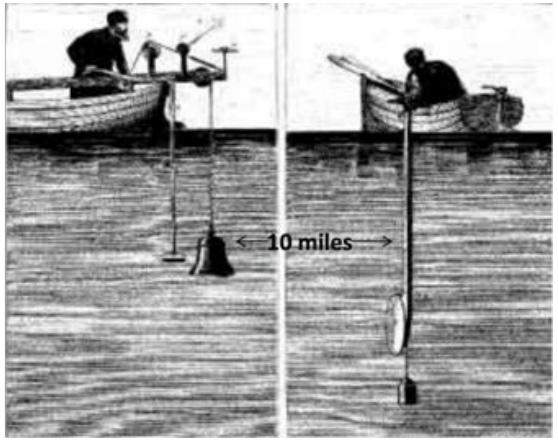
Ultrasound Imaging

ใช้คลื่นเสียง → ไม่อันตราย

- US is defined as any sound wave above 20KHz

1794-Lazzaro Spallanzani - Physiologist
First to study US physics by deducing bats used to US to navigate by echolocation

1826-Jean Daniel Colladon - Physicist
Uses church bell (early transducer) under water to calculate speed of sound through water prove sound traveled faster through water than air.



1880-Pierre&Jacques Curie
discover the Piezo-Electric Effect (ability of certain materials to generate an electric charge in response to applied mechanical stress.

US Imaging Technology

1942-Karl Dussik - Neurologist

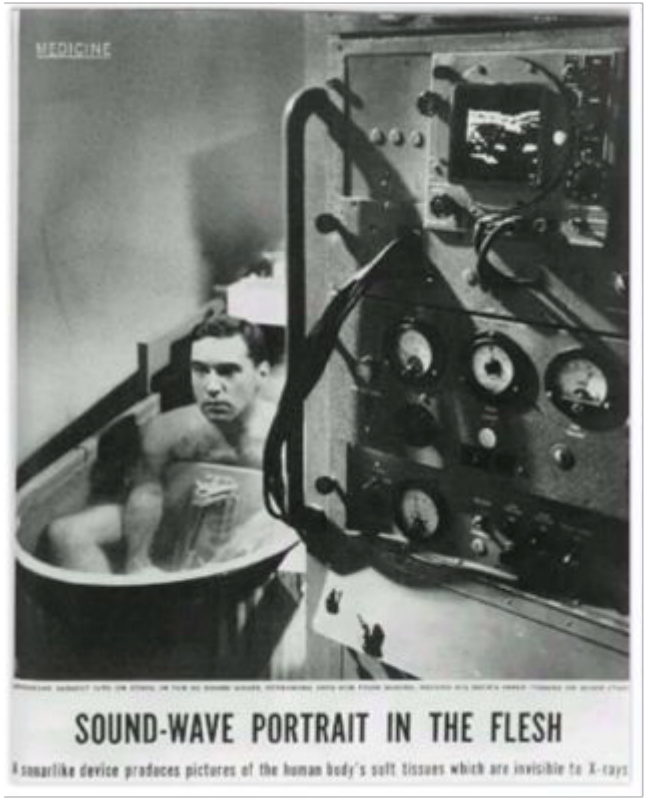
First physician to use US for medical diagnosis

1948-George Ludwig - MD

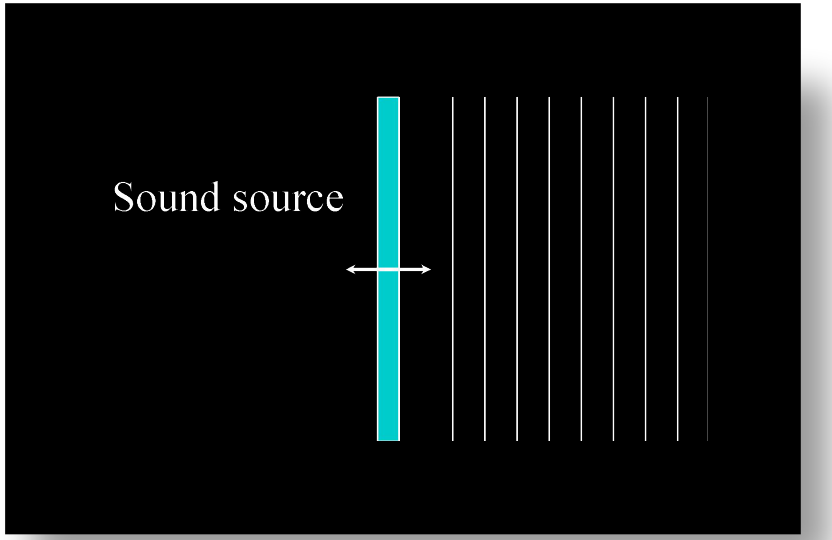
First described the use of US to diagnose gallstones

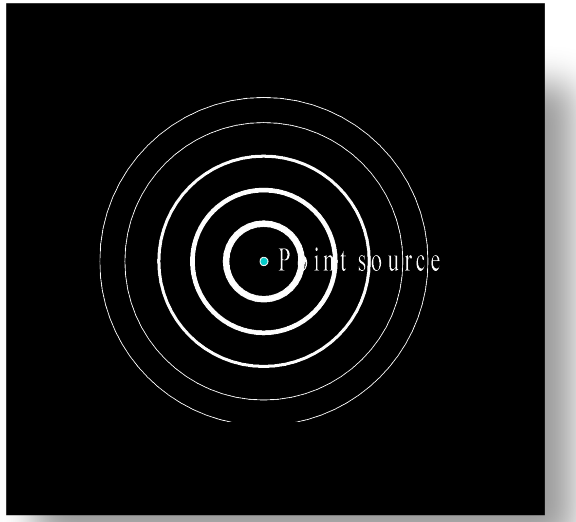
1958-Ian Donald

Pioneers in OB-GYN



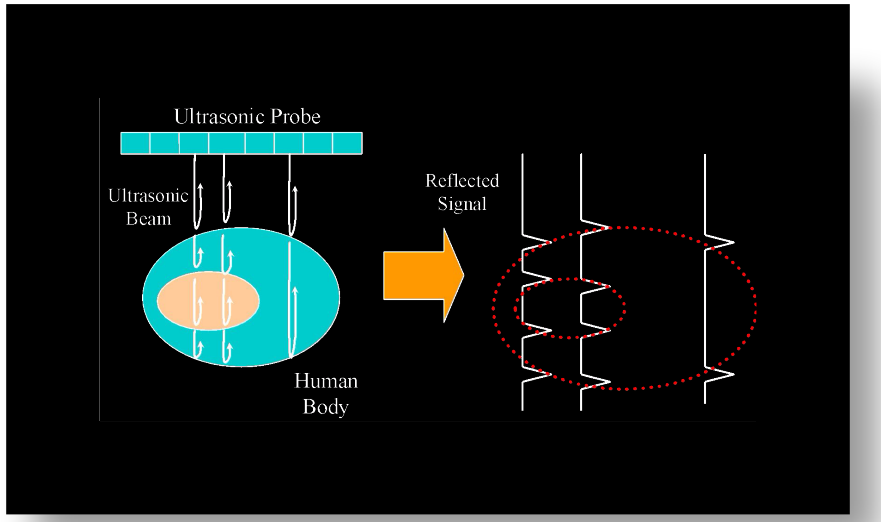
Principle of US Imaging





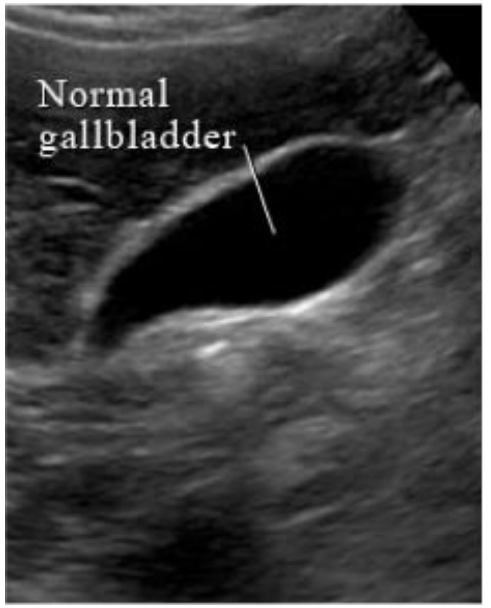
US equipment assumes that sound velocity is constant in the body.

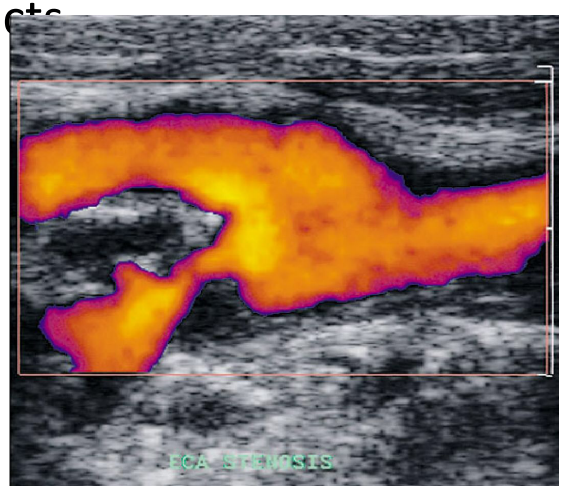
Body tissue	Acoustic impedance (10^6 Rayls)
Air	0.0004
Lung	0.18
Fat	1.34
Liver	1.65
Blood	1.65
Kidney	1.63
Muscle	1.71
Bone	7.8



Features of US Imaging

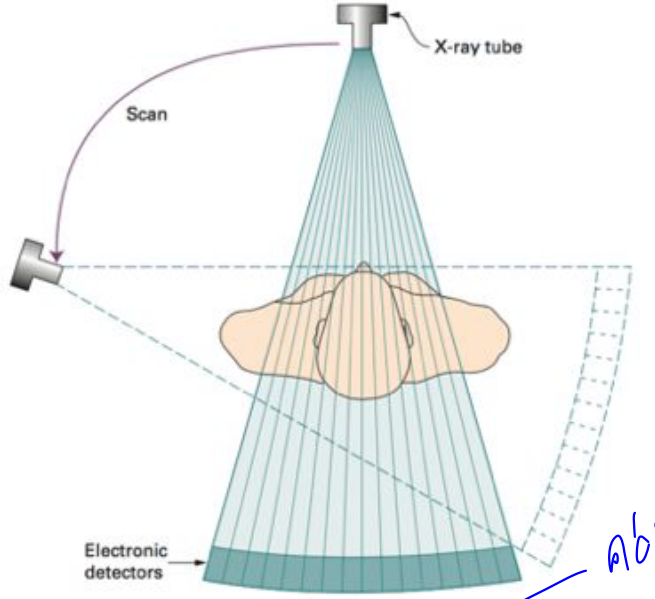
- Resolution:
  - direction of pulse propagation, pulse width 1-2mm
  - direction of scanning: beam width 2-3mm
  - **low resolution** and low SNR in deep region
- Ability of **imaging soft tissue**
- imaging in **real time**
- Doppler image
- Artefacts

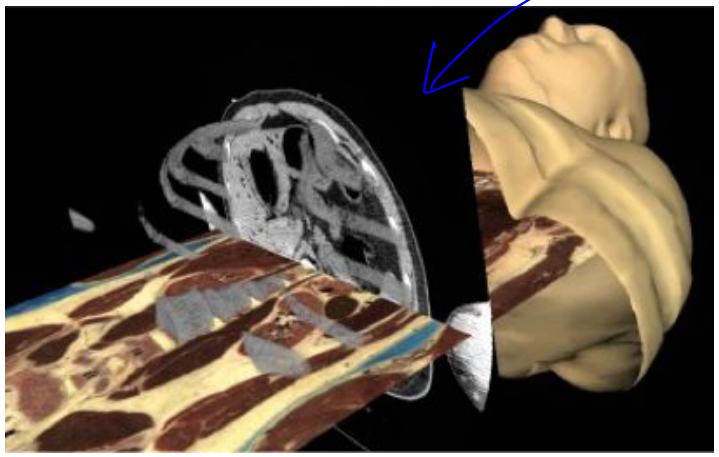


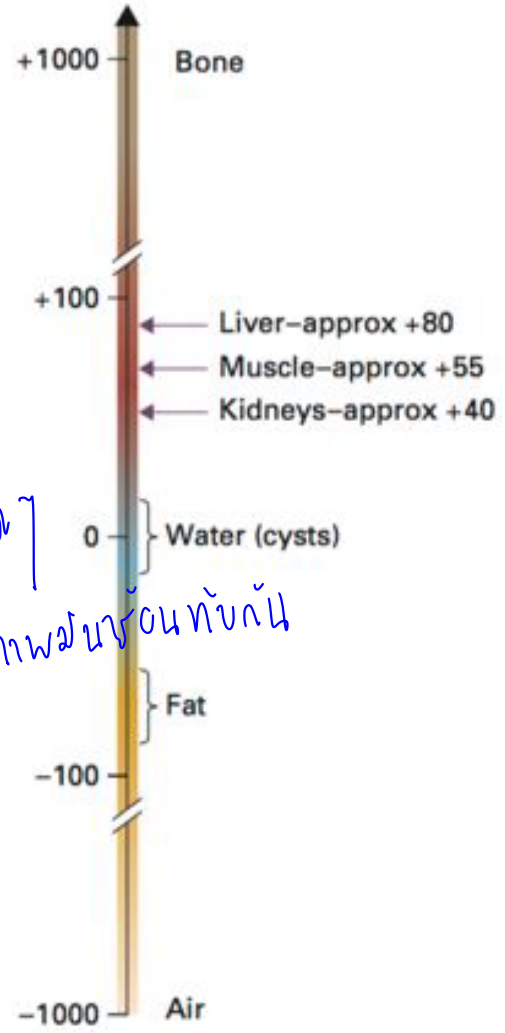


Color flow mapping shows simultaneous amplitude (US) and velocity information (doppler)

Computed Tomography (CT)



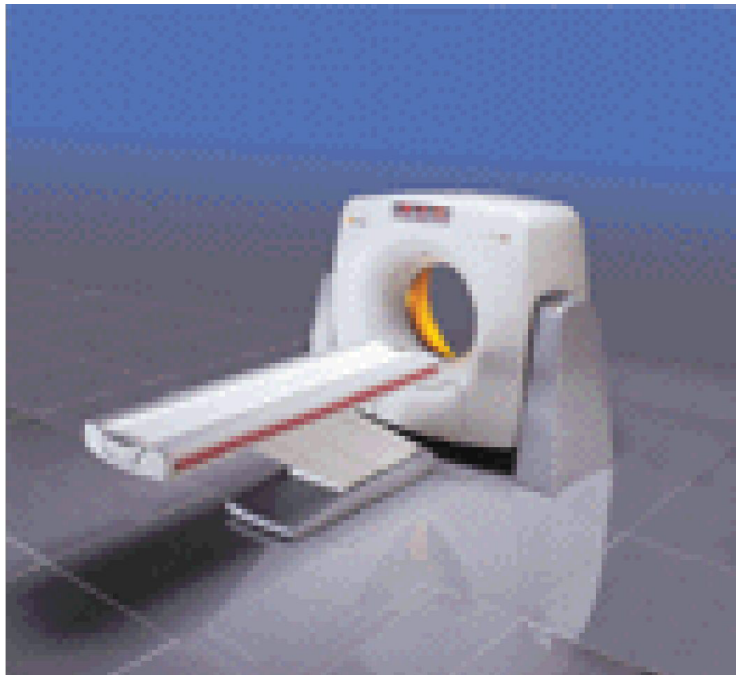




Tomo: slice/level (Greek)
Graphe: draw

CT Imaging (continue)

C-arm

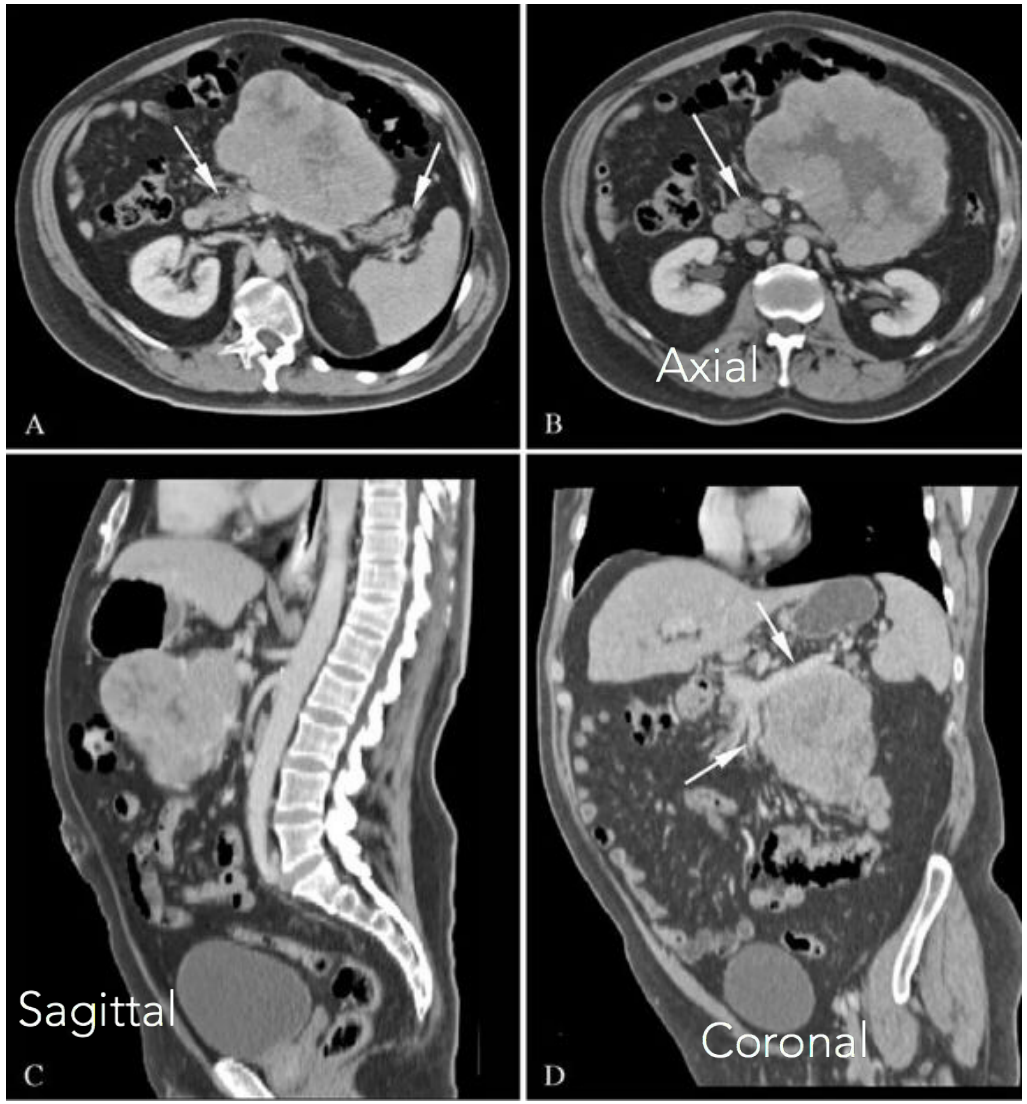


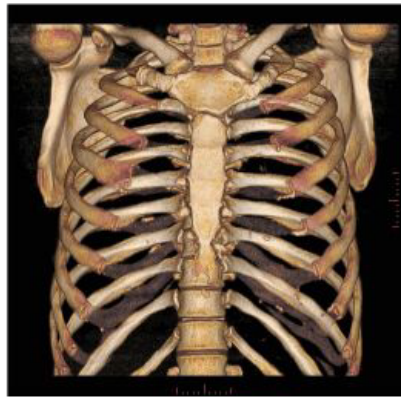
CT

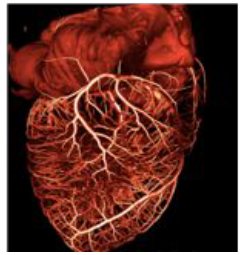
Micro-CT

~CAT Scan
(computerized
Axial tomography)

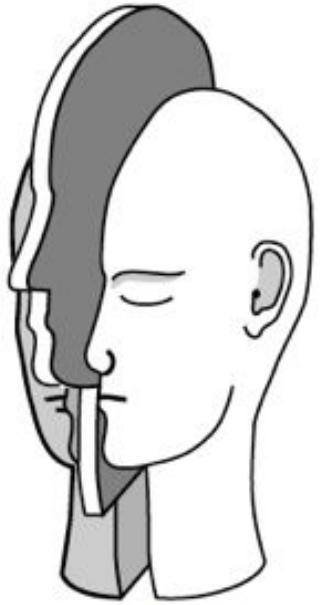
3D Nature of CT







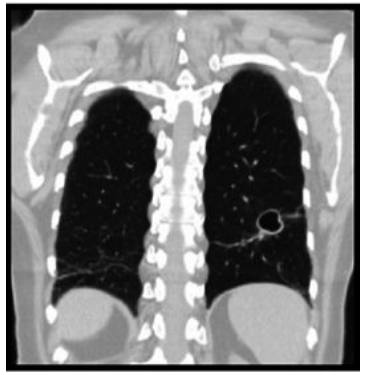
3D View Terminology

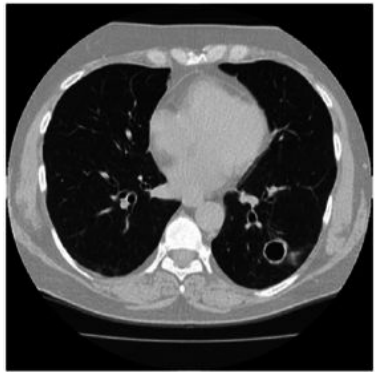


A Sagittal

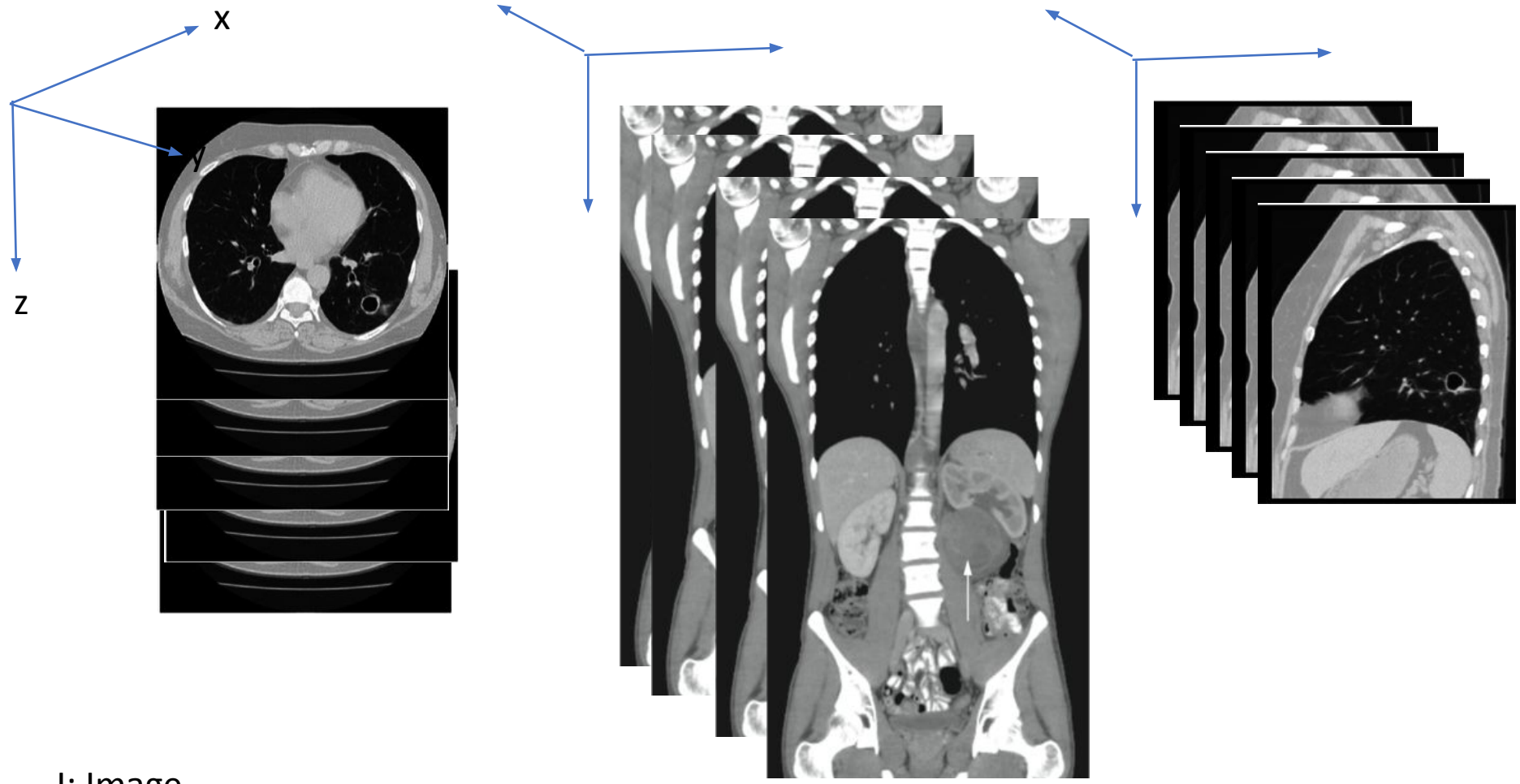
B Coronal

C Axial





3D Images



I: Image

I(x,y,z) denotes intensity value at pixel location x,y,z

Note also that whatever you see on the left is right part of the body!

Magnetic Resonance Imaging (MRI)

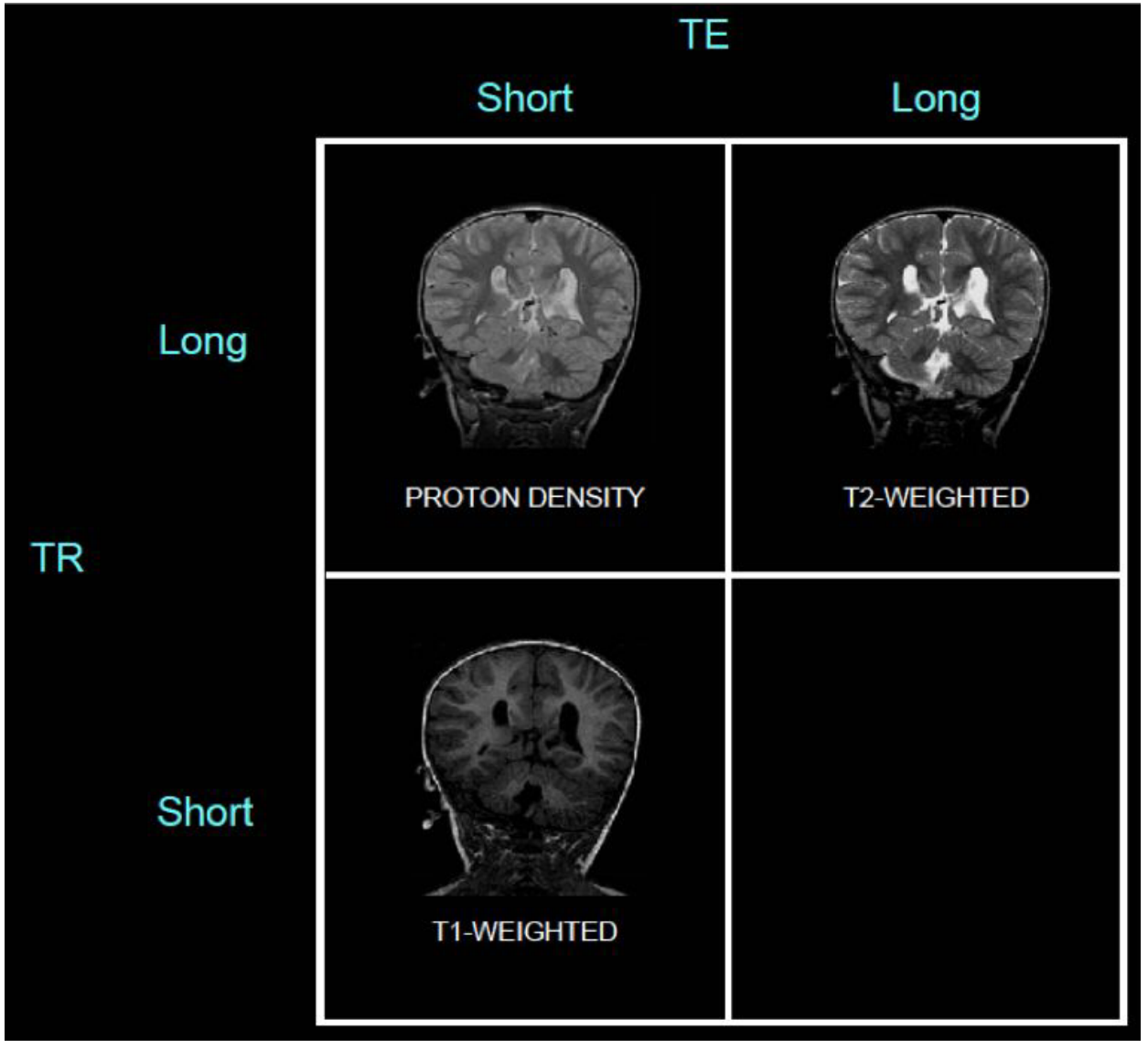
- 1882-Nichola Tesla
- Discovered rotating magnetic field
- 1971-Paul Lauterbur NOBEL PRIZE
- First invented MRI
- Late 1970-Sir Peter Mansfield (Nottingham) NOBEL PRIZE
- Developed mathematical techniques to create clearer images and also in minutes rather than hours as Lauterbur did.

ดูเนื้อเยื่อต่างๆได้ดี

- CT is more widely used than MRI.
- **MRI does not have ionizing-radiation.**
- MRI has excellent soft tissue contrast, while CT is preferred for lung and bone imaging.
- CT is fast (few seconds), while MRI is slow (sparse MRI ~5-10 mins, abdomen or brain may take 30-40 mins).

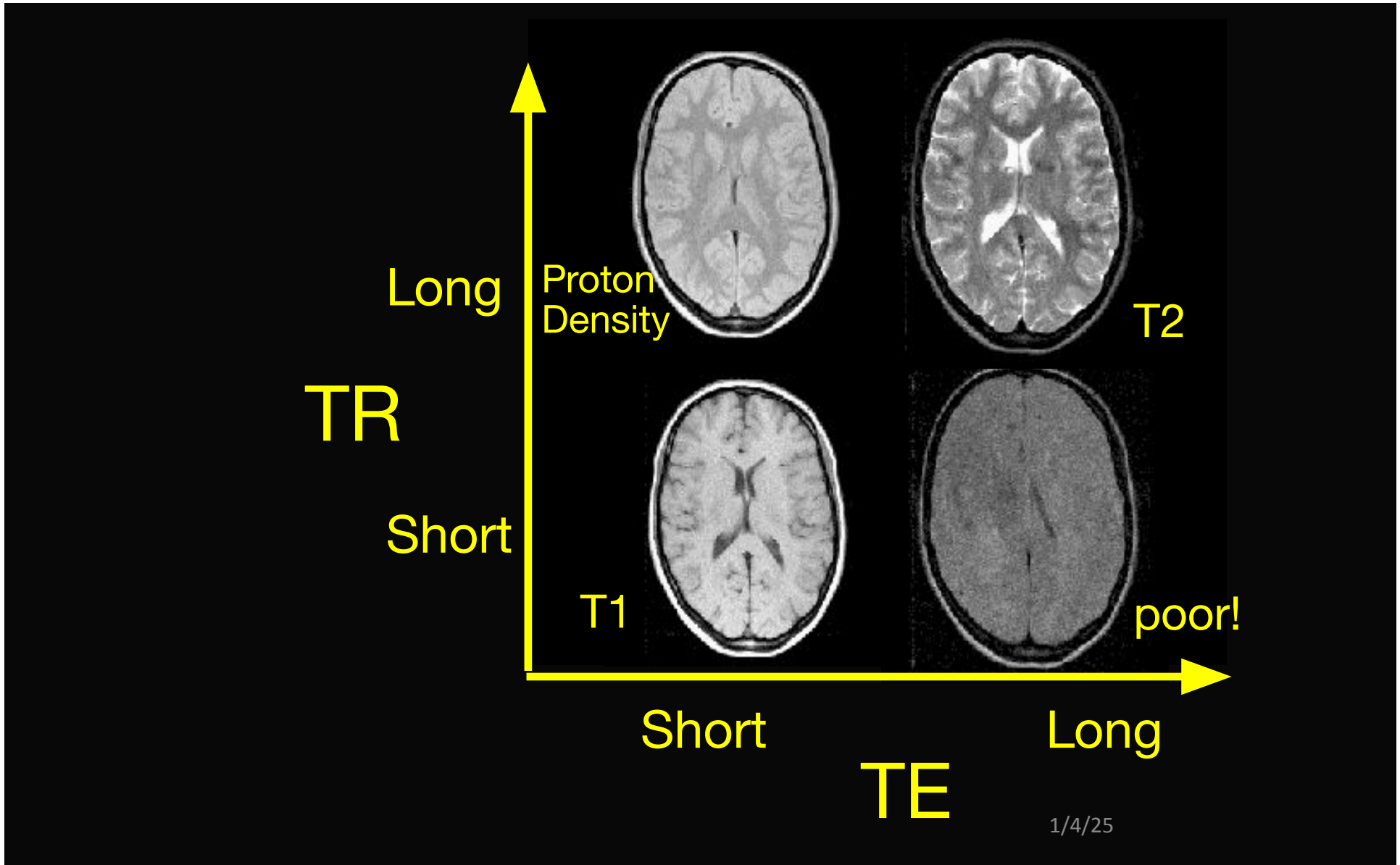
Types of MRI

→ ใช้คลื่นแม่เหล็กไฟฟ้า → ละเอียดกว่า X-ray
↘ เป็นภาพ 3 มิติ

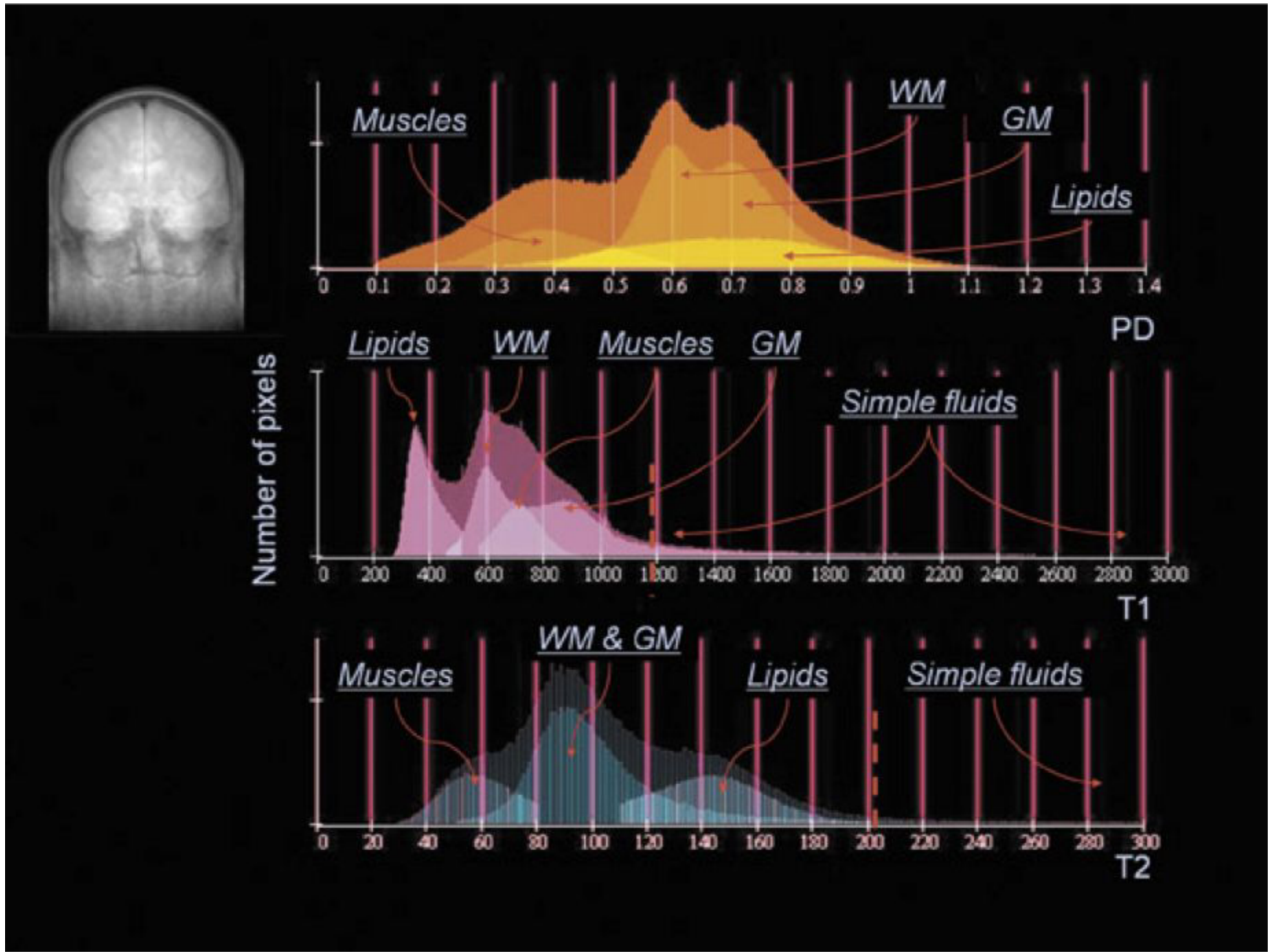


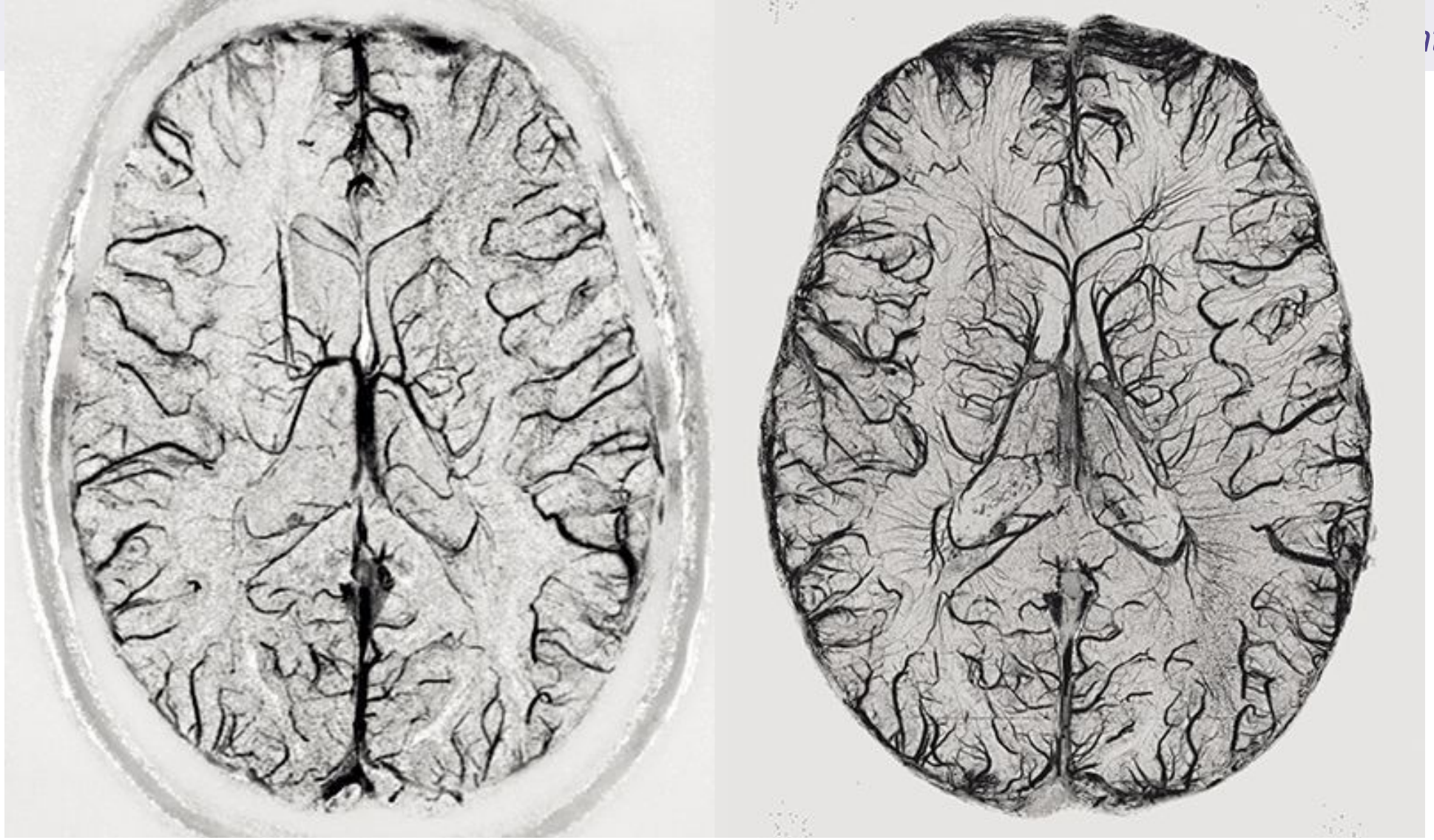
รวม

Image contrast summary: TR, TE



Brain MRI


Muscles
WM
GM
Lipids
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 1.1 1.2 1.3 1.4
PD
Lipids
WM
Muscles
GM
Simple fluids
Number of pixels
0 200 400 600 800 1000 1200 1400 1600 1800 2000 2200 2400 2600 2800 3000
T1
WM & GM
Muscles
Lipids
Simple fluids
0 20 40 60 80 100 120 140 160 180 200 220 240 260 280 300
T2



The brain of a volunteer is imaged using a 3-T (left) and 9.4-T (right) magnetic resonance imaging machine.Credit: Rolf Pohmann/Max-Planck-Institute for Biological Cybernetics
(https://www.nature.com/articles/d41586-018-07182-7)

Shallow Comparison of Imaging Methods

	Chest	Abdomen	Head/Neck	Cardiovascular	Skeletal/muscular
CT	gold standard	Need contrast for excellency, widely used	Good for trauma	Gold standard	Gold standard
US	no use except heart or P.Effusion	Problems with gas	Poor	Poor	Elastography
Nuclear	Extensive use in heart and therapy in lung	CT or MRI is merged	PET	Perfusion	bone marrow
MRI	growing cardiac applications	Increased role of MRI	Gold standard	Will replace ct in near future	Excellent

PART A4 · 0:55–1:10

Sampling and quantization: what 8 bits throw away

A pixel is a sample of a continuous world, stored at a chosen precision. Watch the same photo at 8, 5, 3, and 1 bits, then remember your CT scans are 12-bit.

SOURCE DECK

Sampling and Quantization, 6.003 Signals and Systems, Fall 2011, Lecture 22 · Dennis Freeman, MIT OpenCourseWare.
https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/ License: CC BY-NC-SA 4.0 · attribution page retained at section end

6.003: Signals and Systems

Sampling and Quantization

November 29, 2011

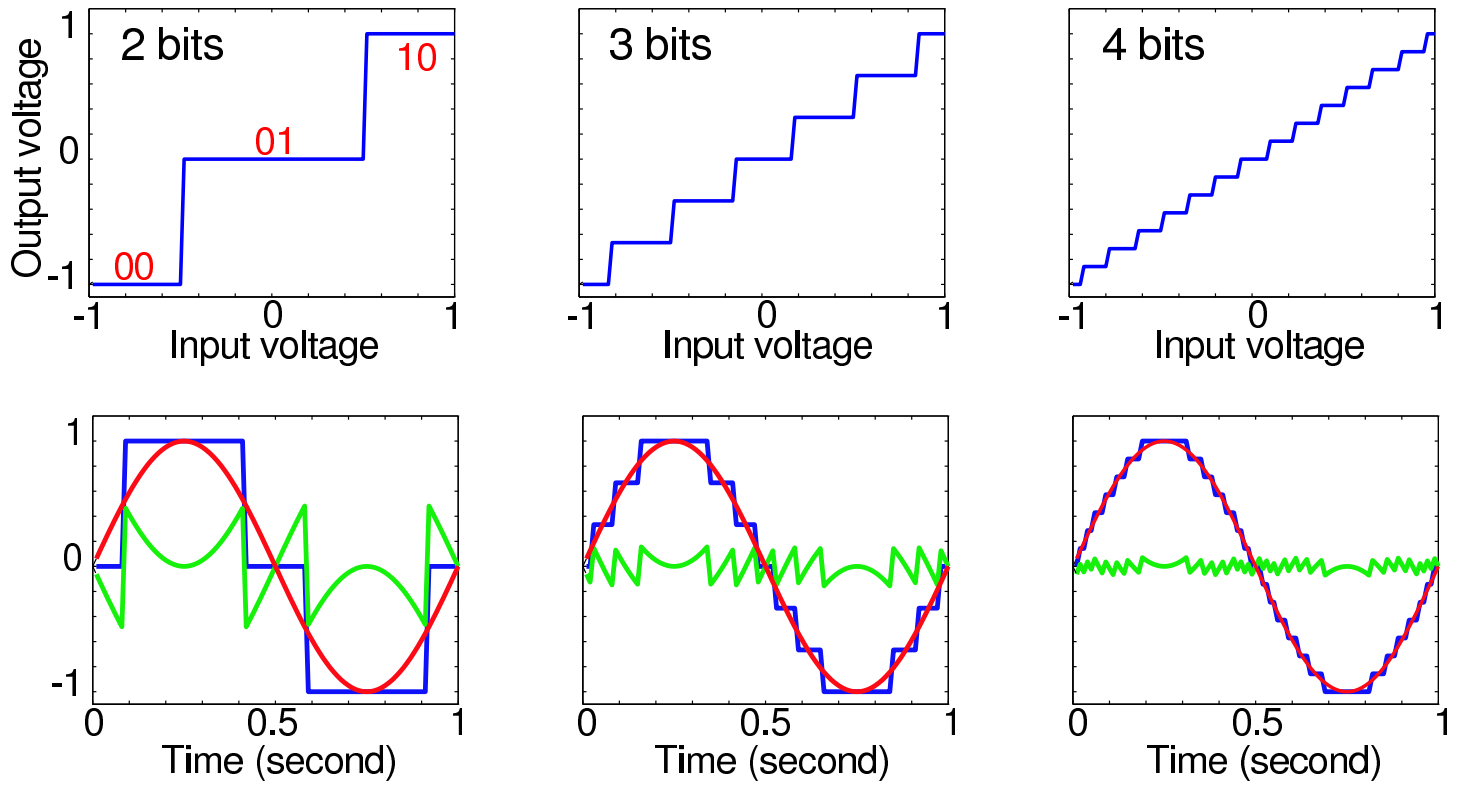
Last Time: Sampling

Sampling allows the use of modern digital electronics to process, record, transmit, store, and retrieve CT signals.

- audio: MP3, CD, cell phone
- pictures: digital camera, printer
- video: DVD
- everything on the web

Quantization

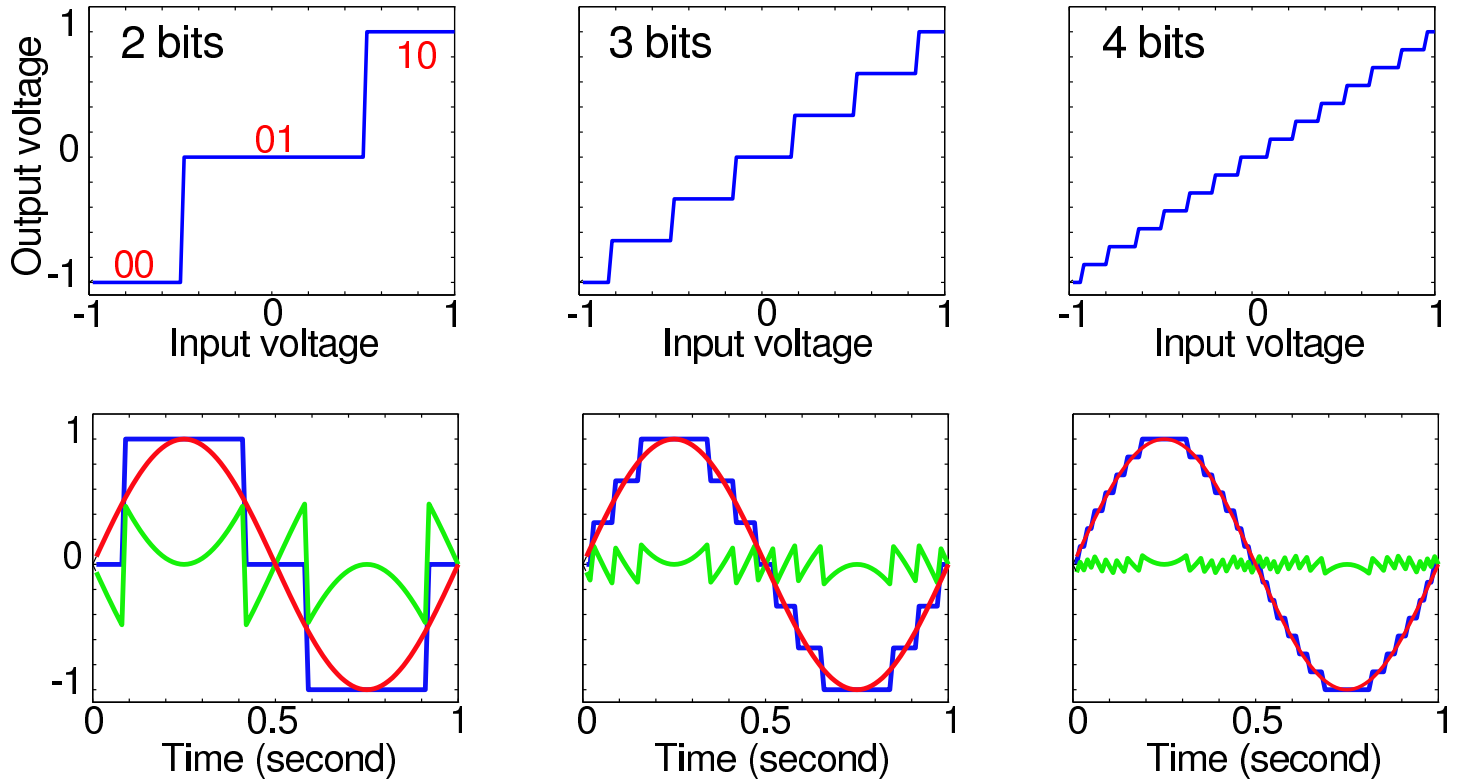
We measure discrete amplitudes in bits.



Bit rate = (# bits/sample)×(# samples/sec)

Quantization

We measure discrete amplitudes in bits.



Example: audio CD

$$2\,\text{channels} \times 16\,\frac{\text{bits}}{\text{sample}} \times 44,100\,\frac{\text{samples}}{\text{sec}} \times 60\,\frac{\text{sec}}{\text{min}} \times 74\,\text{min} \approx 6.3\,\text{G bits}$$

$$\approx 0.78\,\text{G bytes}$$

Quantizing Images

Converting an image from a continuous representation to a discrete representation involves the same sort of issues.

This image has 280×280 pixels, with brightness quantized to 8 bits.

Quantizing Images

8 bit image

7 bit image

Quantizing Images

8 bit image

5 bit image

Quantizing Images

8 bit image

3 bit image

Quantizing Images

Cross stitch

8 bit image

1 bit image

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6.003 Signals and Systems

Fall 2011

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PART A5 · 1:10–1:30

Aliasing: how careless downsampling invents patterns

Throw away every second row and column and you do not get a smaller copy, you get moire stripes that were never in the scene. The fix: smooth first, then sample.

SOURCE DECK

Image pyramids and frequency domain, 16-385 Computer Vision, Fall 2023, Lecture 3 · Ioannis Gkioulekas, Carnegie Mellon University.

http://16385.courses.cs.cmu.edu/fall2023/

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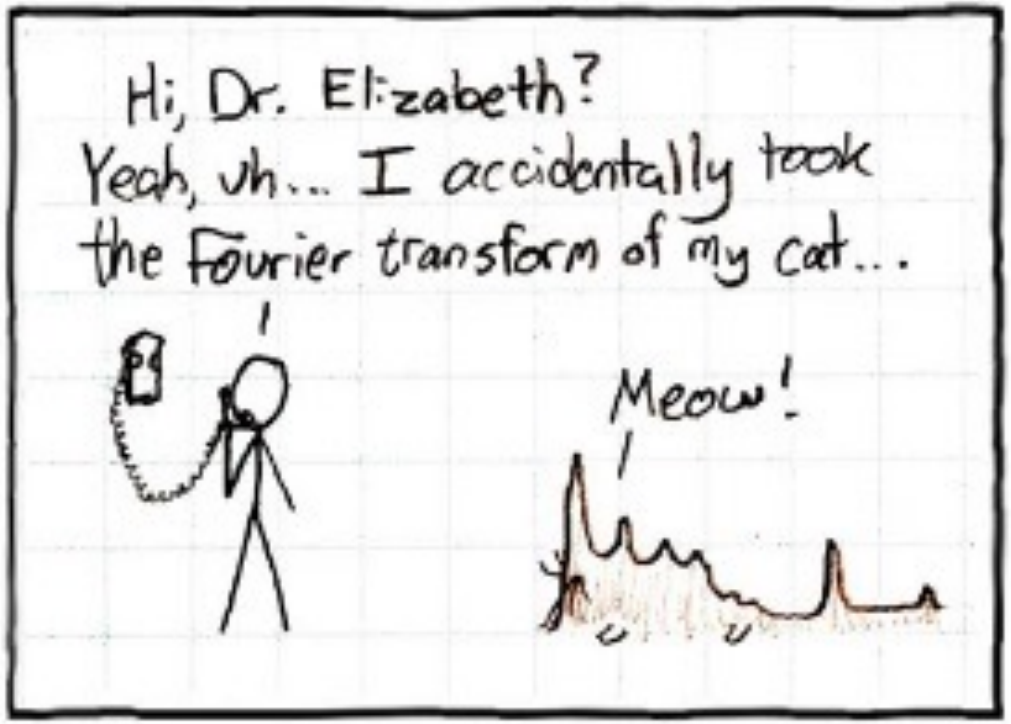
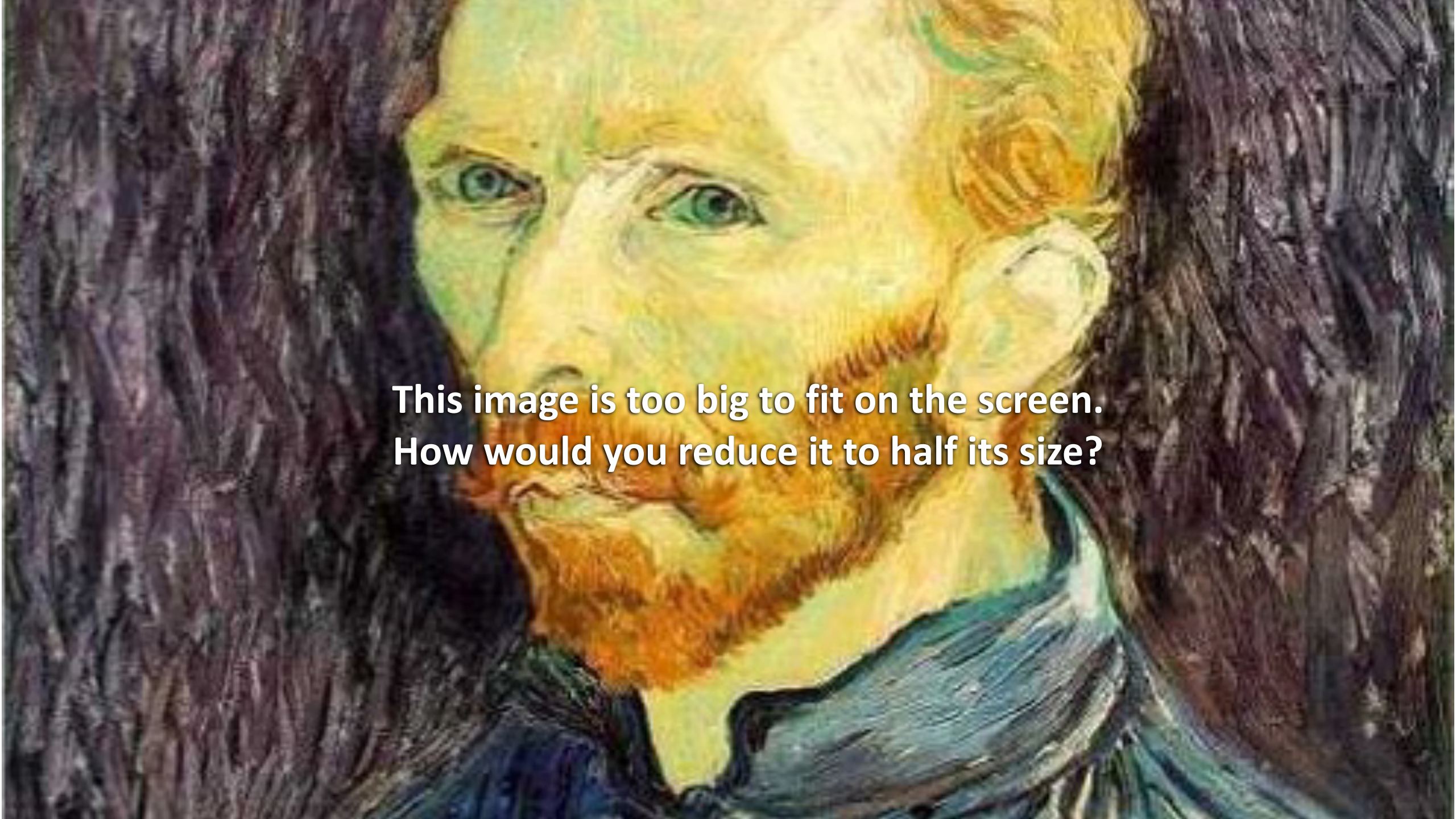
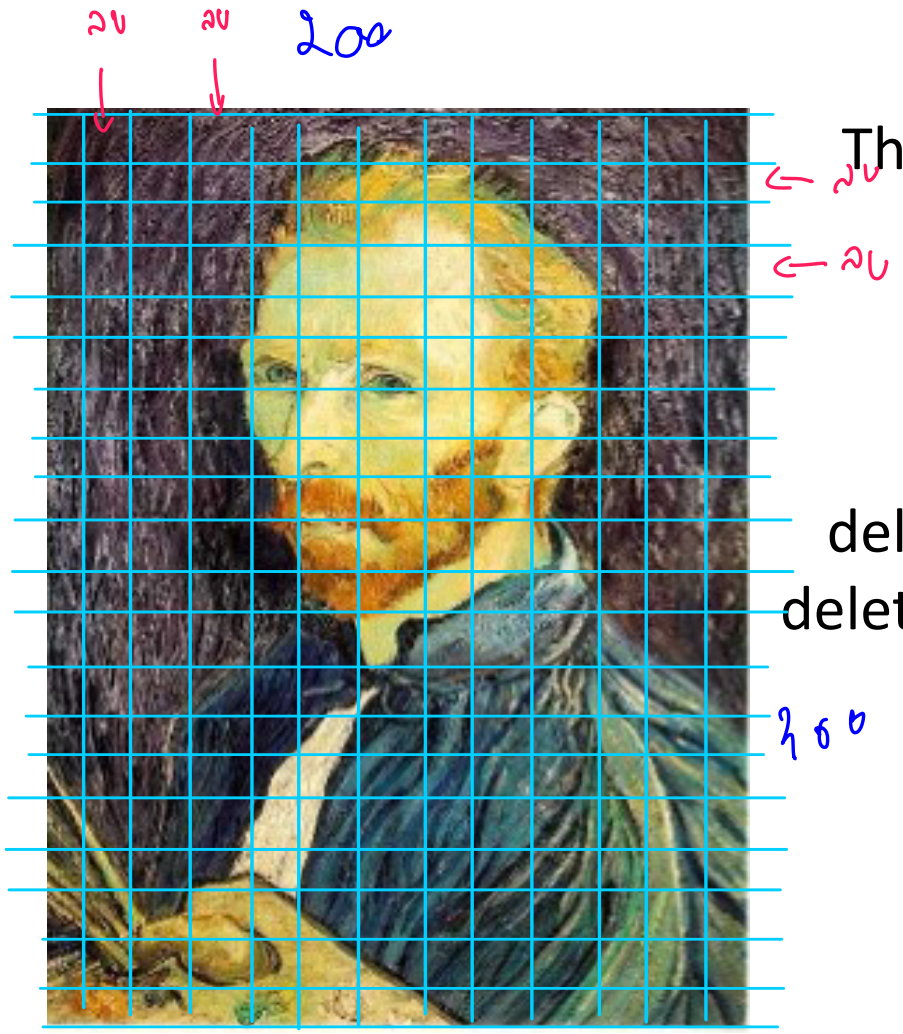
Image pyramids and frequency domain




This image is too big to fit on the screen.
How would you reduce it to half its size?

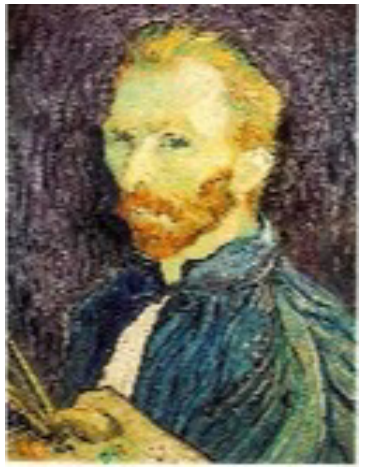
Naïve image downsampling



1/2

Throw away half the rows and columns

delete even rows
delete even columns



1/4

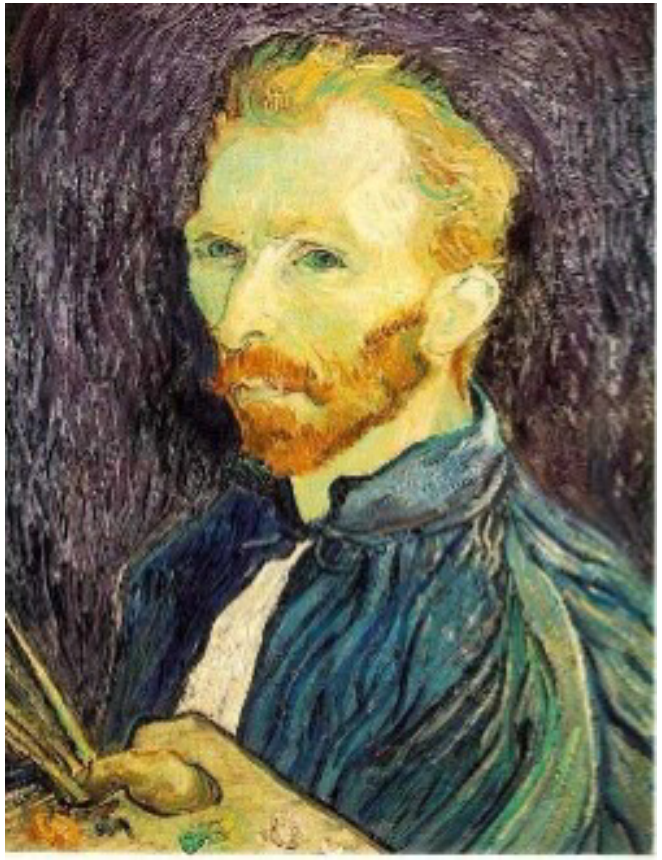
delete even rows
delete even columns

1/8

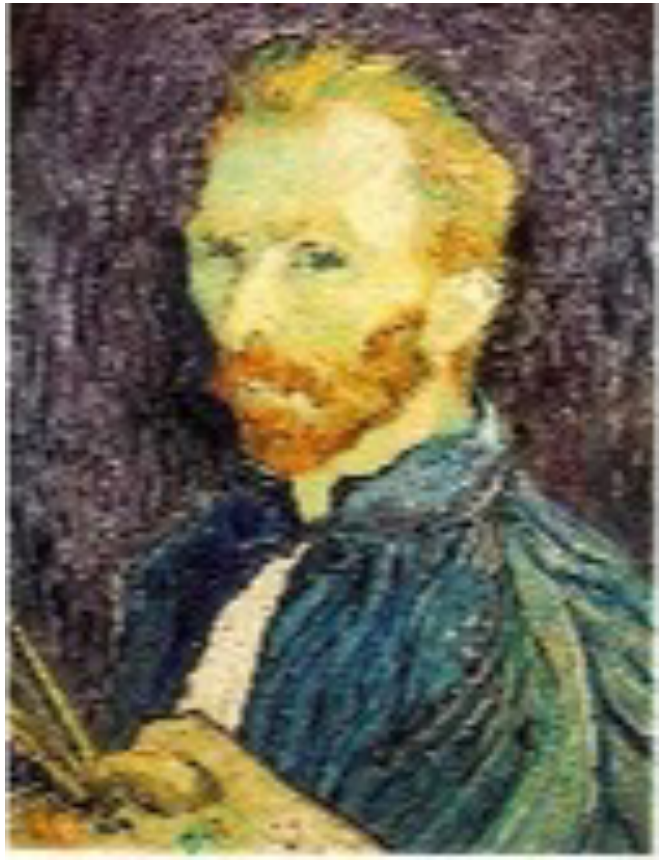
What is the problem with this approach?

Naïve image downsampling

ปัญหา ⇒ ภาพไม่ชัด เบลอ



1/2

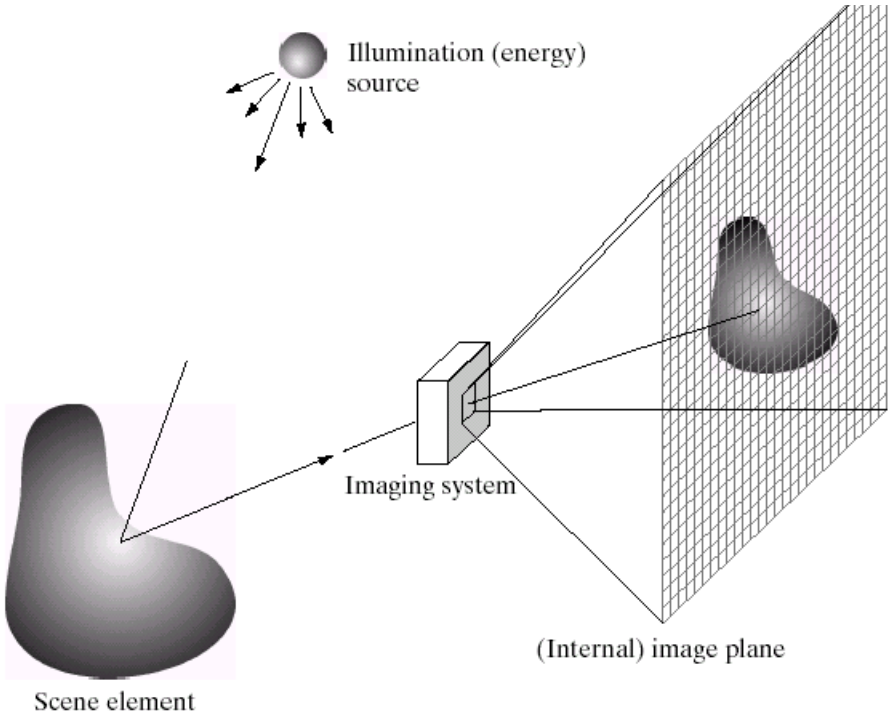


1/4 (2x zoom)

1/8 (4x zoom)

Why is the 1/8 image so pixelated (and do you know what this effect is called)?

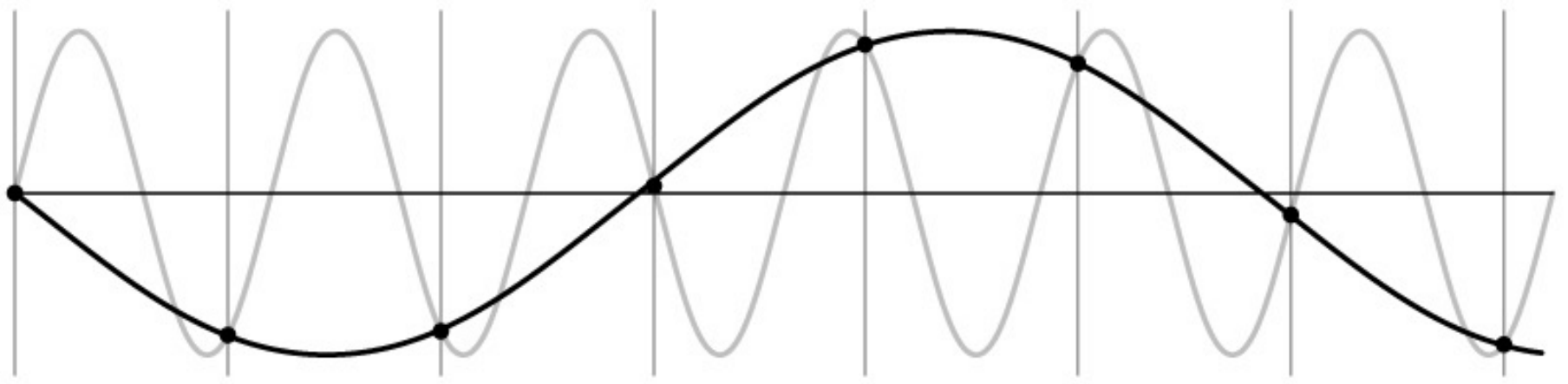
Reminder



Images are a *discrete*, or *sampled*, representation of a *continuous* world

Aliasing

Fancy term for: *Undersampling can disguise a signal as one of a lower frequency*

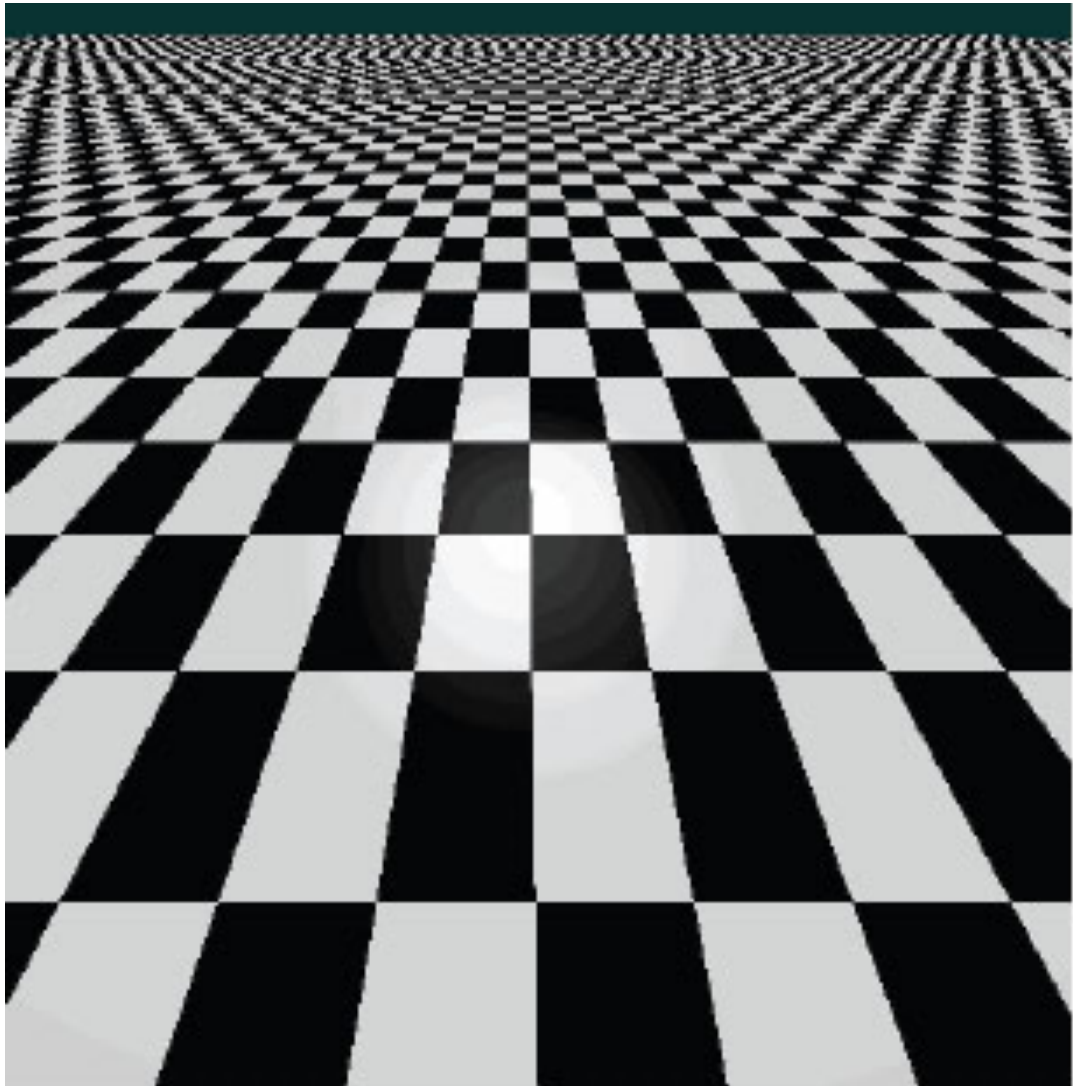


Unsurprising effect: information is lost.

Surprising effect: can confuse the signal with one of *lower* frequency.

Note: we could always confuse the signal with one of *higher* frequency.

Aliasing in textures



Aliasing in photographs

This is also known as “moire”

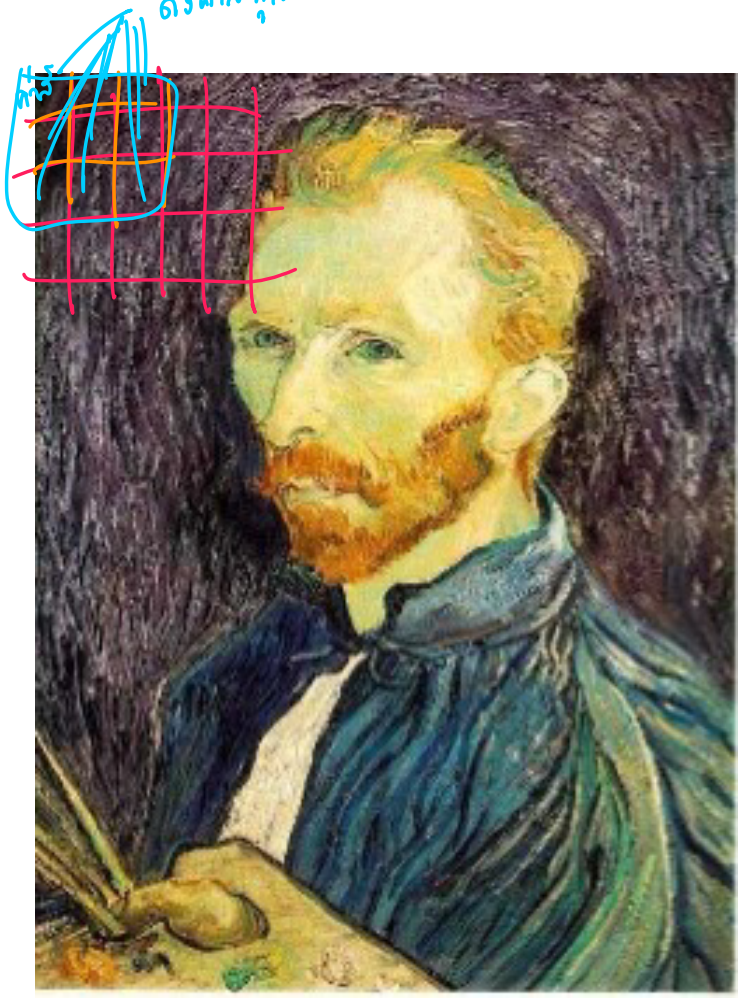
Anti-aliasing

Question 1: How much smoothing is needed to avoid aliasing?

Question 2: How many samples are needed to avoid aliasing?

Answer to both: Enough to reach the Nyquist limit. (We'll see what this means soon.)

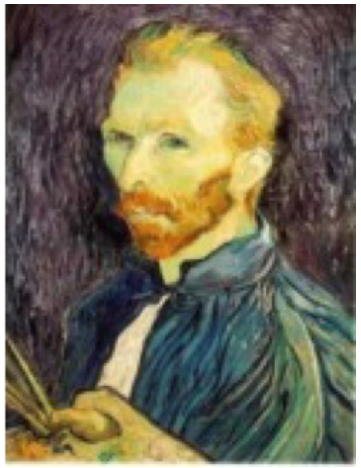
Better image downsampling



1/2

Apply a smoothing filter first, then throw away half the rows and columns

Gaussian filter
delete even rows
delete even columns

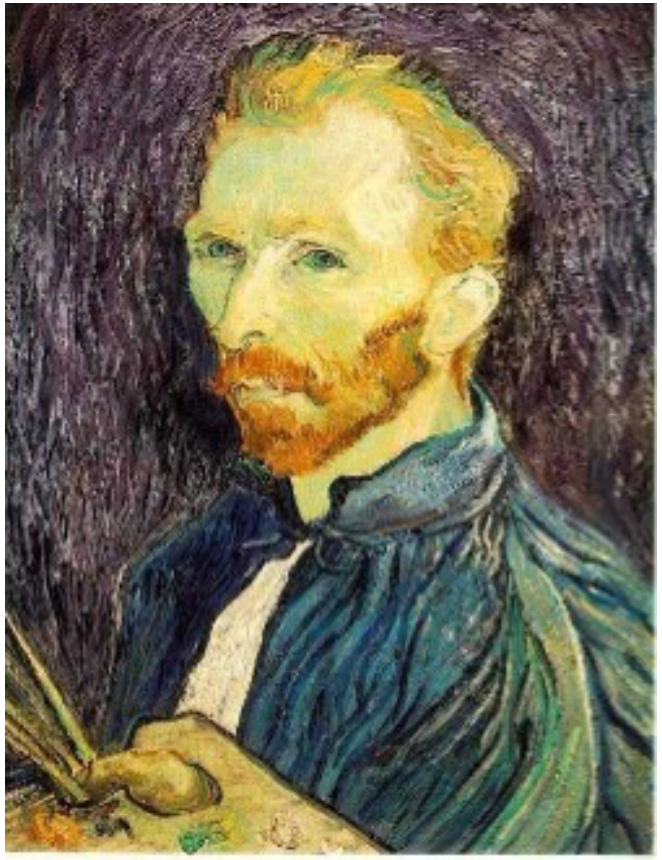


1/4

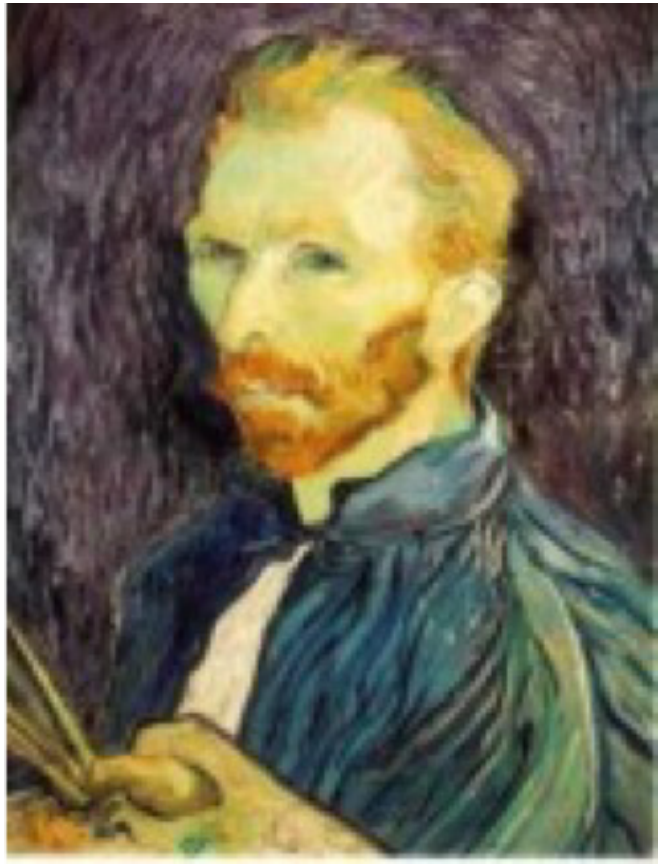
Gaussian filter
delete even rows
delete even columns

1/8

Better image downsampling



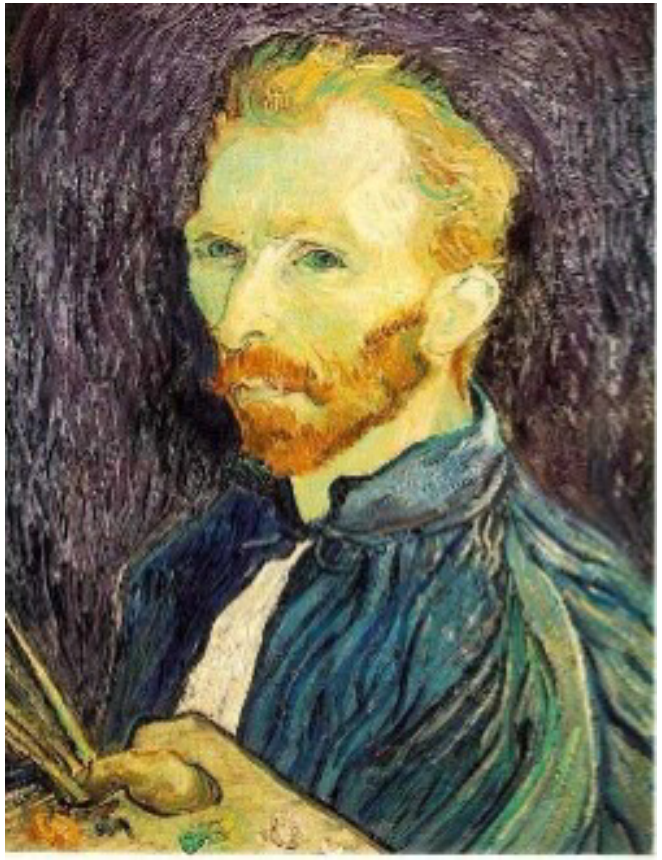
1/2



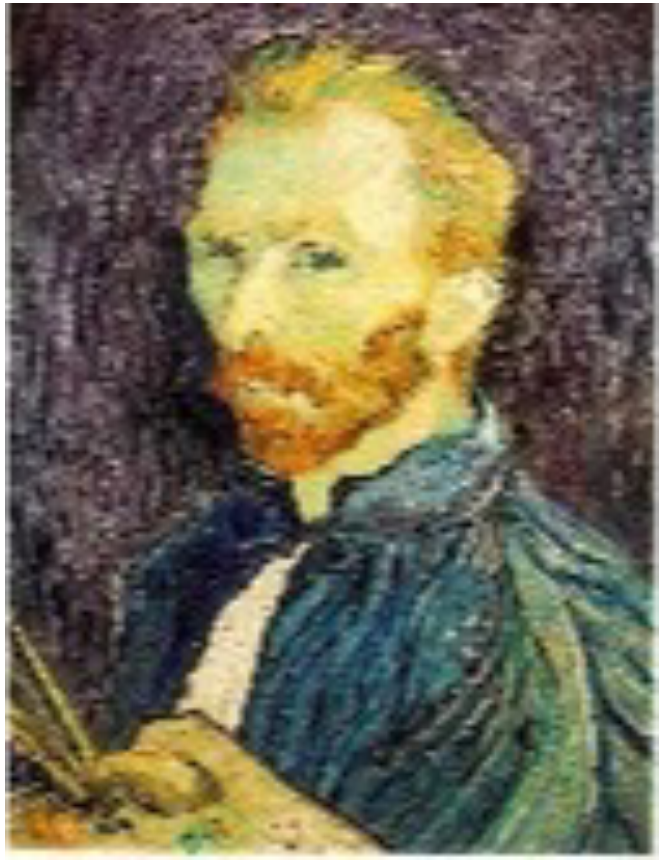
1/4 (2x zoom)

1/8 (4x zoom)

Naïve image downsampling



1/2



1/4 (2x zoom)

1/8 (4x zoom)

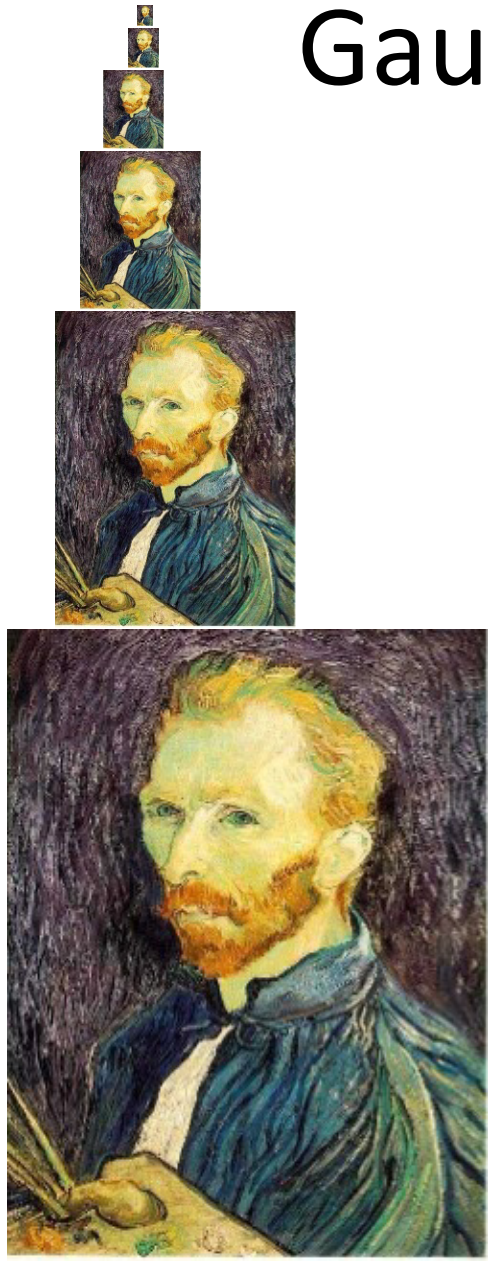
The Nyquist-Shannon sampling theorem

A continuous signal can be perfectly reconstructed from its discrete version using linear interpolation, if sampling occurred with frequency:

$$f_s \geq 2 f_{\max}$$ ← This is called the Nyquist frequency

Equivalent reformulation: When downsampling, aliasing does not occur if samples are taken at the Nyquist frequency or higher.

Gaussian image pyramid



The name of this sequence of subsampled images

PART B · 1:40–2:30

Representation and color: what exactly is in the array

Grayscale and RGB tensors, DICOM and voxels, then color sensing: why three cameras photograph three different colors of the same balloon, and what the in-camera pipeline already did to your dataset.

SOURCE DECKS

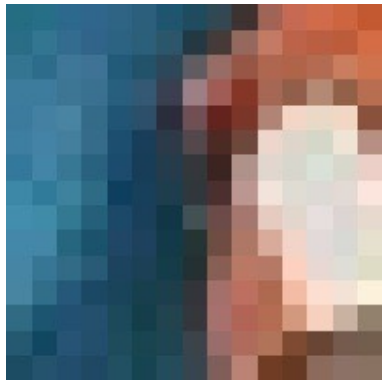
Image filtering (L2) and **Digital photography** (L24–25), CMU 16-385 Fall 2023 · **Lectures 1–2**, Bagci Lab BME 495 · plus authored color-space slides.

each source credited on its own pages · shown unmodified · selected pages

What is an image?

A (color) image is a 3D tensor of numbers.

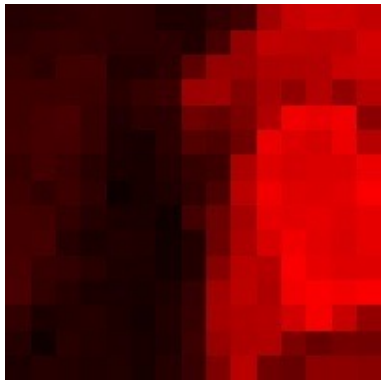
What is an image?



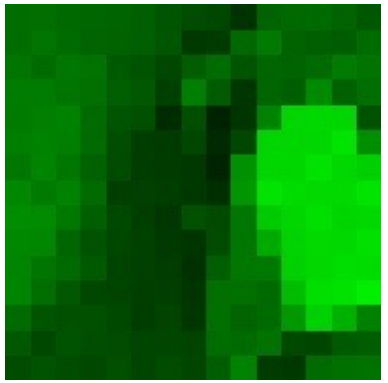
color image patch

How many bits are the intensity values?

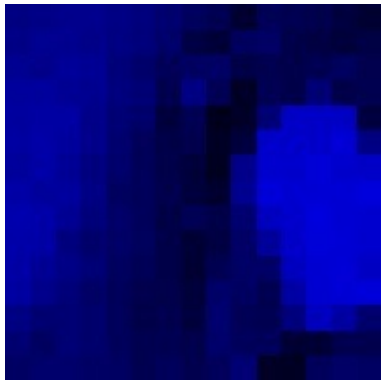
red



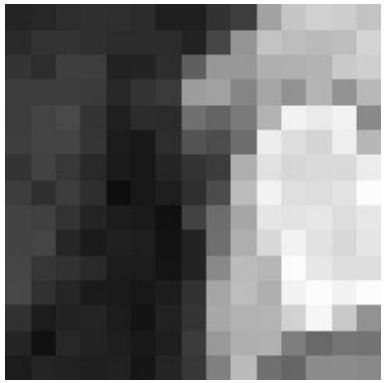
green

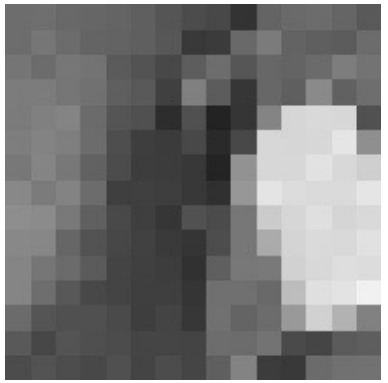


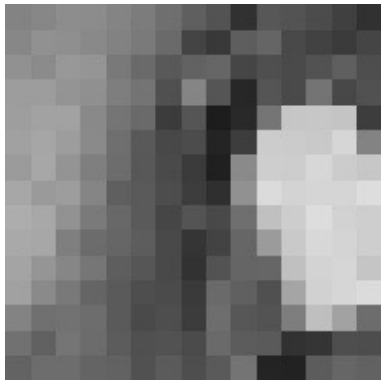
blue



colorized for visualization



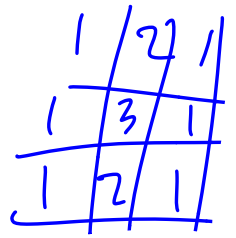




actual intensity values per channel

Each channel is a 2D array of numbers.

Sobel filter.



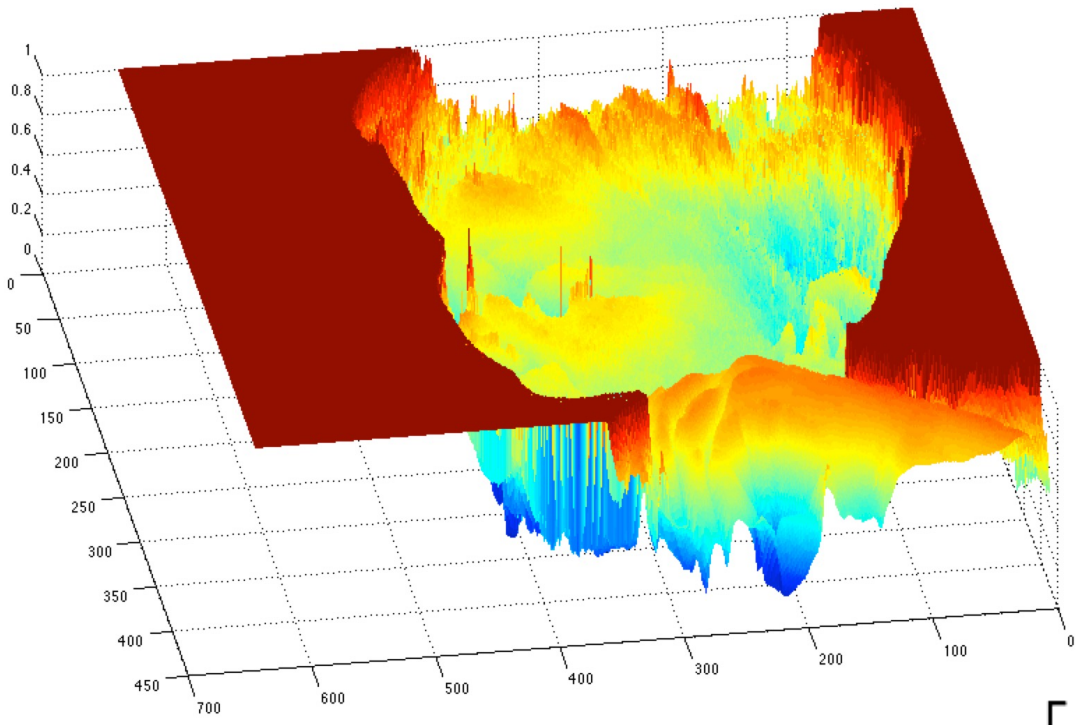
เอาค่าสีไปคูณเลขนำ้นหก แล้วบวกกันหาร 9

What is an image?

grayscale image

What is the range of the image function f?

$f(\boldsymbol{x})$



domain $\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix}$

A (grayscale) image is a 2D function.

Examples of point processing

original

darken

ลดค่าสีทุกจุด

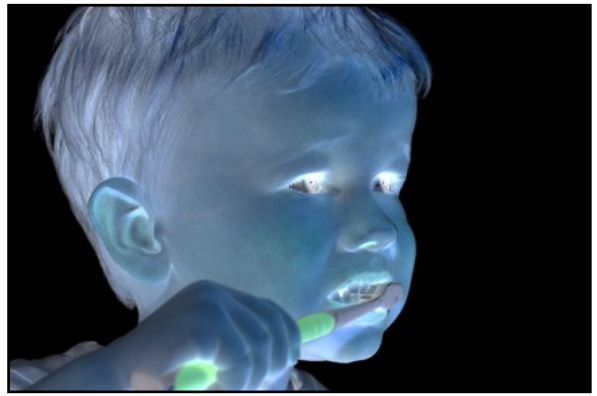
lower contrast

ปรับค่า contrast ให้ไปทางดำ

non-linear lower contrast

ดำ ○ ขาว

invert



lighten

raise contrast

non-linear raise contrast

Many other types of point processing

camera output

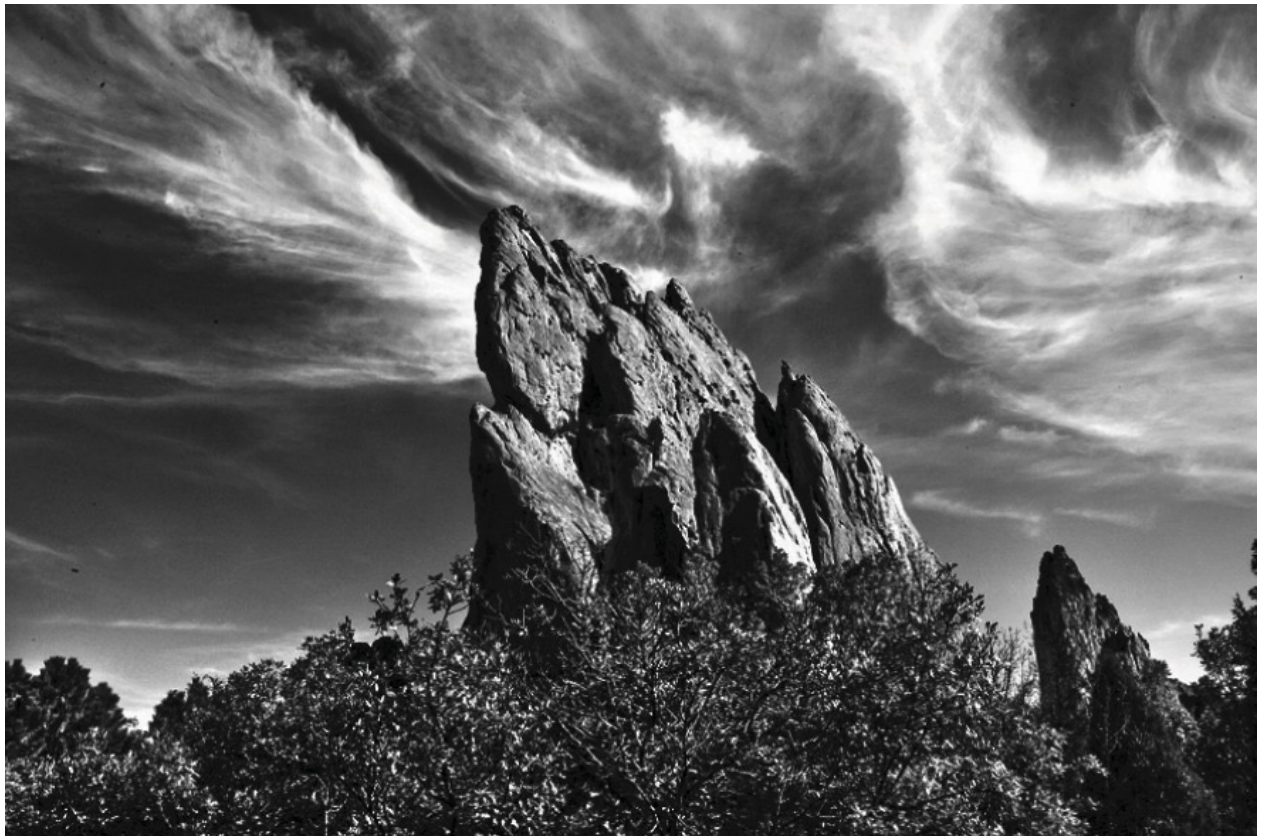


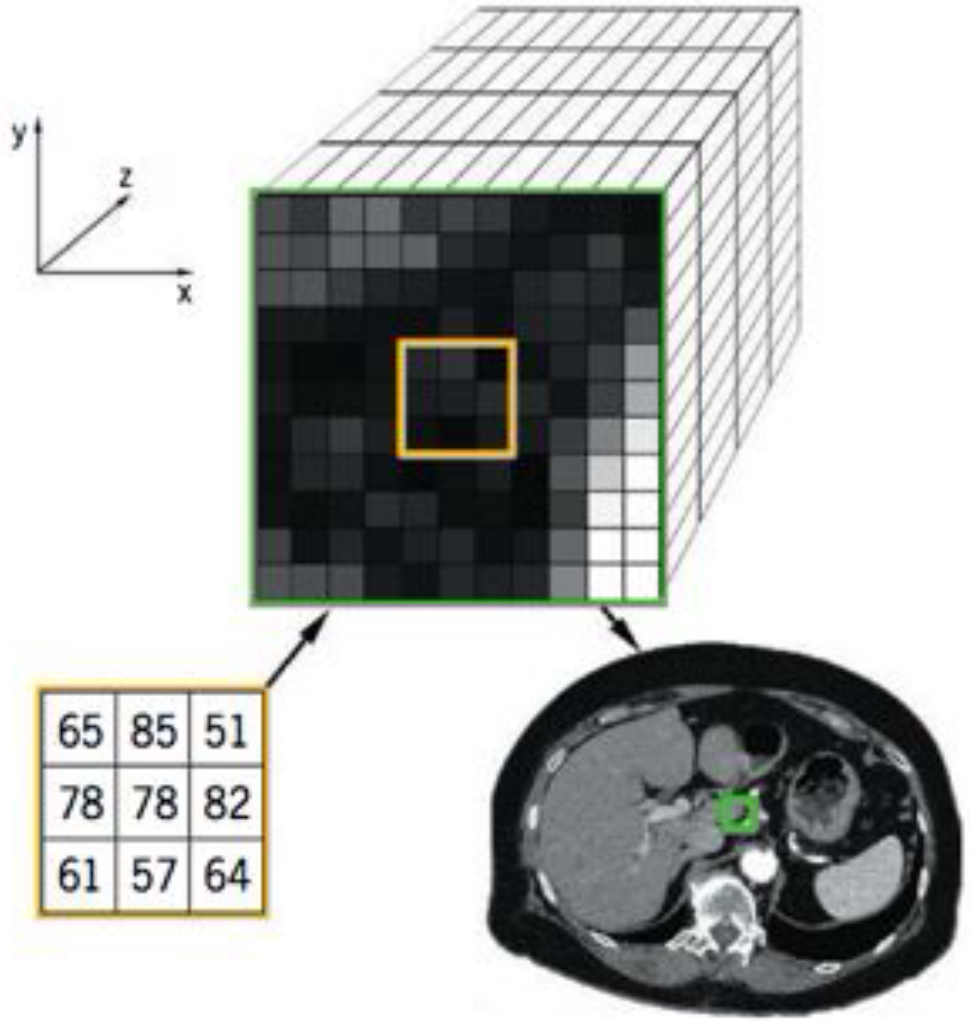
image after stylistic tonemapping

[Bae et al., SIGGRAPH 2006]

Medical Image Formats

- Dicom
- Nifti
- Analyze (img/hdr)
- Raw data
- …

Digital Images

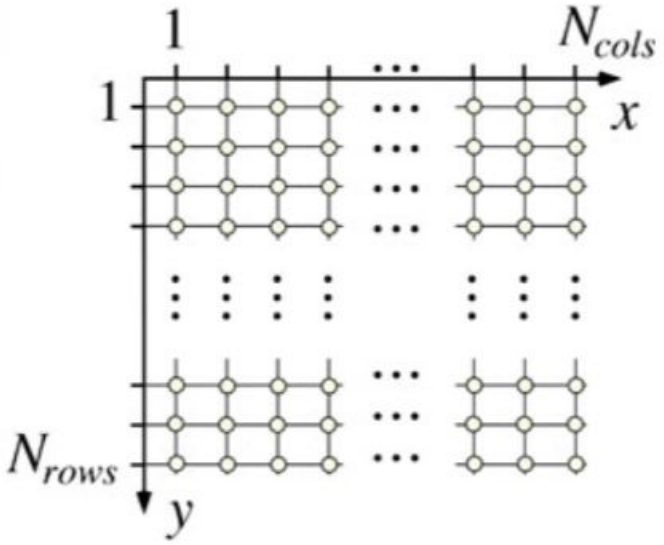


0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

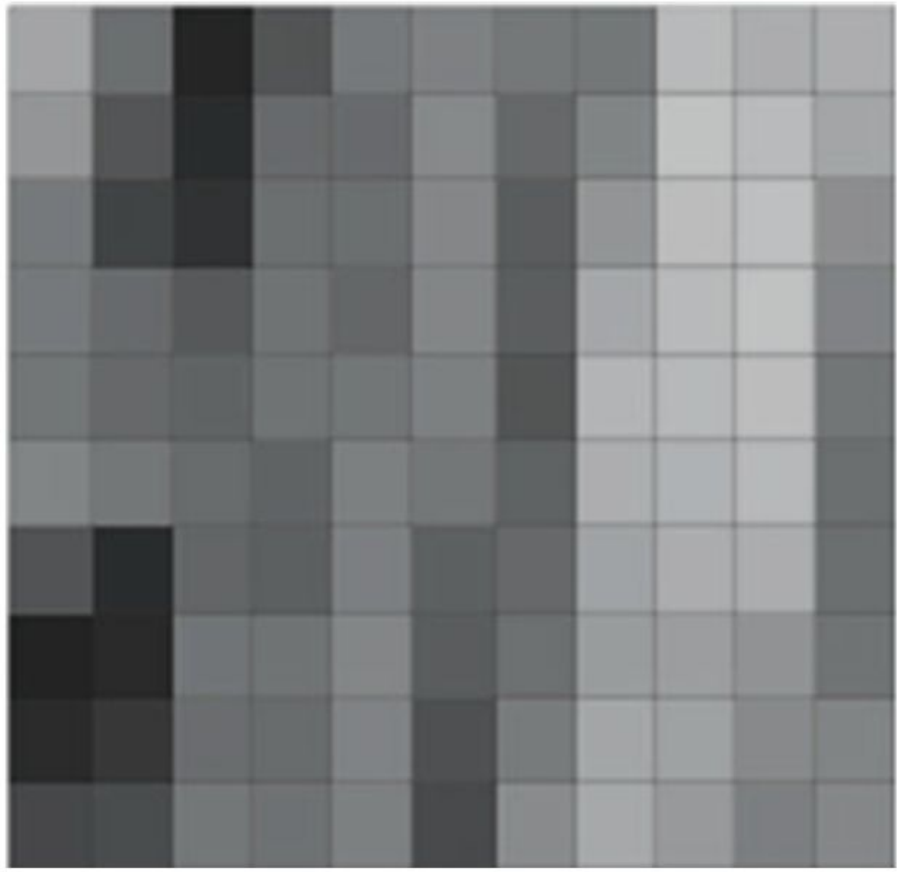
What computer sees!

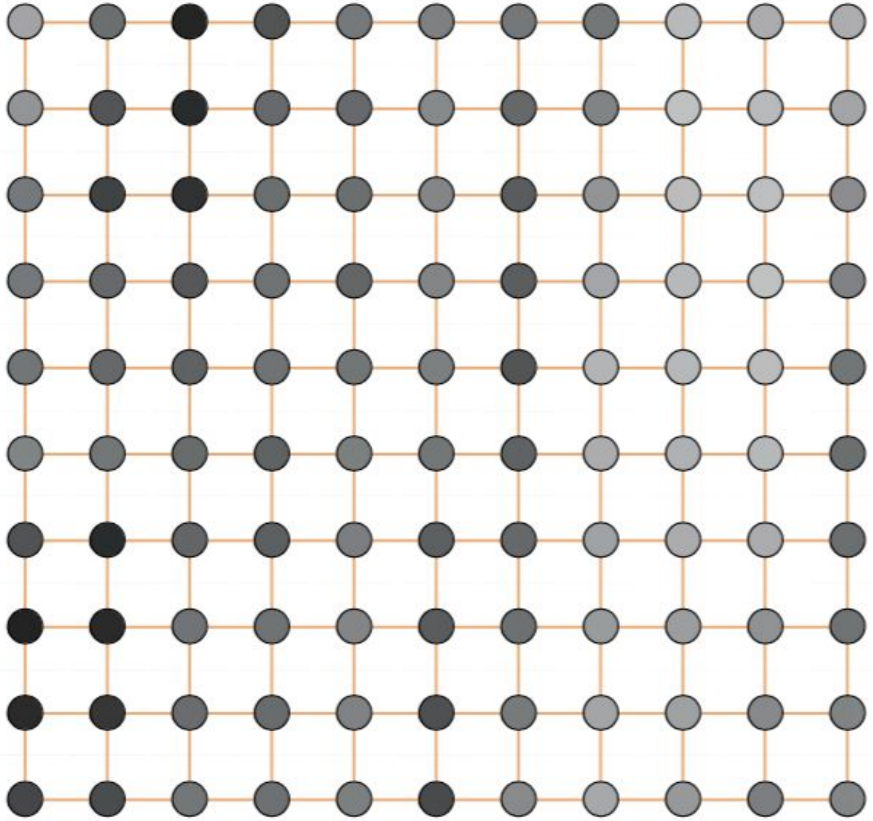
Digital Images

- Definition: A digital image is defined by *integrating* and *sampling* continuous (analog) data in a spatial domain [Klette, 2014].



Picture Elements (Pixels), Volume Elements (Voxels)





PIXELS are ATOMIC ELEMENTS of an image.
In late 1960s, terminology ‘pixel’ was introduced by a group of scientist at JPL in California!

Image Types

- A scalar image has integer values

$$u \in \{0, 1, ..., 2^a - 1\}$$

a: level (bit)

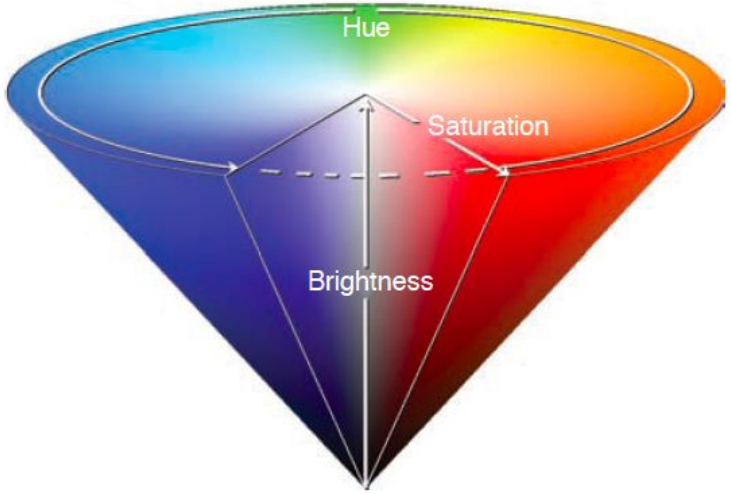
Ex. If 8 bit (a=8), image spans from 0 to 255

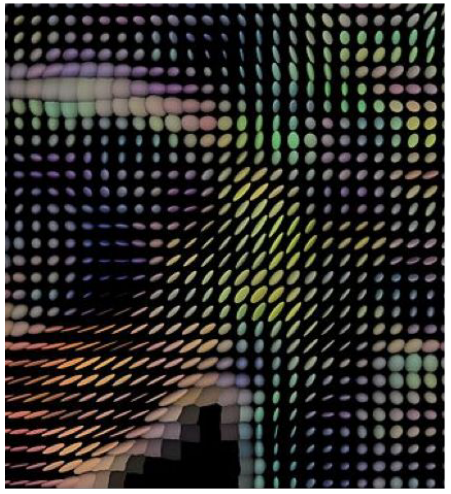
0 black

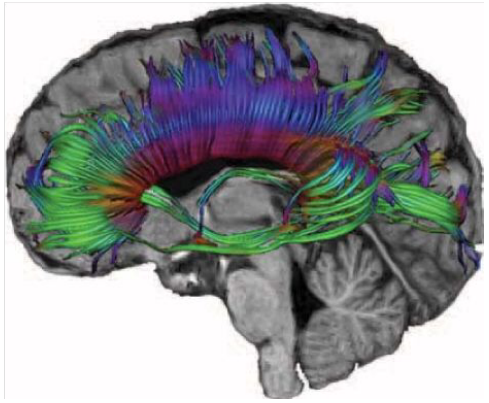
255 white

Ex. If 1 bit (a=1), it is binary image, 0 and 1 only.

Image Types-Color

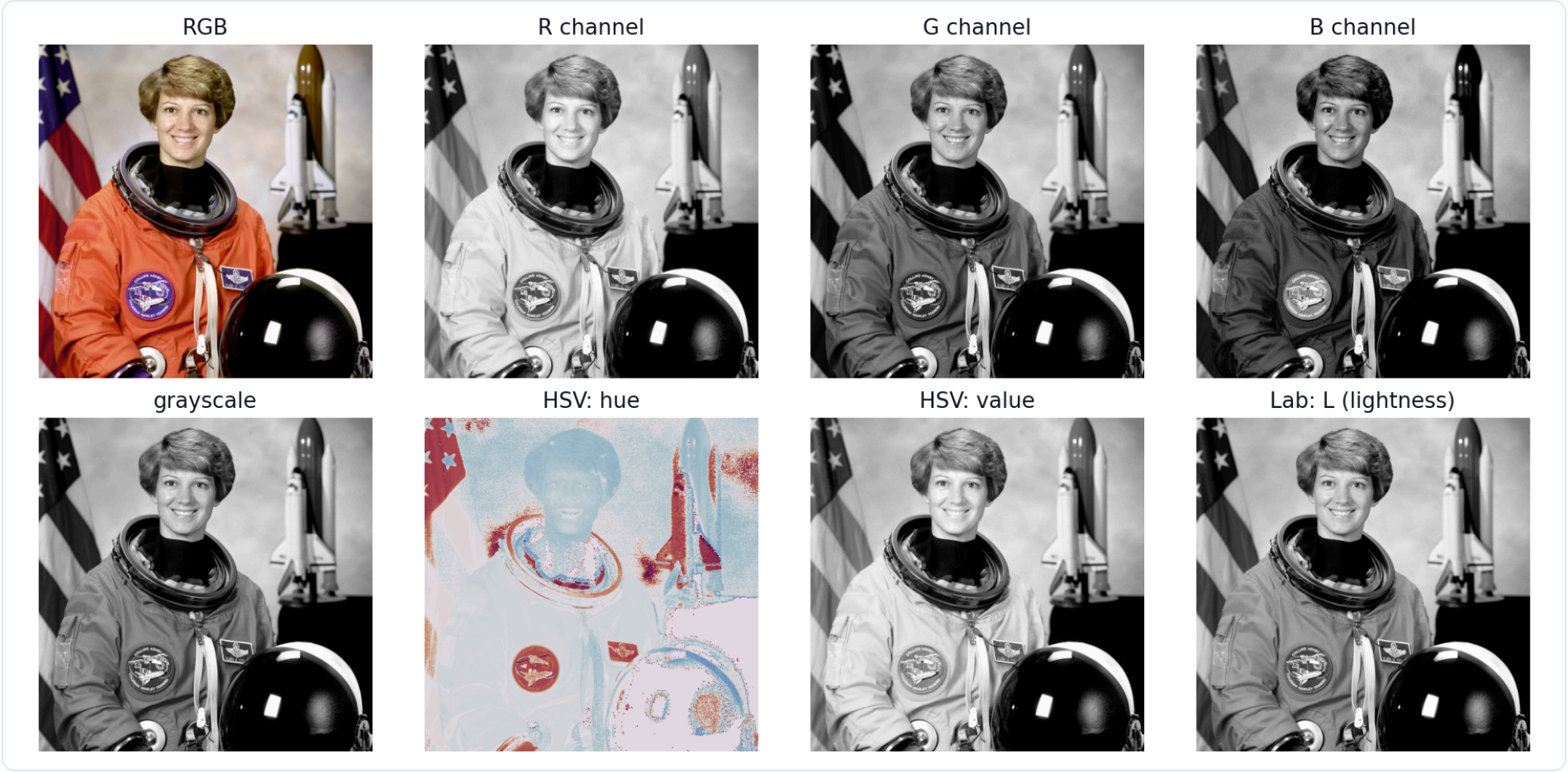






- Image has three channels (bands), each channel spans a-bit values.
- RGB, Hue-Saturation-Brightness

One image, several coordinate systems



scikit-image astronaut sample. RGB channels each look like a plausible grayscale photo; HSV separates what color from how bright; Lab makes distances roughly perceptual.

Conversions are one line, bugs are one line too

```python
# OpenCV loads BGR, not RGB. The classic bug:
img = cv2.imread("scan.png")          # BGR!
rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(rgb, cv2.COLOR_RGB2GRAY)
hsv  = cv2.cvtColor(rgb, cv2.COLOR_RGB2HSV)

# scikit-image: floats in [0, 1], RGB order
from skimage import color
gray = color.rgb2gray(rgb)   # 0.2125 R + 0.7154 G + 0
lab  = color.rgb2lab(rgb)
```

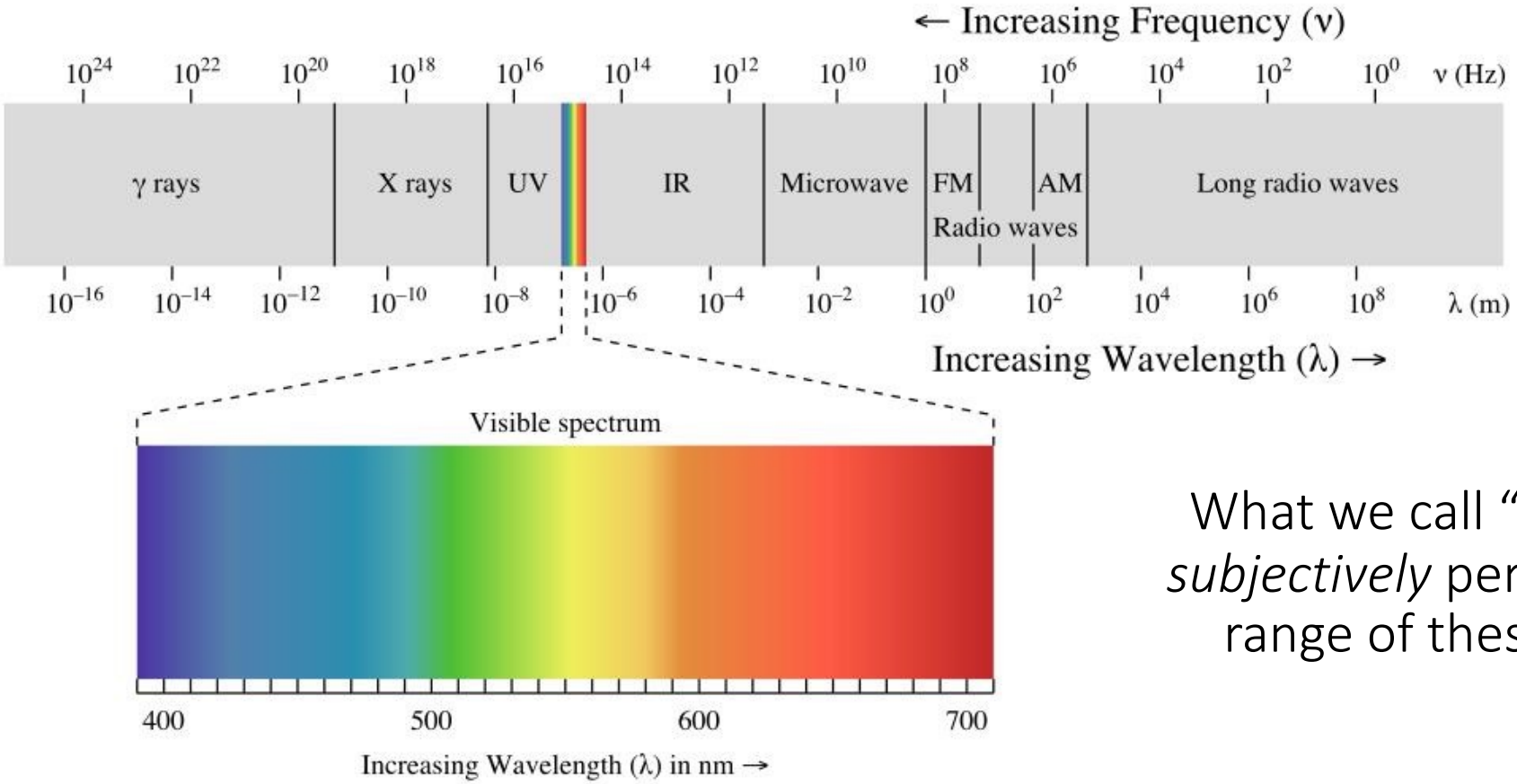
- **Channel order is a convention**, not a property of the file: OpenCV is BGR, PIL and scikit-image are RGB, PyTorch tensors are `C,H,W`, TensorFlow is `H,W,C`.
- Grayscale conversion is a **weighted** sum matched to human luminance, and it is **lossy**: a decision, not a default.
- HSV is handy for color thresholds; **Lab** underlies stain normalization in histopathology (course topic 3.13).

Mixed-up channel order will not crash anything. It just quietly costs you accuracy.

Color sensing in cameras

Color is an artifact of human perception

- "Color" is not an *objective* physical property of light (electromagnetic radiation).
- Instead, light is characterized by its wavelength.



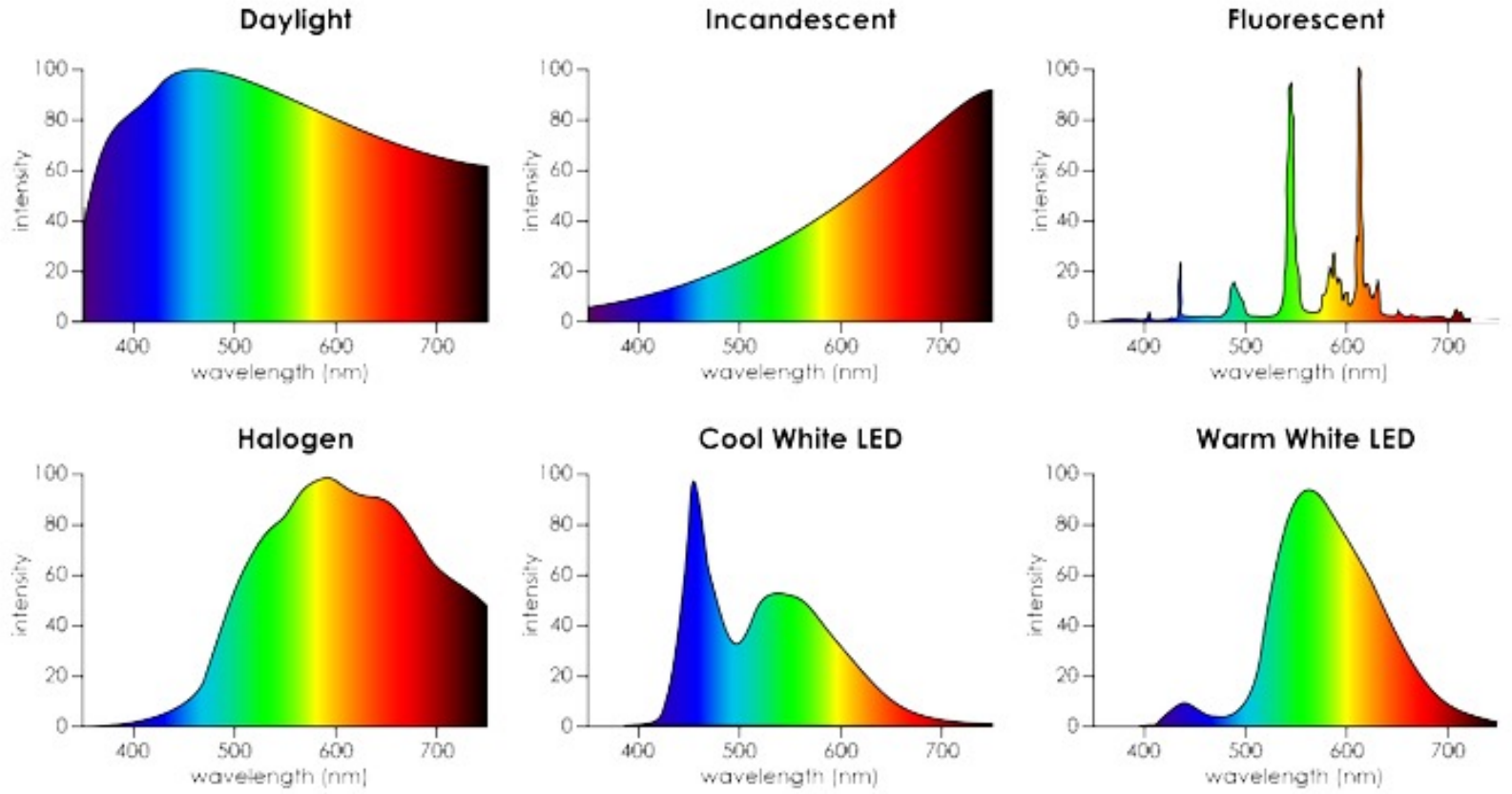
electromagnetic spectrum

What we call "color" is how we *subjectively* perceive a very small range of these wavelengths.

Spectral Power Distribution (SPD)

- Most types of light “contain” more than one wavelengths.
- We can describe light based on the distribution of power over different wavelengths.

We call our sensation of all of these distributions “white”.



Spectral Sensitivity Function (SSF)

- *Any* light sensor (digital or not) has different sensitivity to different wavelengths.
- This is described by the sensor's *spectral sensitivity function* $f(\lambda)$.
- When measuring light with some SPD $\Phi(\lambda)$, the sensor produces a *scalar* response:

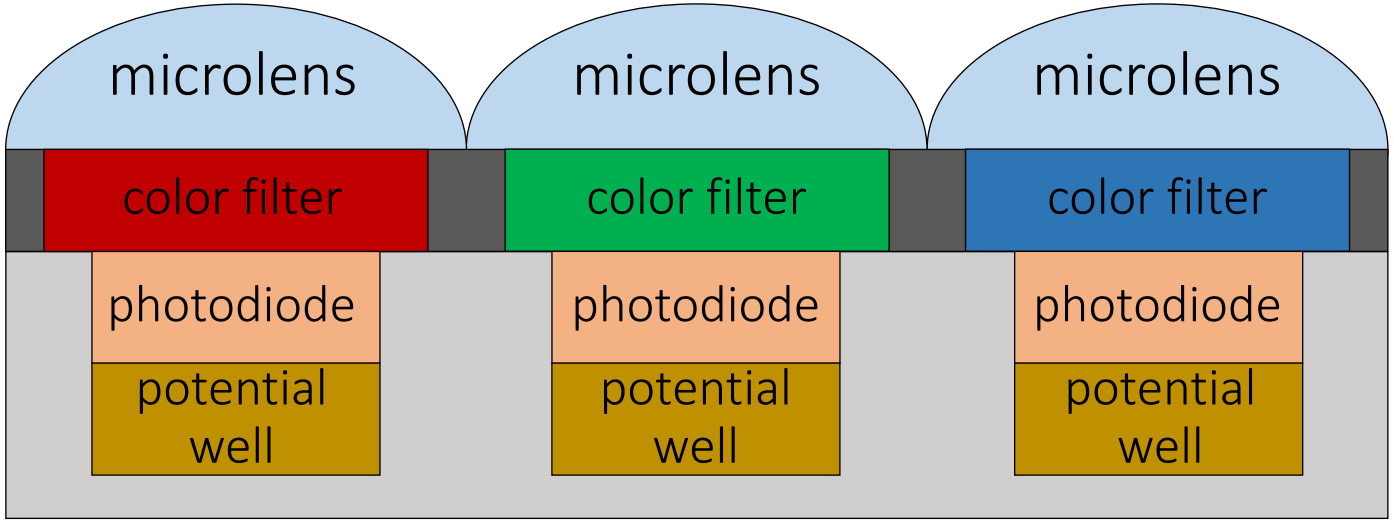
light SPD sensor SSF

sensor response $\longrightarrow$ $$R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$$

Weighted combination of light's SPD: light contributes more at wavelengths where the sensor has higher sensitivity.

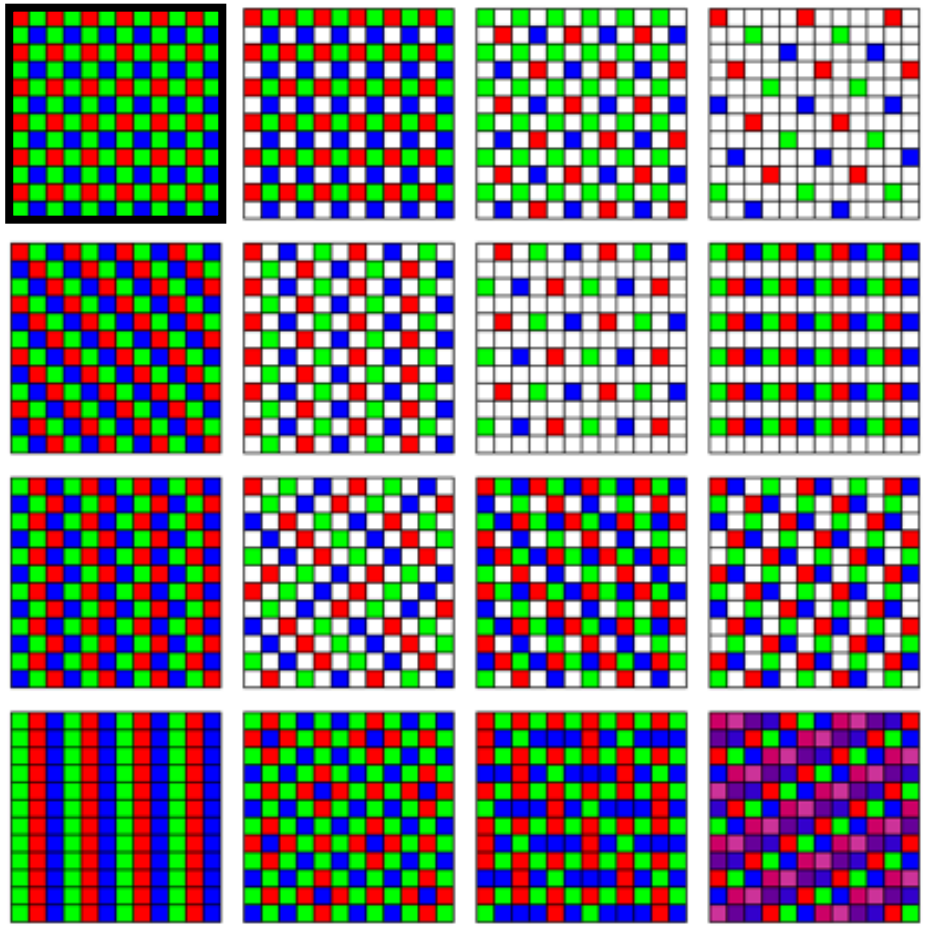
Color filter arrays (CFA)

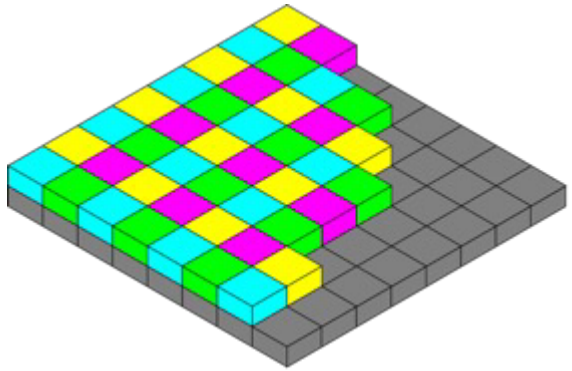
- To measure color with a digital sensor, mimic cone cells of human vision system.
- “Cones” correspond to pixels that are covered by different color filters, each with its own spectral sensitivity function.



Many different CFAs

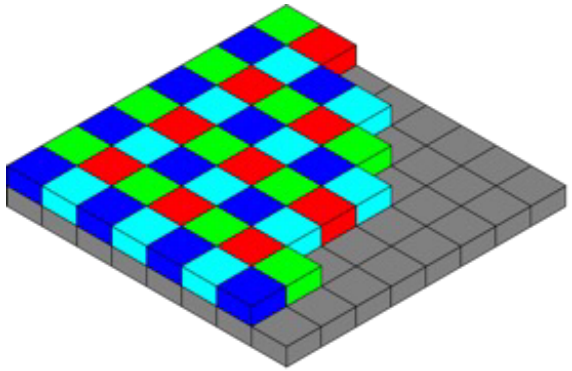
Finding the “best” CFA mosaic is an active research area.





CYGM

Canon IXUS, Powershot



RGBE

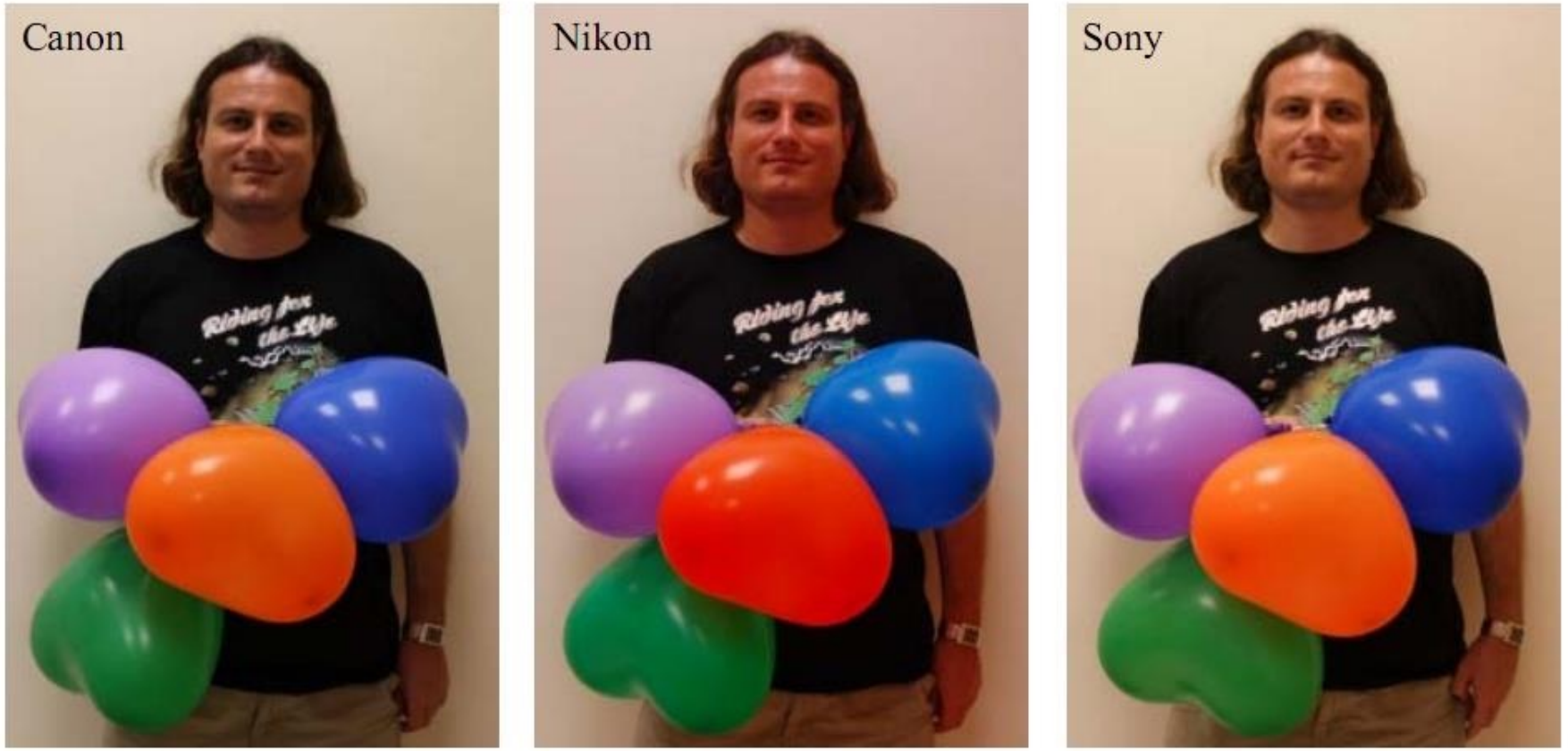
Sony Cyber-shot

How would you go about designing your own CFA? What criteria would you consider?

Many different spectral sensitivity functions

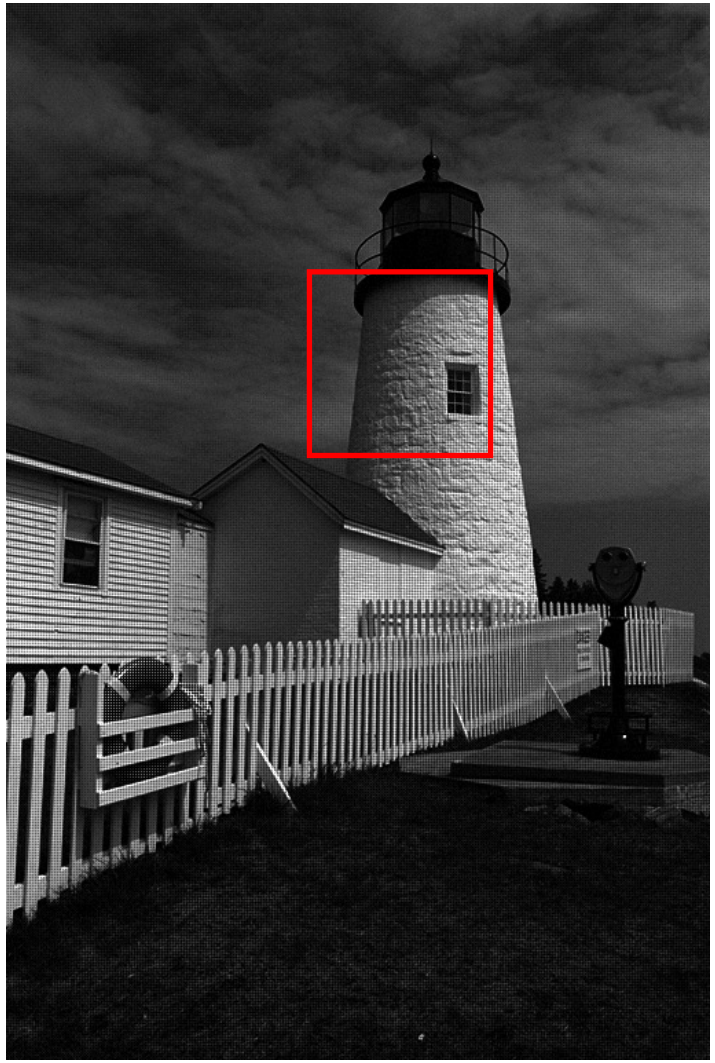
Each camera has its more or less unique, and most of the time *secret*, SSF.

- Makes it very difficult to correctly reproduce the color of sensor measurements.

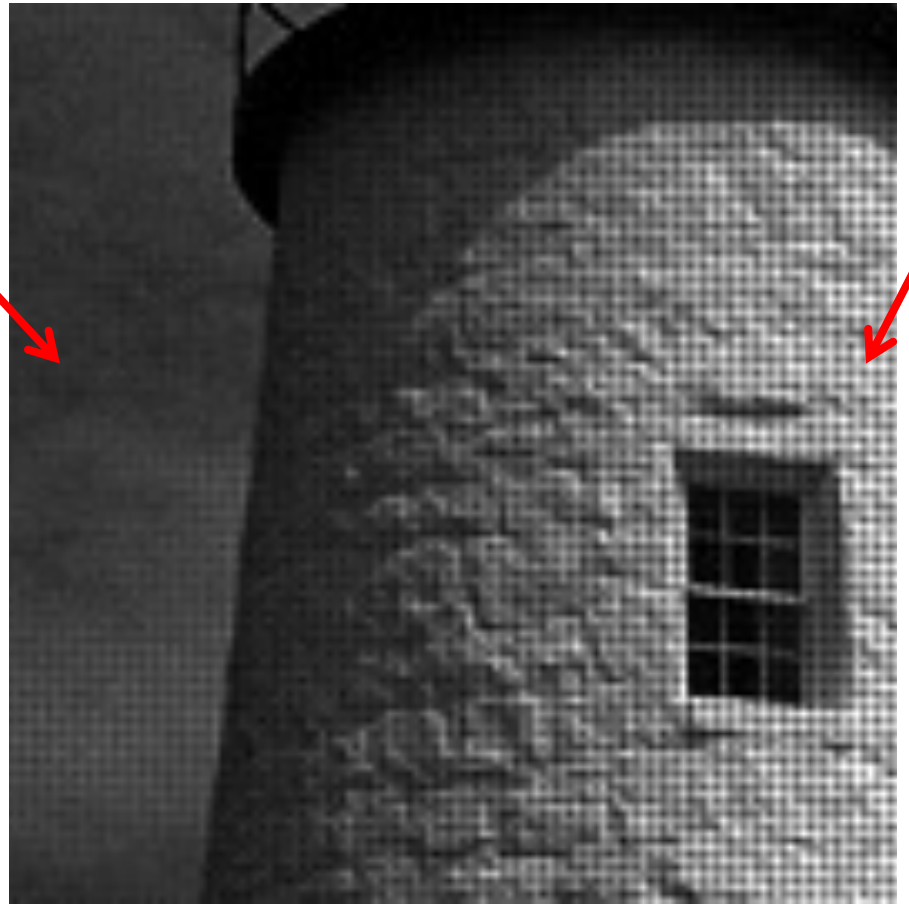


Images of the same scene captured using 3 different cameras with identical **sRGB** settings.

After all of this, what does an image look like?



lots of noise



mosaicking artifacts

- Kind of disappointing.
- We call this the *RAW* image.

PART B, CONTINUED · INSIDE THE CAMERA

Your JPEG was already preprocessed, by the camera

White balance, demosaicing, denoising, gamma. Before you write a single line of Python, an in-camera pipeline has transformed every pixel. Know what it did.

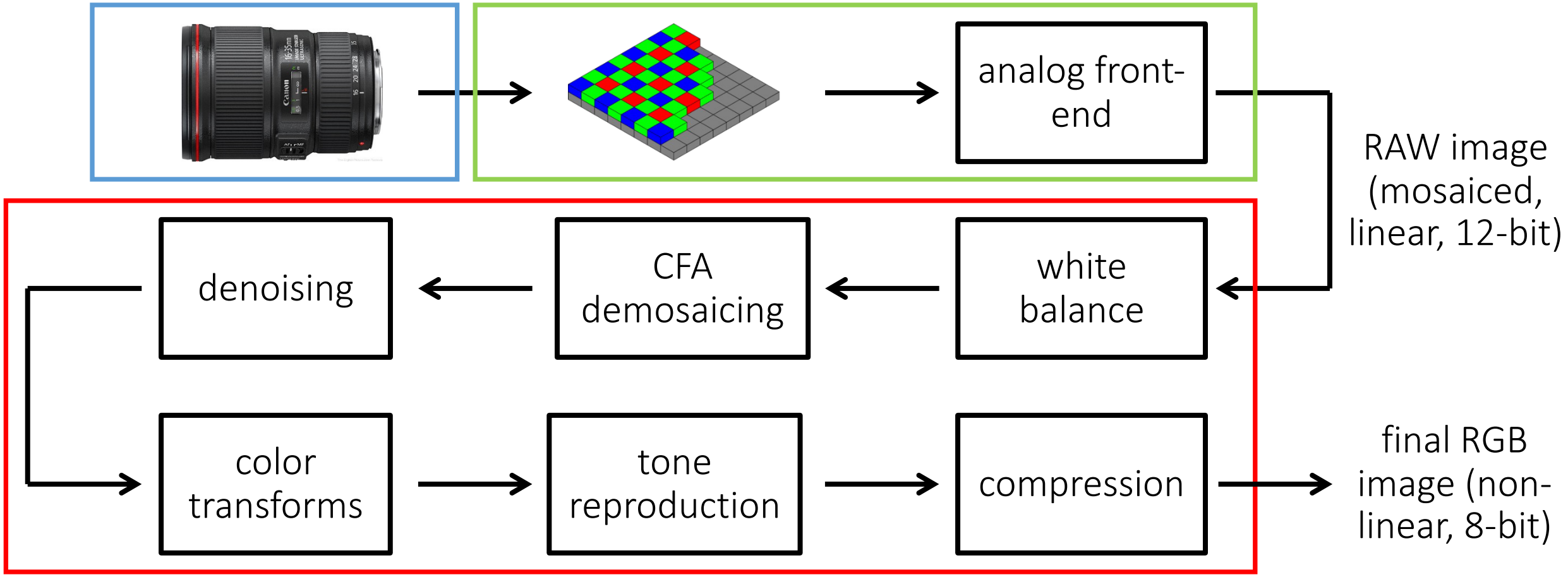
SOURCE DECK

Digital photography, 16-385 Computer Vision, Fall 2023, Lectures 24–25 · Ioannis Gkioulekas, Carnegie Mellon University.
http://16385.courses.cs.cmu.edu/fall2023/ © CMU and credited original authors · shown unmodified · selected pages

The in-camera image processing pipeline

The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



White balancing

Human visual system has *chromatic adaptation*:

- We can perceive white (and other colors) correctly under different light sources.
- Cameras cannot do that (there is no “camera perception”).

White balancing: The process of removing color casts so that colors that we would *perceive* as white are *rendered* as white in final image.

different whites

image captured under fluorescent

image white-balanced to daylight

Automatic white balancing

Grey world assumption:

- Compute per-channel average.
- Normalize each channel by its average.
- Normalize by green channel average.

white-balanced RGB $\rightarrow$ $\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} G_{avg}/R_{avg} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & G_{avg}/B_{avg} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$ $\leftarrow$ sensor RGB

White world assumption:

- Compute per-channel maximum.
- Normalize each channel by its maximum.
- Normalize by green channel maximum.

white-balanced RGB $\rightarrow$ $\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} G_{max}/R_{max} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & G_{max}/B_{max} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$ $\leftarrow$ sensor RGB

Automatic white balancing example

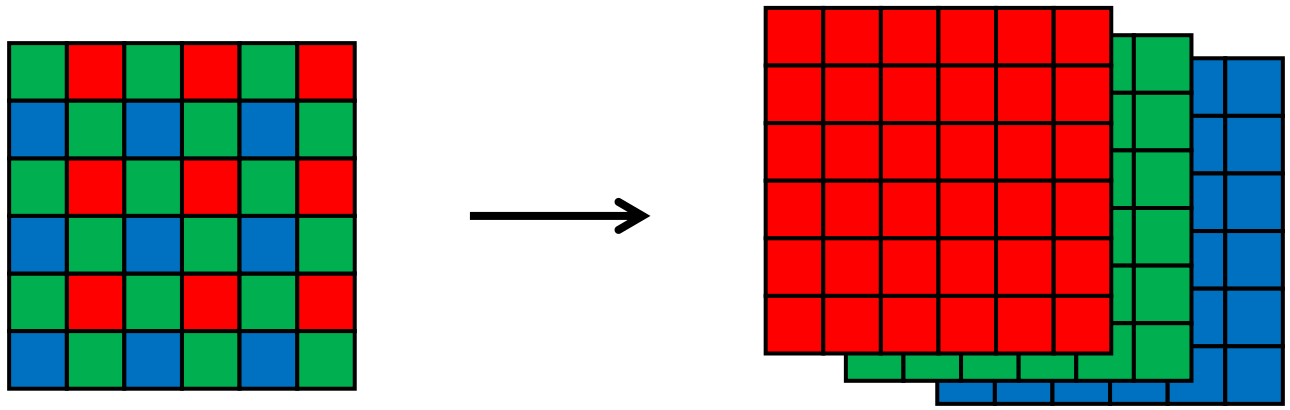
input image

grey world

white world

CFA demosaicing

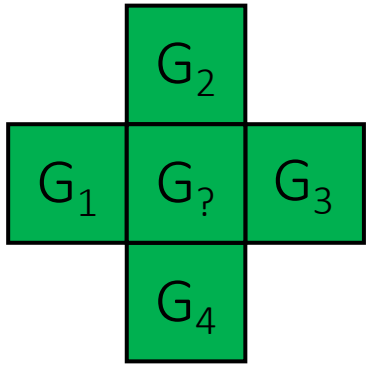
Produce full RGB image from mosaiced sensor output.



Any ideas on how to do this?

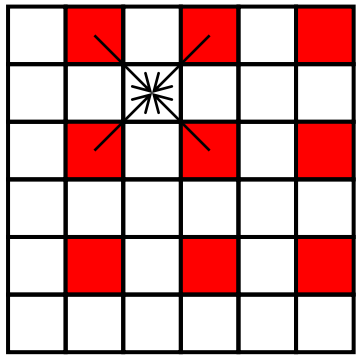
Demosaicing by bilinear interpolation

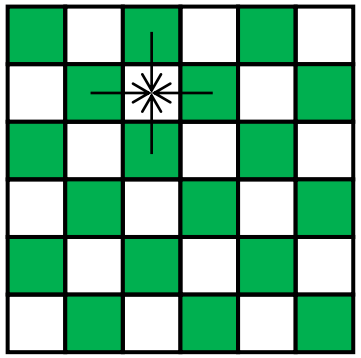
Bilinear interpolation: Simply average your 4 neighbors.

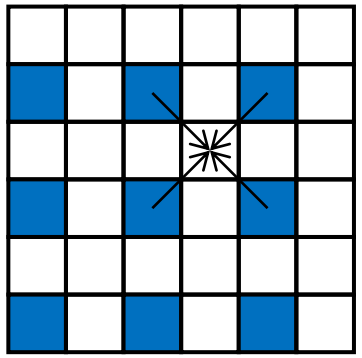


$$G_? = \frac{G_1 + G_2 + G_3 + G_4}{4}$$

Neighborhood changes for different channels:

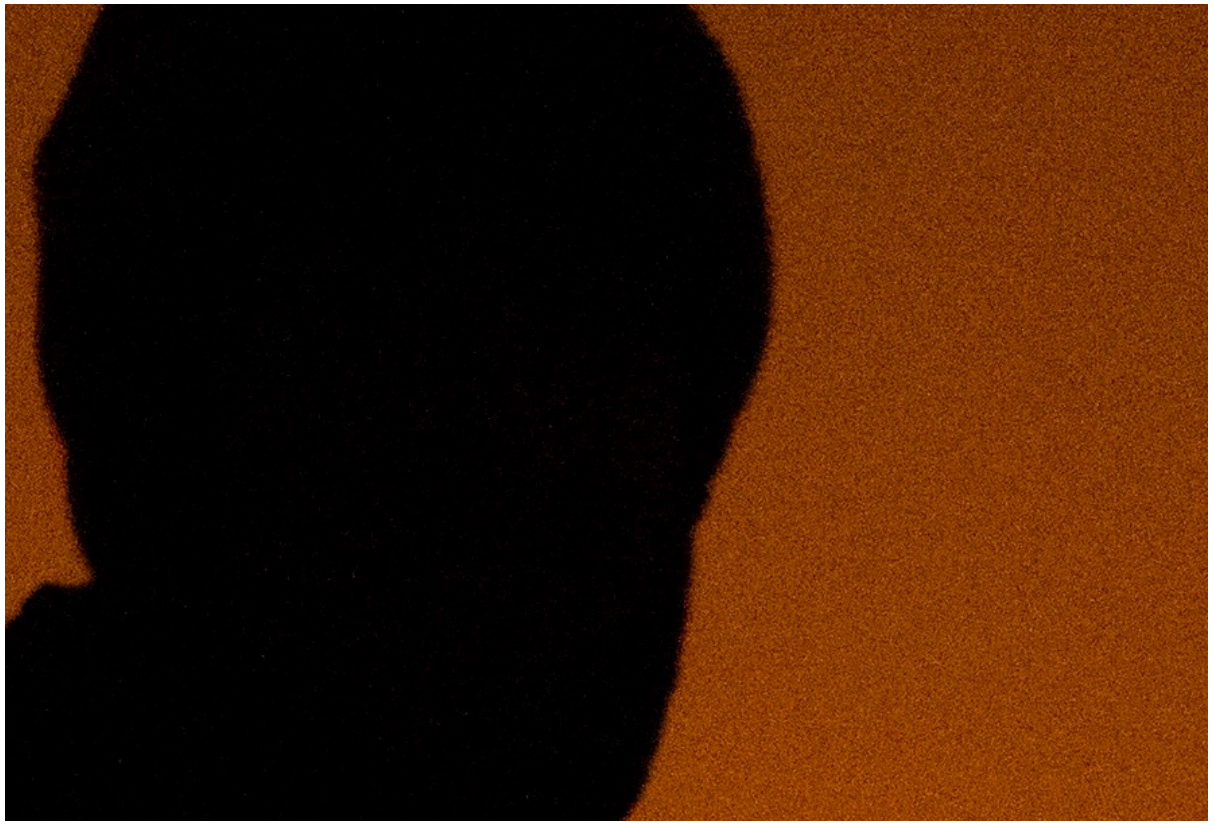






Noise in images

Can be very pronounced in low-light images.



Three types of sensor noise

1) (Photon) shot noise:

- Photon arrival rates are a random process (Poisson distribution).
- The brighter the scene, the larger the variance of the distribution.

2) Dark-shot noise:

- Emitted electrons due to thermal activity (becomes worse as sensor gets hotter.)

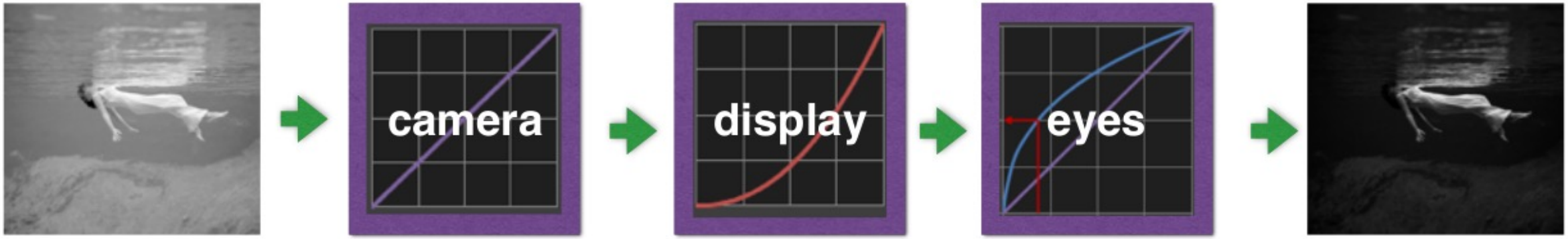
3) Read noise:

- Caused by read-out and AFE electronics (e.g., gain, A/D converter).

Bright scene and large pixels: photon shot noise is the main noise source.

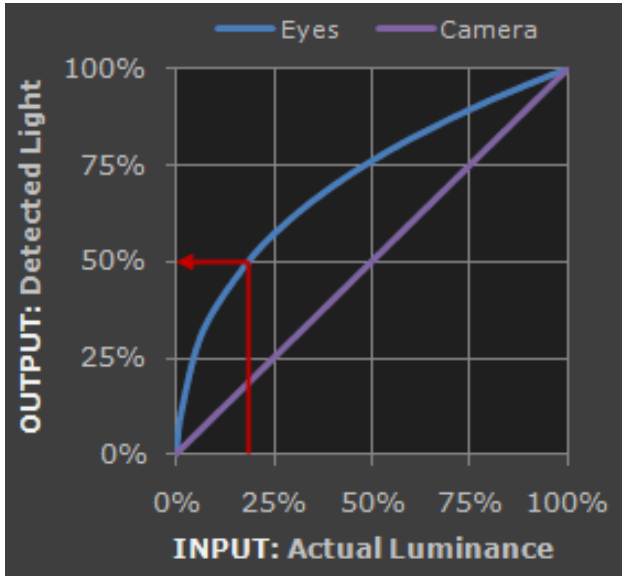
Tone reproduction

- Also known as gamma correction.
- Without tone reproduction, images look very dark.



Why does this happen?

Perceived vs measured brightness by human eye



We have already seen that sensor response is linear.

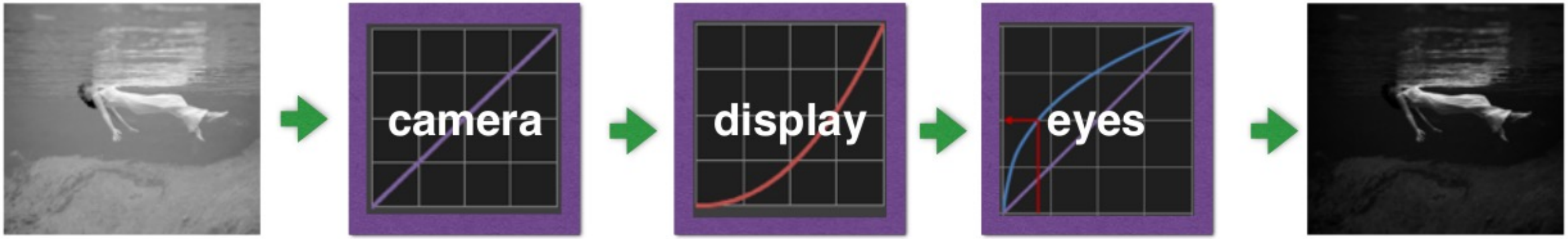
Human-eye *response* (measured brightness) is also linear.

However, human-eye *perception* (perceived brightness) is *non-linear*:

- More sensitive to dark tones.
- Approximately a Gamma function.

Tone reproduction

- Because of mismatch in displays and human eye perception, images look very dark.



How do we fix this?

Do I ever need to use RAW?

Emphatic yes!

- Every time you use a physics-based computer vision algorithm, you need *linear measurements of radiance*.
- Examples: photometric stereo, shape from shading, image-based relighting, illumination estimation, anything to do with light transport and inverse rendering, etc.
- Applying the algorithms on non-linear (i.e., not RAW) images will produce completely invalid results.

Take-home messages

The values of pixels in a photograph and the values output by your camera's sensor are two very different things.

The relationship between the two is complicated and unknown, and we often need to account for it when doing computer vision.

PART C · 2:30–3:00

First preprocessing: resize and normalize

Two operations open every image pipeline. Both look trivial. Both quietly decide what information survives to reach the model. Class 2 builds on exactly these.

AUTHORED

Written for this course; examples use **OpenCV**, **scikit-image**, and **torchvision** public APIs with scikit-image sample images.

docs.opencv.org · scikit-image.org · docs.pytorch.org/vision

Naive downsampling aliases, exactly as CMU warned

Keeping every 4th pixel invents jagged texture; smoothing before resampling keeps the content. The Van Gogh moire from Part A5, now on your own preprocessing step.

Say what you mean: interpolation flags

```
# OpenCV: pick the interpolation on purpose
small = cv2.resize(img, (224, 224),
        interpolation=cv2.INTER_AREA)   # shrink
big   = cv2.resize(img, (896, 896),
        interpolation=cv2.INTER_CUBIC)  # enlarge

# scikit-image: anti-aliasing on by default when shrin
from skimage.transform import resize
small = resize(img, (224, 224), anti_aliasing=True)

# torchvision: antialias=True is the modern default
t = v2.Resize((224, 224), antialias=True)
```

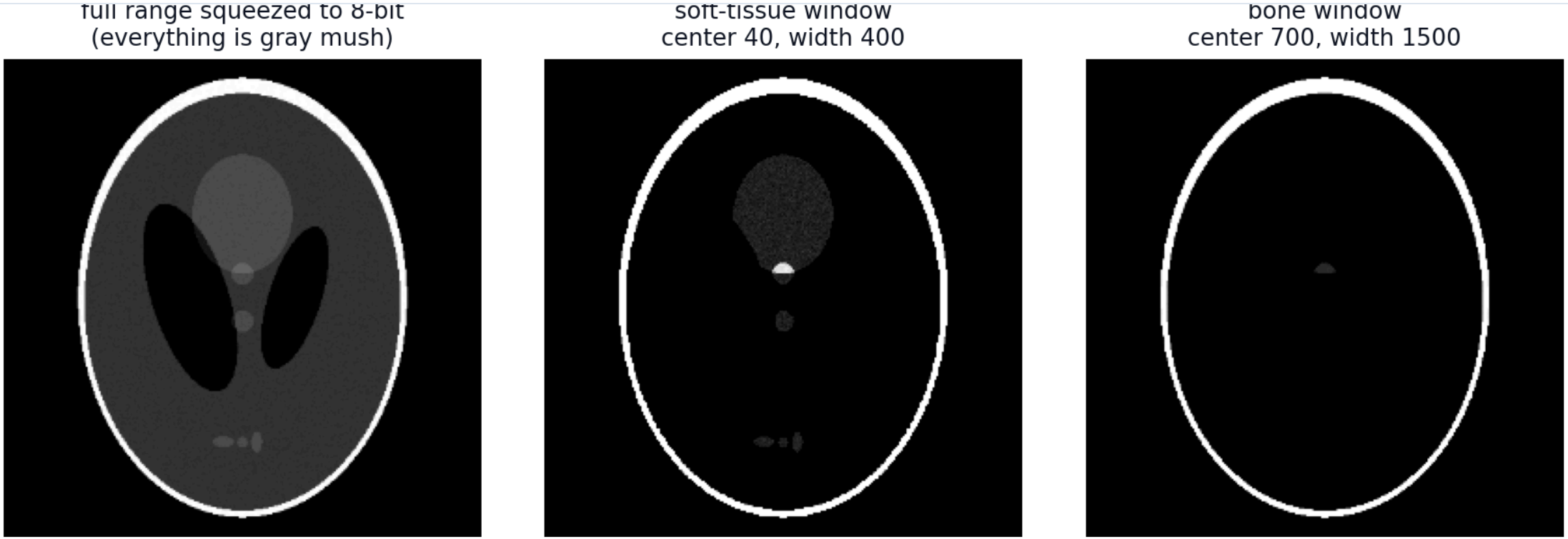
- **Shrinking**: use area averaging or an anti-aliased resampler, never plain nearest neighbor.
- **Enlarging**: bilinear or bicubic; nothing invents real detail, it only interpolates.
- **Nearest neighbor still has one job**: label masks in segmentation, where averaging would create fake classes.
- Backends differ slightly (OpenCV vs PIL vs torch); **keep one resize path** between training and deployment.

Aspect ratio, target size, and medical spacing

- **Squash or letterbox?** Forcing 224×224 distorts shapes; padding to a square keeps geometry but wastes pixels. Detection pipelines usually letterbox; classification often tolerates a mild squash.
- **Target size is a budget**: bigger inputs cost quadratic compute and memory. Choose from the model family and GPU, then keep it fixed and documented.
- **Medical volumes are sampled in millimetres, not pixels.** Two CT scans with equal array shape can cover different anatomy per voxel. Resample to a common **voxel spacing** first (MONAI `Spacingd`, Class 2), and record it.
- **Write the numbers down.** Interpolation method, output size, padding color, spacing: they belong in your paper's preprocessing paragraph (session 4 clinic asked exactly this).

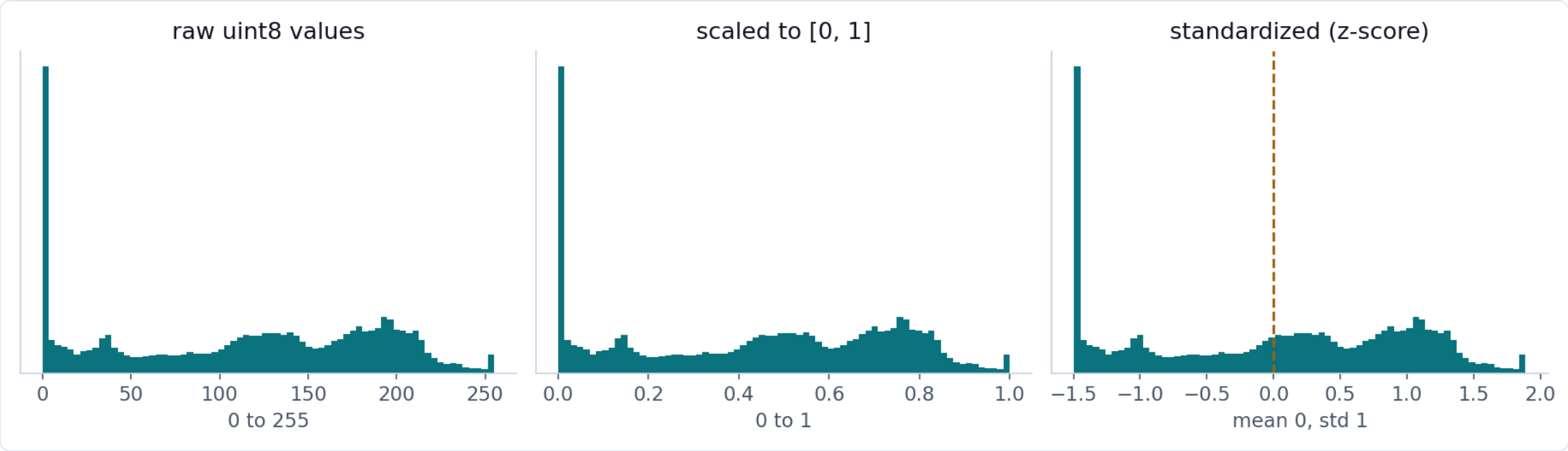
Reproducibility check: could a colleague re-create your tensors from the methods section alone?

12-bit scans do not fit in 8-bit habits



Simulated CT-style intensities: squeezing the full range into one 8-bit image hides tissue; windowing (center, width) chooses which contrast survives. Real CT stores Hounsfield units far beyond 255.

Same picture, three number ranges



- Dividing by 255 changes units, not shape; z-scoring recenters the distribution where optimizers like it. **Why models care: Class 2, hour 2.**
- The question we deliberately left open: **whose mean and std?** This image's? The dataset's? ImageNet's? That choice is a leakage question, and it opens Class 2.

Before class 2

- **Task.** Take three images from your own project (or phone). For each: downsample 4× naively and with anti-aliasing; reduce to 4-bit depth; convert to grayscale and HSV. Bring the ugliest artifact you produced and explain its cause in one sentence.
- **Medical option.** If you work with CT or MRI: load one volume, report voxel spacing, and show one slice at two window settings.
- **Reading.** Szeliski, *Computer Vision* 2nd ed., chapter 2 (image formation), free PDF at `szeliski.org/Book` · CS231n notes on data preprocessing at `cs231n.github.io/neural-networks-2`.
- **Question to sleep on.** Your camera already white-balanced, demosaicked, denoised, and gamma-compressed every photo. Is your "raw data" raw?

NEXT TIME · CLASS 2

Pre-processing before and during training

- HOUR 1 Classical operations: filtering, denoising, histogram equalization and CLAHE, medical intensity work.
- HOUR 2 Preprocessing for the model: zero-centering, per-channel statistics, the training-data-only rule, MRI intensity standardization, pretrained-model statistics.
- HOUR 3 Preprocessing during training: augmentation from flips to MixUp, on-the-fly pipelines in torchvision, Albumentations, Keras, and MONAI.

Bring the take-home artifact: we will diagnose a few live.

Source decks used in class 1

1 **16-385 Computer Vision, Fall 2023, Carnegie Mellon University**, Ioannis Gkioulekas. Lecture 9 (Geometric camera models), Lectures 24–25 (Digital photography), Lecture 3 (Image pyramids and frequency domain), Lecture 2 (Image filtering).

http://16385.courses.cs.cmu.edu/fall2023/lectures

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2 **MIT OpenCourseWare, 6.003 Signals and Systems, Fall 2011**, Dennis Freeman. Lecture 22, Sampling and Quantization.

https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/

License CC BY-NC-SA 4.0; the deck's own OCW attribution page is retained at the end of part A4.

3 **BME 495 Deep Learning for Biomedical Imaging, Winter 2025, Northwestern University**, Ulas Bagci. Lectures 1–2, introduction to medical imaging and modalities.

https://bagcilab.com/courses/

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Libraries, sample data, and further reading

4 **Authored slides** (marked in the footer) were written for this course. Demo figures were computed with **scikit-image** (sample images: astronaut, moon, Shepp-Logan phantom) and **matplotlib**; code examples use public APIs of **OpenCV**, **scikit-image**, and **torchvision**.
`scikit-image.org · docs.opencv.org · docs.pytorch.org/vision`

5 **Szeliski, Computer Vision: Algorithms and Applications, 2nd edition** (2022). Chapters 2 and 3 are the textbook backbone for parts A and B; free personal-use PDF from the author.
`https://szeliski.org/Book/` (link only, per the author's terms)

6 **CS231n course notes**, data preprocessing section (MIT license), background reading for part C and the bridge into class 2.
`https://cs231n.github.io/neural-networks-2/`

7 **Compilation note.** Curated by T. Intharah for the Research Master's module Data Preparation for ML and AI, Khon Kaen University, August 2026. For classroom use; each source retains its own license.